PATWARI

Recruitment Guide

According to the syllabus of Patwari (BPSC)

MCQs
(General Science)

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ATOMIC STRUCTURE AND RADIOACTIVITY

What do you call the positively charged particles found in the center of an atom? (Source: General science text book, BBISE)

(a) **Protons**
(b) Electrons
(c) Neutrons
(d) Ions

What are the negatively charged particles that orbit the atom in specific shells?
(a) Protons
(b) **Electrons**
(c) Neutrons
(d) Nucleus

The number of electrons is equal to what in a neutral atom? (Source: General science text book, BBISE)
(a) The number of neutrons
(b) The mass number
(c) **The atomic number (number of protons)**
(d) The volume of the nucleus

What do you call the number of protons present in the nucleus of an atom? (Source: General science text book, BBISE)
(a) Mass number
(b) **Atomic number**
(c) Number of electrons
(d) Number of neutrons

The mass number (A) of an atom is equal to the sum of what? (Source: General science text book, BBISE)
(a) The number of protons and electrons
(b) The number of neutrons and electrons
(c) **The number of protons and neutrons**
(d) The number of protons only

What is the number of neutrons in Helium? (Source: General science text book, BBISE)
(a) 1
(b) **2**
(c) 4
(d) 5

What is the mass number of Uranium? (Source: General science text book, BBISE)
(a) 92
(b) 146
(c) **238**
(d) 237

What are atoms of the same element that have the same atomic number but different mass numbers called? (Source: General science text book, BBISE)
(a) Isobars
(b) Isotones
(c) **Isotopes**
(d) Ions

How many neutrons are in the nucleus of the hydrogen isotope Protium? (Source: General science text book, BBISE)
(a) **0**
(b) 1
(c) 2
(d) 3

How many neutrons are in the nucleus of the hydrogen isotope Deuterium? (Source: General science text book, BBISE)

(a) 0
(b) **1**
(c) 2
(d) 3

How many neutrons are in the nucleus of the hydrogen isotope Tritium? (Source: General science text book, BBISE)
(a) 0
(b) 1
(c) **2**
(d) 3

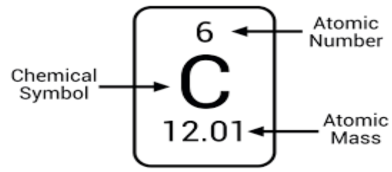
How many naturally occurring isotopes of Tin are there? (Source: General science text book, BBISE)
(a) 2
(b) 3
(c) 5
(d) **10**

n which part of the atom does most of its mass concentrate? (Source: General science text book, BBISE)
(a) In the shells
(b) In the orbits
(c) **In the nucleus**
(d) In the empty space

In a neutral atom, if the number of protons is 'Z', what is the number of electrons? (Source: General science text book, BBISE)
(a) Z - 1
(b) Z + 1
(c) **Z**
(d) A - Z

If the mass number of an element is 'A' and the atomic number is 'Z', how do you calculate the number of neutrons? (Source: General science text book, BBISE)
(a) A + Z
(b) Z - A
(c) **A - Z**
(d) A * Z

What symbol is generally used to represent the atomic number? (Source: General science text book, BBISE)
(a) A
(b) N
(c) **Z**
(d) M



What symbol is generally used to represent the mass number? (Source: General science text book, BBISE)
(a) **A**
(b) Z
(c) N
(d) P

If an element has an atomic number (Z) of 79, what is the number of protons and electrons in its neutral atom? (Source: General science text book, BBISE)
(a) 92 protons, 92 electrons
(b) **79 protons, 79 electrons**
(c) 79 protons, 118 neutrons
(d) 118 protons, 79 electrons

How many neutrons does Carbon (C) have? (Source: General science text book, BBISE)
(a) 4
(b) 6
(c) **6**
(d) 12

According to the table provided in the text, how many protons does Silver (Ag) have? (Source: General science text book, BBISE)
(a) 47
(b) 61

(c) **47**
(d) 108

How is the mass number (A) of an atom calculated? (Source: General science text book, BBISE)
(a) Protons + Electrons
(b) Neutrons + Electrons
(c) **Protons + Neutrons**
(d) Protons only.

According to the table, what is the number of protons for gold? (Source: General science text book, BBISE)
(a) 64
(b) 118
(c) 29
(d) **79**

What is the mass number of the Uranium isotope that has 146 neutrons? (Source: General science text book, BBISE)
(a) 92
(b) **238**
(c) 235
(d) 234
Explanation: The table shows Uranium with 92 protons and 146 neutrons.

$$92 + 146 = 238$$

What causes the difference in mass (different mass numbers) among isotopes of the same element? (Source: General science text book, BBISE)
(a) Different number of protons
(b) Different number of electrons
(c) **Different number of neutrons**
(d) Different atomic number

Most isotopes are found in what form? (Source: General science text book, BBISE)
(a) Artificially created
(b) Synthetically produced
(c) **Naturally occurring**
(d) Only in labs

How many total isotopes of Uranium are mentioned in the text book? (Source: General science text book, BBISE)
(a) One
(b) Two
(c) **Three**
(d) Ten

Which element mentioned in the text book has ten isotopes? (Source: General science text book, BBISE)
(a) Hydrogen
(b) Carbon
(c) Uranium
(d) **Tin**

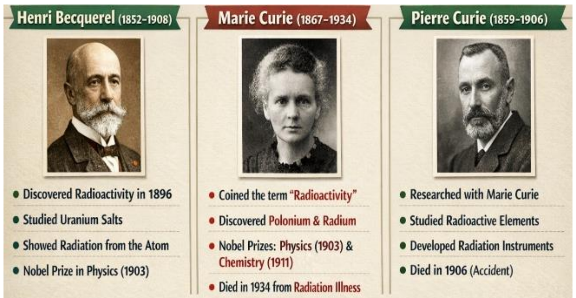
What is the common characteristic among all isotopes of a single element? (Source: General science text book, BBISE)
(a) Their mass number
(b) Their number of neutrons
(c) Their atomic mass
(d) **Their atomic number**
Explanation: Isotopes are defined as having the *same* atomic number but differing in mass number/neutron number.

According to the table, which element has a neutron-to-proton ratio of exactly 1:1? (Source: General science text book, BBISE)
(a) Silver
(b) Uranium
(c) **Carbon**
(d) Helium

Who accidentally discovered the phenomenon of radioactivity in 1896? (Source: General science text book, BBISE)
(a) Marie Curie

(b) **Henry Becquerel**
(c) Ernest Rutherford
(d) Niels Bohr

Explanation: The text book states, in 1896 AD, a **French physicist Henry Becquerel** accidentally discovered the phenomenon of radioactivity."



What material did Henry Becquerel use when he discovered radioactivity? (Source: General science text book, BBISE)
(a) Lead
(b) Aluminum
(c) **Uranium salt**
(d) Potassium

What observation led Becquerel to hypothesize that invisible rays were being emitted? (Source: General science text book, BBISE)
(a) The sample became hot
(b) The sample glowed in the dark
(c) **The uranium left marks on the photographic plate**
(d) The air around the sample ionized

What term is used for elements that emit invisible rays from their atoms? (Source: General science text book, BBISE)
(a) Stable elements
(b) Transuranic elements
(c) **Radioactive elements**
(d) non-radioactive elements

Why do radioactive elements transform into new elements? (Source: General science text book, BBISE)
(a) They lose electrons
(b) **They emit radioactive rays/particles**
(c) They gain protons
(d) They change their chemical state

Which type of atoms are defined as those that change into new atoms over time or are radioactive? (Source: General science text book, BBISE)
(a) Stable atoms
(b) Lead atoms
(c) **Unstable atoms**
(d) Man-made atoms

Which atoms are classified as "stable atoms"? Source: General science text book, BBISE)
(a) Any atom heavier than lead
(b) All radioactive atoms
(c) **All remaining atoms (that do not change over time)**
(d) Only laboratory-made atoms

Which naturally occurring element is generally the heaviest *stable* element mentioned, with all elements heavier than it being radioactive? (Source: General science text book, BBISE)
(a) Uranium
(b) Potassium
(c) **Lead**

Which of the following elements has natural isotopes that are radioactive? (Source: General science text book, BBISE)
(a) Tin
(b) Aluminum
(c) **Potassium**
(d) Lead

Explanation: Potassium, Carbon, and Oxygen are elements, whose some isotopes are radioactive.

What kind of elements did scientists begin creating after the discovery of radioactivity? (Source: General science text book, BBISE)

(a) Naturally occurring elements
(b) Stable elements
(c) **Elements not naturally found (transuranic)**
(d) Only alpha particles

Which of these is an example of a laboratory-made (synthetic) element mentioned in the text book? (Source: General science text book, BBISE)

(a) Uranium
(b) Lead
(c) Potassium
(d) **Plutonium**

How many types of rays/particles emerge from the nucleus of unstable atoms? (Source: General science text book, BBISE)

(a) Two
(b) **Three**
(c) Four
(d) Five

What are the three types of nuclear radiations called? (Source: General science text book, BBISE)

(a) Protons, Neutrons, Electrons
(b) X-rays, Gamma rays, Beta rays
(c) **Alpha particles, Beta particles, and Gamma radiations**
(d) Alpha rays, Beta rays, and Neutron rays

What is the composition of Alpha (α) particles? (Source: General science text book, BBISE)

(a) High-speed electrons
(b) Electromagnetic radiation
(c) **Helium nucleus**
(d) Neutrons

What is the composition of Beta (β) particles? (Source: General science text book, BBISE)

(a) Helium nucleus
(b) **High-speed electron**
(c) Electromagnetic radiation
(d) Protons

What is the composition of Gamma (γ) radiation? (Source: General science text book, BBISE)

(a) High-speed electrons
(b) Neutrons
(c) Protons
(d) **Electromagnetic radiation**

Which type of radiation listed has the lowest penetrating power? (Source: General science text book, BBISE)

(a) Beta particles
(b) Gamma rays
(c) Neutrons
(d) **Alpha particles**

Explanation: Alpha rays are stopped by "Paper/Skin".

Which material is capable of stopping Beta (β) particles? (Source: General science text book, BBISE)

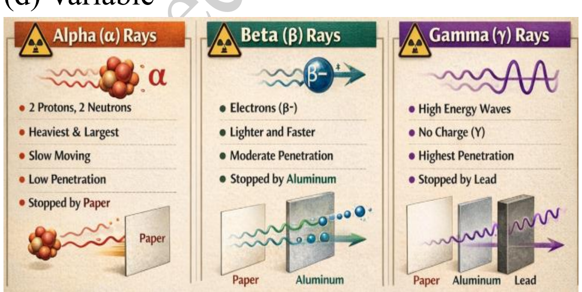
(a) Paper
(b) **Aluminum**
(c) skin
(d) thin gold foil

Which material is needed to effectively block Gamma (γ) radiation? (Source: General science text book, BBISE)
(a) Paper/Skin
(b) Aluminum
(c) **Lead and concrete**
(d) Water

What is the approximate speed of Alpha particles compared to the speed of light? (Source: General science text book, BBISE)
(a) One-third
(b) Half
(c) Equal to
(d) **One-tenth**

Alpha particles consist of how many protons and neutrons? (Source: General science text book, BBISE)
(a) 1 proton, 1 neutron
(b) 1 proton, 2 neutrons
(c) 2 protons, 1 neutron
(d) **2 protons, 2 neutrons**

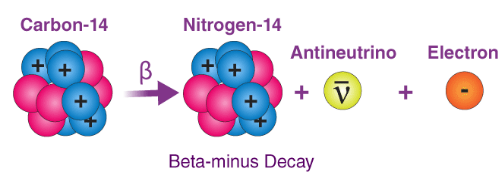
What kind of charge do Alpha particles carry? (Source: General science text book, BBISE)
(a) Negative
(b) Neutral
(c) **Positive**
(d) Variable



How far can alpha rays typically travel in the air? (Source: General science text book, BBISE)
(a) Several meters
(b) **Only a few centimeters**
(c) Over a kilometer
(d) Indefinitely

Beta particles carry what type of charge? (Source: General science text book, BBISE)
(a) Positive
(b) Neutral
(c) **Negative**
(d) Variable

Beta particles are physically high-speed what? (Source: General science text book, BBISE)
(a) Protons
(b) Neutrons
(c) **Electrons**
(d) Helium nuclei



How far can beta particles typically travel in the air? (Source: General science text book, BBISE)
(a) A few centimeters
(b) Only 2 meters
(c) **Up to a few meters**
(d) Tens of kilometers

When beta particles are stopped by a heavy metallic barrier, what phenomenon is produced as a result? (Source: General science text book, BBISE)
(a) Alpha rays
(b) Gamma rays
(c) **X-rays**

What kind of waves are Gamma radiation? (Source: General science text book, BBISE)
(a) Sound waves
(b) Radio waves
(c) **Electromagnetic waves**
(d) Water waves

Gamma radiations possess which characteristics regarding wavelength and energy? (Source: General science text book, BBISE)
(a) Long wavelength, low energy
(b) Long wavelength, high energy
(c) **Very short wavelength, very high energy**
(d) Very short wavelength, low energy
Explanation: Their wavelength is very short and their energy is very high.

At what speed do Gamma radiations travel? (Source: General science text book, BBISE)
(a) Speed of sound
(b) 1/10th speed of light
(c) Slower than light
(d) **Speed of light**

What general term is used for isotopes that emit radiation? (Source: General science text book, BBISE)
(a) Stable isotopes
(b) Heavy isotopes
(c) **Radioisotopes**
(d) Neutral isotopes
Explanation: Such isotopes that are emitting radioactive rays are called radioisotopes.

What device is explicitly mentioned as being used to detect the presence of tagged atoms in living organisms? (Source: General science text book, BBISE)
(a) Microscope
(b) Telescope
(c) X-ray machine
(d) **Geiger counter**

Which specific radioisotope is used for treating "Goiter/Deep Tumor"? (Source: General science text book, BBISE)
(a) **Iodine-131**
(b) Cobalt-60
(c) Phosphorus-32
(d) Iron-56

Which specific radioisotope is listed for treating "heart diseases"? (Source: General science text book, BBISE)
(a) I-131
(b) Co-60
(c) **Na-24**
(d) P-32

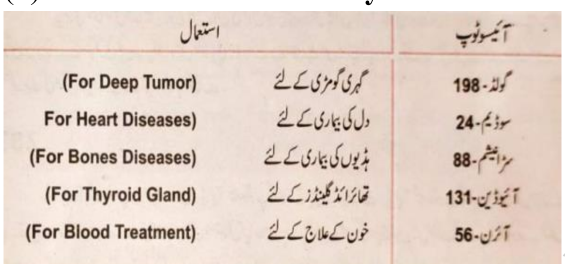
Which specific radioisotope is listed for treating "Bones Diseases"? (Source: General science text book, BBISE)
(a) Na-24
(b) Co-60
(c) I-131
(d) **P-32**

Which specific radioisotope is listed for treating "Blood Treatment"? (Source: General science text book, BBISE)
(a) **Fe-56**
(b) P-32
(c) I-131
(d) Co-60

What general use of Iodine-131 (I-131) is listed besides Goiter/Tumor treatment? (Source: General science text book,

BBISE)
(a) Heart diseases
(b) Blood treatment
(c) Bone diseases
(d) **Thyroid gland treatment**
Explanation: I-131 is used for both Goiter and Thyroid Gland issues

How are the strong radioactive rays used in treatment (like for cancer)? (Source: General science text book, BBISE)
(a) Injected directly into the bloodstream
(b) Tagged to roots of plants
(c) Used to check thickness of materials
(d) **Focused to burn/destroy diseased cells**



استعمال		آئیسوٹوپ
(For Deep Tumor)	گہری گومڑی کے لئے	گولڈ-198
For Heart Diseases)	دل کی بیماری کے لئے	سوڈیم-24
(For Bones Diseases)	ہڈیوں کی بیماری کے لئے	سٹرانشیم-88
(For Thyroid Gland)	تھائرائڈ گلینڈز کے لئے	آئیوڈین-131
(For Blood Treatment)	خون کے علاج کے لئے	آئرن-56

In agriculture, which element is mentioned as being tracked when absorbed through a plant's roots? (Source: General science text book, BBISE)
(a) Sodium
(b) Iron
(c) Cobalt
(d) **Phosphorus**
Explanation: The text book provides an example where plant roots absorb P-32 (Phosphorus 32).

Which type of radiation has the highest penetrating power and requires lead shielding? (Source: General science text book, BBISE)
(a) Alpha
(b) Beta
(c) **Gamma**
(d) Neutron
Explanation: The diagram indicates Gamma rays penetrate further than Alpha or Beta and require lead shielding.

Why are Gamma rays more penetrating than Alpha or Beta particles? (Source: General science text book, BBISE)
(a) They move slower
(b) They are charged particles
(c) **They have very high energy and no mass**
(d) They are helium nuclei
Explanation: They are high-energy electromagnetic waves that travel at the speed of light.

What is the key difference between Beta particles and Alpha particles in terms of composition? (Source: General science text book, BBISE)
(a) Alpha is EM radiation, Beta is a particle
(b) Alpha is an electron, Beta is a nucleus
(c) **Alpha is a Helium nucleus, Beta is an electron**
(d) Alpha has negative charge, Beta has positive charge

What initiates the fission process in a Uranium-235 nucleus? (Source: General science text book, BBISE)

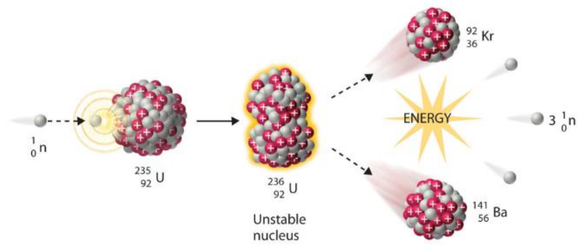
(a) A fast neutron
(b) A proton
(c) **A slow neutron**
(d) A gamma ray
Explanation: The text book begins: "When a slow neutron enters the nucleus of Uranium 235, the Uranium atom splits..."

What is the process of a heavy nucleus splitting into lighter nuclei called? (Source: General science text book, BBISE)
(a) Fusion
(b) Decay
(c) **Fission**
(d) Chain Reaction

How many new neutrons are typically emitted during a single fission event of Uranium-235? (Source: General science text book, BBISE)
(a) One
(b) Two
(c) **Three**
(d) Four

What do the emitted neutrons do after the initial fission event? (Source: General science text book, BBISE)
(a) They slow down and stop
(b) They combine to form a nucleus
(c) **They break the nuclei of other Uranium atoms in sequence**
(d) They turn into gamma rays

What is the continuous process of sequential breaking of uranium nuclei called? (Source: General science text book, BBISE)
(a) Fusion process
(b) Fission process
(c) **Chain Reaction**
(d) Decay process
Explanation: "Since there are millions of atoms in a one cubic centimeter piece of uranium, fission in all these atoms occurs so fast that it seems like a continuous process. This continuous process is called a Chain Reaction."



What two new elements does a Uranium atom transform into after it splits? (Source: General science text book, BBISE)
(a) Krypton and Sodium
(b) Uranium-238 and Lead
(c) Helium and Barium
(d) **Barium-141 and Krypton-92**
Explanation: "After the Uranium atom splits, it transforms into two new elements, Barium-141 and Krypton-92."

What else is emitted during the transformation into Barium and Krypton, besides the main nuclei? (Source: General science text book, BBISE)
(a) Protons
(b) Electrons
(c) Alpha particles
(d) **Three neutrons**

What principle explains how mass is converted into energy during nuclear fission? (Source: General science text book, BBISE)
(a) Law of Conservation of Mass
(b) The First Law of Thermodynamics
(c) Newton's Third Law
(d) **Einstein's Mass-Energy Equivalence Theory**

During fission, the total mass of the resulting Barium, Krypton, and three neutrons is different from the original

Uranium nucleus. What happens to this mass difference? (Source: General science text book, BBISE)
(a) It is destroyed
(b) It is converted into light
(c) It becomes heat
(d) **It appears in the form of energy**

The energy released during fission consists of what two components? (Source: General science text book, BBISE)
(a) Light and Sound
(b) **Kinetic energy of particles and Gamma rays**
(c) Heat and Radio waves
(d) Alpha and Beta particles

What must be done to make this energy usable for practical purposes? (Source: General science text book, BBISE)
(a) Increase the amount of Uranium
(b) Start the fusion process
(c) **Control the chain reaction**
(d) Stop the emission of gamma rays

What is the consequence if the chain reaction is NOT controlled? (Source: General science text book, BBISE)
(a) The energy produced will be small
(b) Only gamma rays will be produced
(c) The reaction will stop immediately
(d) **The immense energy produced will destroy everything in that place**

The destruction caused by an atomic bomb is due to what? (Source: General science text book, BBISE)
(a) The sound waves it produces
(b) The alpha particles released
(c) **The immense energy produced during the chain reaction**

What is the result of controlling the chain reaction? (Source: General science text book, BBISE)
(a) More atomic bombs are made
(b) The energy is wasted
(c) **This immense energy is being used for useful work**
(d) Only X-rays are produced

Why does a single cubic centimeter of Uranium sustain a rapid, continuous chain reaction? (Source: General science text book, BBISE)
(a) It is very dense material
(b) It is easy to enrich Uranium
(c) **It contains millions of atoms, allowing neutrons to find targets easily**
(d) It produces only two neutrons per fission fast that it seems like a continuous process."

What specific isotope of Uranium is primarily discussed as undergoing fission with a slow neutron? (Source: General science text book, BBISE)
(a) Uranium-238
(b) Uranium-234
(c) **Uranium-235**
(d) Uranium-239

What must balance when considering Einstein's mass-energy equivalence theory during fission? (Source: General science text book, BBISE)
(a) Protons and neutrons must be equal
(b) The sum of masses on both sides of the reaction
(c) **The original mass of Uranium equals the final mass plus the energy released**
(d) The number of gamma rays and neutrons

Why do the newly emitted neutrons need to be slowed down during a controlled

chain reaction? (Source: General science text book, BBISE)
(a) They absorb too much energy otherwise
(b) **They are too fast to collide with other nuclei effectively**
(c) They turn into protons if they remain fast
(d) They would escape the reactor immediately
Explanation: Because these neutrons are very high energy, they become very fast. To control the process, it is necessary to slow down their speed.

What substance is commonly used as a moderator to slow down fast neutrons in a nuclear reactor? (Source: General science text book, BBISE)
(a) Cadmium
(b) Uranium-235
(c) **Graphite/Heavy water**
(d) Lead

What is the composition of the "Heavy Water D_2O" used as a moderator? (Source: General science text book, BBISE)
(a) It is highly enriched with Hydrogen
(b) It is made of regular Hydrogen isotopes
(c) **It is made of the Hydrogen isotope Deuterium**
(d) It is a mixture of water and graphite
Explanation: The molecule of heavy water is made from the Hydrogen isotope Deuterium."

What is the general term for substances used to slow down the speed of neutrons in a nuclear reactor? (Source: General science text book, BBISE)
(a) Absorbers
(b) Fissile materials
(c) Reflectors
(d) **Moderators**

Which element is used in rods to completely absorb excess neutrons to help control the chain reaction rate? (Source: General science text book, BBISE)
(a) Graphite
(b) Uranium
(c) Hydrogen
(d) **Cadmium**
Explanation: Cadmium is an element that absorbs neutrons. Cadmium rods are placed between the moderators to achieve this control.

Why is it necessary to stop the chain reaction at will within a nuclear reactor? (Source: General science text book, BBISE)
(a) To increase the number of neutrons
(b) **To prevent the reaction from accelerating beyond control**
(c) To produce X-rays
(d) To keep the neutrons moving fast
Explanation: This control is necessary "so that the process can be stopped as needed" and to ensure safe operation.

What happens if the amount of Uranium-235 available for the chain reaction is below a certain threshold (critical mass)? (Source: General science text book, BBISE)
(a) The reaction speeds up uncontrollably
(b) More neutrons are produced
(c) **The chain reaction cannot be sustained**
(d) The heavy water boils immediately
Explanation: "It is also necessary that a sufficient amount of Uranium is present to sustain the chain reaction." If the quantity is

too small, neutrons escape and the reaction dies out.

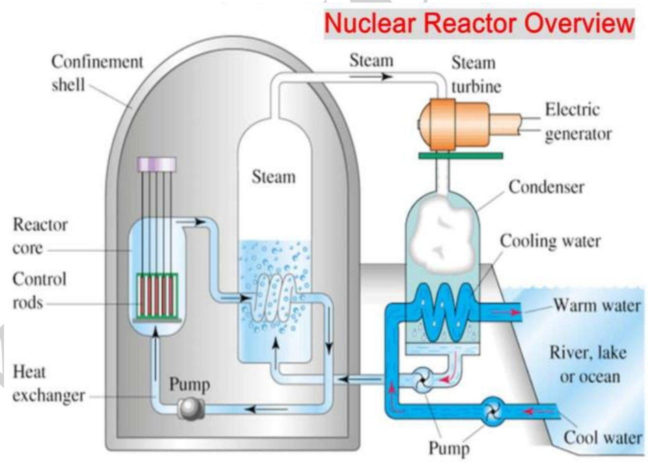
What is the name given to the minimum amount of Uranium required to sustain a chain reaction? (Source: General science text book, BBISE)
(a) Maximum Mass
(b) Fissile Mass
(c) **Critical Mass**
(d) Supercritical Mass

What type of nuclear reactor technology is installed at the Institute of Nuclear Science and Technology (PINSTECH) near Islamabad? (Source: General science text book, BBISE)
(a) Pressurized Water Reactor (PWR)
(b) Boiling Water Reactor (BWR)
(c) Fast Neutron Reactor
(d) **Small research reactor**

What location near Islamabad uses heavy water to slow down neutrons in its reactor? (Source: General science text book, BBISE)
(a) Karachi Nuclear Power Plant
(b) Chashma Power Plant
(c) **PINSTECH (Institute of Nuclear Science and Technology)**
(d) Dera Ghazi Khan

What is the main reason that neutrons can get "lost" or "wasted" if they are too fast? (Source: General science text book, BBISE)
(a) They run out of energy
(b) They are absorbed by the heavy water
(c) **They don't collide with other Uranium nuclei**
(d) They are absorbed by the cadmium rods
Explanation: "If on their way they do not collide with another atom's nucleus, they are wasted."

The total fission of one kilogram of Uranium-235 produces energy equivalent to burning approximately how much coal? (Source: General science text book, BBISE)
(a) 30 kilograms
(b) 300 kilograms
(c) 30,000 kilograms
(d) **3 million kilograms**



How many major components does a nuclear reactor have? (Source: General science text book, BBISE)
(a) Two
(b) Three
(c) **Four**
(d) Five

What is the first major component of a nuclear reactor? (Source: General science text book, BBISE)
(a) A moderator material
(b) A neutron absorber
(c) An energy absorber
(d) **A fuel that can undergo fission**

What is the purpose of the second major component of a nuclear reactor? (Source:

General science text book, BBISE)
(a) To absorb energy
(b) To absorb neutrons
(c) **To slow down fast neutrons**
(d) To produce fast neutrons

What is the function of the third major component of a nuclear reactor? (Source: General science text book, BBISE)
(a) To slow down neutrons
(b) To generate electricity
(c) **To absorb neutrons and stop fission as needed**
(d) To absorb energy

What is the role of the fourth major component of a nuclear reactor? (Source: General science text book, BBISE)
(a) To generate steam
(b) To control the reaction
(c) **To absorb energy produced in the reactor and convert it into a useful form**
(d) To provide fuel rods

Which metal is commonly used as fuel in nuclear reactors? (Source: General science text book, BBISE)
(a) Cadmium
(b) Lead
(c) **Uranium**
(d) Graphite

What substance is used to absorb energy and transfer it to the turbine? (Source: General science text book, BBISE)
(a) Heavy water
(b) Cadmium
(c) Uranium
(d) **Water**

Where are the fuel rods located within the reactor core? (Source: General science text book, BBISE)
(a) Outside the pressure vessel
(b) Between the control rods
(c) **Inside the core and between the moderator**
(d) Near the generator

How do the neutron-absorbing rods move within the reactor? (Source: General science text book, BBISE)
(a) Horizontally
(b) Diagonally
(c) Stay stationary
(d) **Up and down between the moderator**

What flows through the pipes to absorb energy? (Source: General science text book, BBISE)
(a) Steam
(b) Fuel
(c) Neutrons
(d) **Water**

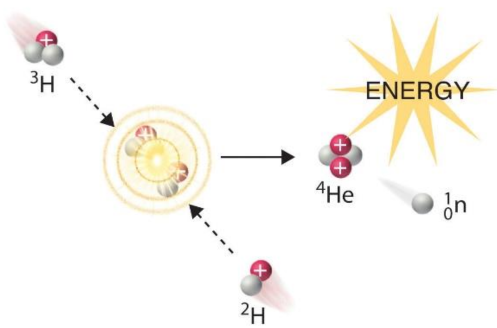
What structure surrounds the reactor core and pipes? (Source: General science text book, BBISE)
(a) A wooden box
(b) A cadmium shield
(c) **A steel pressure vessel**
(d) A generator

What surrounds the steel pressure vessel? (Source: General science text book, BBISE)
(a) A layer of lead
(b) A water tank
(c) A secondary loop
(d) **A block of concrete**

What is the purpose of the concrete block around the pressure vessel? (Source: General science text book, BBISE)

(a) To absorb heat
(b) To absorb neutrons
(c) To provide structural support
(d) **To prevent dangerous radioactive rays from escaping**

In which process do light elements, such as hydrogen, combine to form a relatively heavier element like helium? (Source: General science text book, BBISE)
(a) Fission
(b) Radioactivity
(c) **Fusion**
(d) Chain Reaction
Explanation: In the process of fusion, light elements like hydrogen nuclei combine to form a relatively heavier element, for example a helium nucleus."



How does the amount of mass after a fusion reaction compare to the total mass of the initial light elements? (Source: General science text book, BBISE)
(a) The final mass is exactly the same.
(b) The final mass is greater.
(c) **The final mass is less, with the difference converting to energy.**
(d) The mass is completely destroyed.

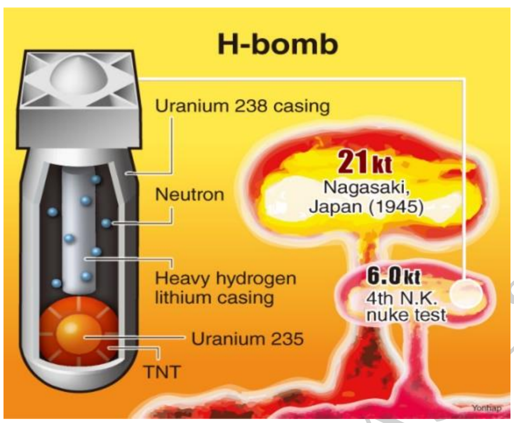
What extremely high temperature is required to start the fusion process involving hydrogen nuclei? (Source: General science text book, BBISE)
(a) Approximately 100 degrees Celsius
(b) Approximately 1,000 degrees Celsius
(c) Approximately one million degrees Celsius
(d) **Approximately ten million degrees Celsius**

What happens during this specific fusion reaction involving four hydrogen nuclei? (Source: General science text book, BBISE)
(a) Energy is absorbed from the surroundings.
(b) A massive explosion occurs instantly.
(c) **Immense energy is released.**
(d) The process stops automatically.

Why is it currently very difficult to sustain the fusion reaction in laboratories? (Source: General science text book, BBISE)
(a) The fuel is too rare to find.
(b) **Creating and maintaining such high temperatures is extremely difficult.**
(c) The resulting helium is radioactive.
(d) We have not achieved success in controlling it yet.
Explanation: "Generating such a high temperature in laboratories and maintaining it for some time in a regular manner is an extremely difficult task."

What energy source, if harnessed successfully, could solve the world's energy problem permanently and cleanly? (Source: General science text book, BBISE)
(a) Coal power
(b) Fission power
(c) **Fusion power**
(d) Solar power

Why is the fuel used in the fusion process considered to be 'unlimited' and clean? (Source: General science text book, BBISE)
(a) It is easily manufactured in labs.
(b) It is produced during fission reactions.
(c) **It uses deuterium, an isotope of hydrogen found abundantly in the world.**
(d) It uses light elements like lithium and lead.
Explanation: "Because the fuel used in the fusion process is deuterium, an isotope of the most abundant element, hydrogen, in the world, which is available in unlimited quantities all over the world."



Where in nature does the fusion process continuously occur? (Source: General science text book, BBISE)
(a) In the Earth's core
(b) In nuclear power plants
(c) In hydrogen bombs on Earth
(d) **In the centers of stars**

The energy released during the ongoing fusion process in stars spreads out in what forms? (Source: General science text book, BBISE)
(a) Heat and sound waves
(b) Alpha and Beta particles
(c) **Light and other radiation waves**
(d) Kinetic energy of particles

What process is used to create non-controlled energy in a hydrogen bomb? (Source: General science text book, BBISE)
(a) Fission only
(b) Stable decay
(c) **Fusion**
(d) Solar reaction

A hydrogen bomb's core (center) contains what device? (Source: General science text book, BBISE)
(a) A fusion reactor
(b) **An atom bomb**
(c) A block of concrete
(d) A sphere of helium

What surrounds the atom bomb core in a hydrogen bomb design, according to the text? (Source: General science text book, BBISE)
(a) Uranium-235
(b) A layer of lead
(c) **A layer of a light element, such as tritium**
(d) A layer of heavy water

What event initiates the fusion reaction inside a hydrogen bomb? (Source: General science text book, BBISE)
(a) A slow neutron is absorbed.
(b) The temperature reaches ten million degrees.
(c) The light element core is compressed.
(d) **The atom bomb core explodes, creating immense heat.**
Explanation: "When the atom bomb explodes, a lot of heat is produced. Due to

this heat, the fusion process starts in the tritium present around the core."

Compared to an atom bomb, a hydrogen bomb is considered to be: (Source: General science text book, BBISE)
(a) Less destructive
(b) Equally destructive
(c) **More destructive**
(d) A different type of element

What happens to some amount of mass during both the fission and fusion processes? (Source: General science text book, BBISE)
(a) It remains constant.
(b) It turns into lead.
(c) It is destroyed completely.
(d) **It is converted into energy.**

Why hasn't the non-controlled fusion reaction been achieved for peaceful energy purposes yet? (Source: General science text book, BBISE)
(a) It only works in stars.
(b) It is too expensive to research.
(c) **Scientists have not managed to contain the reaction for a useful duration.**
(d) The fuel is radioactive and dangerous.
Explanation: "Generating such a high temperature in laboratories and maintaining it for some time in a regular manner is an extremely difficult task."

The fundamental difference between fission and fusion is: (Source: General science text book, BBISE)
(a) Fission is controlled, fusion is not.
(b) Fission uses light elements, fusion uses heavy elements.
(c) **Fission is splitting atoms; fusion is combining atoms.**
(d) Fission happens in stars; fusion happens on Earth.

What element's isotope, Tritium, is used in the creation of a hydrogen bomb? (Source: General science text book, BBISE)
(a) Helium
(b) Lead
(c) Uranium
(d) **Hydrogen**.

A hydrogen bomb relies on what two stages of nuclear reactions? (Source: General science text book, BBISE)
(a) Alpha decay then Beta decay
(b) Fusion only
(c) **Fission (trigger) then Fusion (main reaction)**
(d) Chemical explosion then fission

In which year did Pakistan decide to use nuclear energy for peaceful purposes? (Source: General science text book, BBISE)
(a) 1947
(b) 1950
(c) **1955**
(d) 1972
Explanation: For this purpose, the Pakistan Atomic Energy Commission was established in 1955."

A nuclear power plant was built in Karachi in 1972 with the cooperation of which country? (Source: General science text book, BBISE)
(a) USA
(b) China
(c) **Canada**
(d) Russia
Explanation: "In 1972, a power plant was

built in Karachi with the cooperation of Canada."

What is the total power generation capacity of the Karachi nuclear power plant mentioned in the text? (Source: General science text book, BBISE)
(a) 92 MW
(b) **137 MW**
(c) 500 MW
(d) 104 MW

What type of fuel is used in the Karachi nuclear power plant? (Source: General science text book, BBISE)
(a) Heavy water
(b) Coal
(c) **Enriched uranium**
(d) Deuterium

In which fields, besides energy production, is nuclear technology being promoted in Pakistan? (Source: General science text book, BBISE)
(a) Agriculture and Defense
(b) Medicine and Defense
(c) **Agriculture, Medicine, and Industry**
(d) Industry and Defense

Where is a high-grade Nuclear Institute for Agriculture Research located? (Source: General science text book, BBISE)
(a) Karachi
(b) Islamabad
(c) **Faisalabad**
(d) Peshawar

What institute near Peshawar introduced the use of nuclear radiation for preserving food for longer periods? (Source: General science text book, BBISE)
(a) PINSTECH
(b) PAEC
(c) NIAR
(d) **NIFA (Nuclear Institute for Food and Agriculture)**

What is the biggest "abuse" or inappropriate use of nuclear energy mentioned? (Source: General science text book, BBISE)
(a) Creating medical isotopes
(b) Generating electricity
(c) **Destruction through the atomic bomb and hydrogen bomb**
(d) Agricultural research

What process starts immediately when an atom or hydrogen bomb explodes? (Source: General science text book, BBISE)
(a) Fusion reaction
(b) Chemical reaction
(c) **Nuclear fission chain reaction**
(d) Alpha decay

ELEMENTS CREATED IN LABORATORIES

Which element was the first to be produced artificially? (Source: General science text book, BBISE)

A. Neptunium
B. Plutonium
C. Technetium
D. Promethium

Technetium was synthesized in 1937 by bombarding molybdenum with deuterons.

What is the atomic number of the element Oganesson? (Source: General science text book, BBISE)

A. 114
B. 116
C. 118
D. 120

Oganesson is currently the highest-numbered element on the periodic table and completes the 7th period.

Which synthetic element is used in household smoke detectors? (Source: General science text book, BBISE)

A. Curium
B. Americium-241
C. Berkelium
D. Californium

Elements with an atomic number greater than 92 are generally called what? (Source: General science text book, BBISE)

A. Lanthanides
B. Actinides
C. Trans-uranium elements
D. Noble gases

Any element with an atomic number higher than Uranium (92) is classified as transuranic; most must be created in a lab.

Which element was named after the creator of the periodic table? (Source: General science text book, BBISE)

A. Bohrium
B. Mendelevium
C. Seaborgium
D. Nobelium

Mendelevium (atomic number 101) was synthesized in 1955 and named in honor of Dmitri Mendeleev.

Which element was discovered in the debris of the first hydrogen bomb test? (Source: General science text book, BBISE)

A. Technetium
B. Einsteinium
C. Roentgenium
D. Copernicium

Einsteinium (99) and Fermium (100) were first identified in the fallout of the "Ivy Mike" nuclear test in 1952.

Which synthetic element is named after the physicist who discovered the nucleus? (Source: General science text book, BBISE)

A. Meitnerium
B. Curium
C. Rutherfordium
D. Fermium

Rutherfordium (104) is named in honor of Ernest Rutherford, the father of nuclear physics.

Which element was named after a female scientist? (Source: General science text book, BBISE)

A. Meitnerium
B. Seaborgium
C. Roentgenium
D. Copernicium

Meitnerium (109) is named after Lise Meitner, a pioneer in nuclear fission.

What is the primary decay mode for most superheavy synthetic elements? (Source: General science text book, BBISE)

A. Alpha decay
B. Beta decay
C. Electron capture
D. Gamma emission

Most elements beyond atomic number 100 are unstable and decay by emitting alpha particles (helium nuclei).

Which synthetic element is produced by bombarding Bismuth with Iron ions? (Source: General science text book, BBISE)

A. Copernicium
B. Meitnerium
C. Flerovium
D. Oganesson

Which synthetic element is a member of the noble gas group (Group 18)? (Source: General science text book, BBISE)

A. Tennessine
B. Oganesson
C. Livermorium
D. Flerovium

Which of the following elements is classified as a "Transactinide"? (Source: General science text book, BBISE)

A. Curium
B. Americium
C. Rutherfordium
D. Uranium

Transactinides are elements with atomic numbers 104 to118, following the actinide series.

What target material was used to synthesize Livermorium (116) at the Dubna facility? (Source: General science text book, BBISE)

A. Curium-248
B. Lead-208
C. Plutonium-244
D. Californium-249

Livermorium was produced by bombarding a Curium-248 target with Calcium-48 ions.

Which element was the first "transuranium" element to be discovered? (Source: General science text book, BBISE)

A. Neptunium
B. Plutonium
C. Americium
D. Curium

Neptunium (93) was synthesized in 1940

The "Transfermium Wars" refers to a dispute over what? (Source: General science text book, BBISE)

A. The cost of particle accelerators
B. The naming rights of elements 104-109
C. The safety of nuclear tests
D. The ownership of plutonium

This was a long-standing naming dispute between American, Russian, and German research teams.

Which synthetic element is named after the physicist who developed the theory of relativity? (Source: General science text

book, BBISE)
A. Roentgenium
B. Copernicium
C. Einsteinium
D. Bohrium
Einsteinium (99) was named in honor of Albert Einstein.

Which synthetic element is used as a trigger for nuclear weapons? (Source: General science text book, BBISE)
A. Technetium
B. Plutonium
C. Promethium
D. Nihonium
Plutonium-239 is the primary fissile isotope used in modern nuclear weapon cores.

Which synthetic element is named after the Danish physicist who developed the model of the atom? (Source: General science text book, BBISE)
A. Rutherfordium
B. Bohrium
C. Hassium
D. Meitnerium
Bohrium (107) is named after Niels Bohr, who contributed foundational work to atomic structure and quantum theory.

What is the atomic number of the element Darmstadtium? (Source: General science text book, BBISE)
A. 110
B. 112
C. 114
D. 116
Darmstadtium was first synthesized in 1994 and is named after the city of Darmstadt, Germany.

Which synthetic element is named after the astronomer who proposed the heliocentric model? (Source: General science text book, BBISE)
A. Galilleium
B. Copernicium
C. Keplerium
D. Newtonium
Copernicium (112) was named to honor Nicolaus Copernicus for his contributions to science.

Which of the following is a synthetic radioactive element found in the lanthanide series? (Source: General science text book, BBISE)
A. Neodymium
B. Samarium
C. Promethium
D. Gadolinium
Promethium (61) is the only lanthanide that does not have stable isotopes and is produced in nuclear reactors.

Which element has the longest-lived isotope among the transactinides? (Source: General science text book, BBISE)
A. Dubnium
B. Seaborgium
C. Bohrium
D. Hassium
Dubnium-268 has a half-life of about 28 hours, which is relatively long compared to other superheavy elements.

What is the chemical symbol for Moscovium? (Source: General science text book, BBISE)
A. Mo
B. Ms

C. Mc
D. Mv
Moscovium (115) uses the symbol Mc and is named after the Moscow region in Russia.

Which element was synthesized by bombarding Lead-208 with Nickel-64? (Source: General science text book, BBISE)
A. Flerovium
B. Livermorium
C. Darmstadtium
D. Roentgenium
Darmstadtium-269 was produced using this specific heavy-ion fusion reaction in 1994.

What is the physical state of Oganesson at room temperature predicted to be? (Source: General science text book, BBISE)
A. Solid
B. Liquid
C. Gas
D. Plasma
Oganesson is predicted to be a solid at room temperature despite being in the noble gas group.

Which synthetic element is used to treat certain types of cancer through targeted alpha therapy? (Source: General science text book, BBISE)
A. Technetium
B. Astatine-211
C. Plutonium
D. Fermium

Which synthetic element is named after the laboratory located in the city of Darmstadt, Germany? (Source: General science text book, BBISE)
A. Hassium
B. Bohrium
C. Darmstadtium
D. Meitnerium
Darmstadtium (110) was first synthesized at the GSI Helmholtz Centre for Heavy Ion Research in Darmstadt.

Which synthetic element was named after the Russian nuclear physicist Yuri Oganessian? (Source: General science text book, BBISE)
A. Moscovium
B. Dubnium
C. Oganesson
D. Flerovium
Oganesson (118) is the only element named after a person who was alive at the time of the official naming in 2016.

Which synthetic element is the only one in the lanthanide period (Period 6)? (Source: General science text book, BBISE)
A. Neodymium
B. Promethium
C. Holmium
D. Thulium
Promethium (61) is the only lanthanide that must be artificially produced, as it has no stable isotopes.

Which synthetic element is produced when Americium-243 is bombarded with Calcium-48? (Source: General science text book, BBISE)
A. Nihonium
B. Moscovium
C. Tennessine
D. Oganesson
The fusion of Americium (95) and Calcium (20) creates element 115 (Moscovium).

Which element is named after the physicist who led the Manhattan Project? (Source: General science text book, BBISE)

A. Einsteinium
B. Fermium
C. None of the above
D. Oppenheimerium

There is currently no element named after Robert Oppenheimer.

What is the symbol for the synthetic element Roentgenium? (Source: General science text book, BBISE)

A. Ro
B. Rg
C. Rt
D. Rn

Roentgenium (111) uses the symbol Rg and belongs to Group 11 (the coinage metals).

What is the estimated half-life of Oganesson-294? (Source: General science text book, BBISE)

A. 0.89 milliseconds
B. 10 minutes
C. 5 hours
D. 2 days

Oganesson is extremely short-lived, with a half-life of less than one millisecond.

Which synthetic element has the longest-lived isotope, with a half-life of 2.14 million years? (Source: General science text book, BBISE)

A. Plutonium
B. Neptunium
C. Curium
D. Americium

Neptunium-237 is the most stable isotope of neptunium, significantly longer than most other early synthetic elements.

Which synthetic element is named after the Polish scientist who pioneered radioactivity research? (Source: General science text book, BBISE)

A. Meitnerium
B. Curium
C. Fermium
D. Lawrencium

Curium (96) is named after Marie and Pierre Curie for their extensive research into radioactivity.

Who is credited with the discovery of the electron using cathode ray tube experiments? (Source: General science text book, BBISE)

A. Ernest Rutherford
B. James Chadwick
C. J.J. Thomson
D. John Dalton

Thomson's experiments with cathode rays led to the discovery of the first subatomic particle, the electron, in 1897.

The Gold Foil Experiment led to the discovery of which part of the atom? (Source: General science text book, BBISE)

A. Electron
B. Nucleus
C. Neutron
D. Quark

Rutherford observed alpha particles reflecting at sharp angles, proving the existence of a dense, positively charged center.

What is the maximum number of electrons that can occupy the 'p' subshell? (Source: General science text book, BBISE)
A. 2
B. 6
C. 10
D. 14

Which scientist proposed the "Plum Pudding" model of the atom? (Source: General science text book, BBISE)
A. J.J. Thomson
B. Niels Bohr
C. Robert Millikan
D. Erwin Schrödinger
Thomson envisioned the atom as a sphere of positive charge with electrons embedded like raisins in a pudding.

Isotopes of the same element have the same number of protons but different numbers of:
A. Electrons
B. Neutrons
C. Positrons
D. Quarks

Which particle has the least mass among the following? (Source: General science text book, BBISE)
A. Proton
B. Neutron
C. Electron
D. Alpha particle

The "uncertainty principle," stating that position and momentum cannot both be known precisely, was proposed by:
A. Max Planck
B. Louis de Broglie
C. Werner Heisenberg
D. Albert Einstein

What is the value of Planck's constant (h)? (Source: General science text book, BBISE)
A. 6.626×10^{-34} J·s
B. 3.00×10^{8} m/s
C. 1.602×10^{-19} C
D. 6.022×10^{23} mol^{-1}
Planck's constant relates the energy of a photon to its frequency.

Who discovered the neutron in 1932? (Source: General science text book, BBISE)
A. Ernest Rutherford
B. Enrico Fermi
C. James Chadwick
D. Marie Curie

The region in space where there is a high probability of finding an electron is called:
A. Nucleus
B. Orbital
C. Orbit
D. Shell

Which subatomic particle's mass is used as the standard for 1 atomic mass unit (amu)? (Source: General science text book, BBISE)
A. Electron
B. Hydrogen atom
C. 1/12th of Carbon-12
D. Oxygen-16

Which experiment determined the charge of an electron? (Source: General science text book, BBISE)

A. Cathode Ray Experiment
B. Millikan’s Oil Drop Experiment
C. Alpha Scattering Experiment
D. Photoelectric Effect

What happens to the energy of an electron as it moves further from the nucleus? (Source: General science text book, BBISE)
A. It increases
B. It decreases
C. It remains constant
D. It drops to zero

The wave-particle duality of matter was proposed by:
A. Erwin Schrödinger
B. Louis de Broglie
C. Albert Einstein
D. Max Planck

Which quantum number describes the orientation of an orbital in space? (Source: General science text book, BBISE)
A. Principal
B. Azimuthal
C. Magnetic
D. Spin

What is the shape of an 's' orbital? (Source: General science text book, BBISE)
A. Spherical
B. Dumbbell
C. Cloverleaf
D. Circular

In the symbol $^{40}_{20}Ca$, what does the number 40 represent? (Source: General science text book, BBISE)
A. Atomic number
B. Mass number
C. Number of electrons
D. Number of protons

Who proposed that electrons move in fixed circular orbits around the nucleus? (Source: General science text book, BBISE)
A. J.J. Thomson
B. Niel Bohr
C. John Dalton
D. Erwin Schrödinger
Bohr's 1913 model introduced quantized orbits to explain the stability of atoms.

THE CELLULAR BASES OF LIFE

Structurally, both living and non-living things are made of what? (Source: General science text book, BBISE)

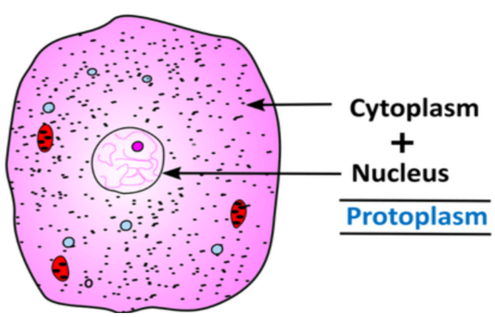
(a) Cells
(b) Protoplasm
(c) **Chemical compounds**
(d) Metabolism

What biological substance comes into existence when chemical compounds organize themselves in living organisms? (Source: General science text book, BBISE)

(a) The Cell
(b) Non-living matter
(c) **Protoplasm**
(d) The chemical compounds themselves

Where is protoplasm found within an organism's basic unit? (Source: General science text book, BBISE)

(a) In chemical compounds
(b) Outside the cell
(c) In non-living things
(d) **In the cell**



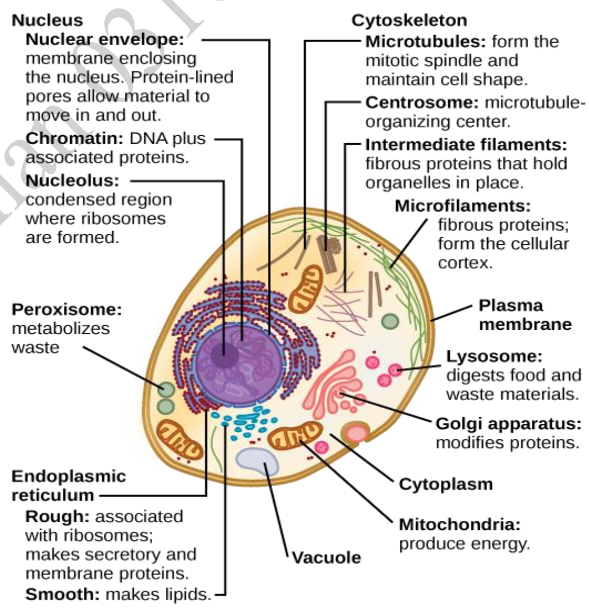
What is impossible without protoplasm? (Source: General science text book, BBISE)

(a) Chemical reactions
(b) Organization of matter
(c) **Life**
(d) Metabolism

The sum of constructive and destructive reactions happening within protoplasm is called what? (Source: General science text book, BBISE)

(a) Protoplasm formation
(b) Sensation
(c) Respiration
(d) **Metabolism**

Explanation: catabolism is a destructive process while anabolism is a constructive process.



If the metabolic process stops, what is the consequence? (Source: General science text book, BBISE)

(a) The organism grows rapidly.
(b) New organisms are produced.
(c) **Life becomes stagnant, and living and non-living cannot be distinguished.**
(d) Movement increases.

The Theory of Spontaneous Generation stated what about life's origin? (Source: General science text book, BBISE)
(a) Life came from outer space.
(b) Life originated from a single cell.
(c) **Living organisms could create themselves spontaneously from non-living matter.**
(d) Life came from chemical compounds only.

According to Aristotle's view cited in the text book, from what could living things arise? (Source: General science text book, BBISE)
(a) Water vapor
(b) Methane and Ammonia
(c) **Dew drops and waste products**
(d) Amino acids

Which scientist initially believed that bacteria could be made from non-living things? (Source: General science text book, BBISE)
(a) Redi
(b) Spallanzani
(c) Miller
(d) **William Harvey**

In the seventeenth century, which Italian scientist proved that living bodies cannot be produced from non-living matter? (Source: General science text book, BBISE)
(a) Spallanzani
(b) Miller
(c) **Redi**
(d) Operion

In 1923, which two scientists proposed a hypothesis about the Earth's early atmosphere? (Source: General science text book, BBISE)
(a) Redi and Spallanzani
(b) Miller and Harvey
(c) **Operion and Haldane**
(d) Aristotle and Miller

What was unique about Earth's early atmosphere according to Operion & Haldane's hypothesis? (Source: General science text book, BBISE)
(a) It had oxygen like today.
(b) It had very cold temperatures.
(c) **It contained methane, ammonia, andwater vapors at very high temperatures.**
(d) It was made entirely of amino acids.

What was formed from these gases reacting at high temperatures in the early atmosphere? (Source: General science text book, BBISE)
(a) The first cell
(b) The first living organism
(c) **Organic compounds**
(d) Oxygen gas

In 1953, who conducted an experiment to verify Operion & Haldane's hypothesis? (Source: General science text book, BBISE)
(a) Operion
(b) Haldane
(c) **Miller**
(d) Redi

Besides other organic compounds, what specific substances did Miller obtain in his 1953 experiment? (Source: General science text book, BBISE)
(a) Nucleic Acids
(b) Proteins
(c) Helium nuclei
(d) **Amino Acids**

Other scientists repeated Miller's experiment with different gases and successfully obtained what crucial biological compound? (Source: General science text book, BBISE)
(a) Amino Acids
(b) Protoplasm
(c) Proteins
(d) **Nucleic Acid**

These compounds (amino acids, nucleic acids) obtained in laboratory settings are known to be present in what vital biological substance today? (Source: General science text book, BBISE)
(a) Water Vapors
(b) Ammonia
(c) Methane
(d) **Protoplasm**

According to the geological timeline, approximately how many billion years ago did the Earth come into existence? (Source: General science text book, BBISE)
(a) 1 billion years
(b) **Several billion years**
(c) 4.6 million years
(d) 10 million years
Explanation: The actual age is estimated at 4.5 to 4.6 billion years.

What was the initial temperature condition of the early Earth immediately after its formation? (Source: General science text book, BBISE)
(a) Very low temperature
(b) **Very high temperature**
(c) Moderate temperature
(d) Fluctuating temperature

As the Earth's temperature gradually decreased, water vapor condensed to form what? (Source: General science text book, BBISE)
(a) The early atmosphere
(b) Methane and ammonia
(c) New organic compounds
(d) **Oceans/Seas**

What gases were believed to be present in the early atmosphere? (Source: General science text book, BBISE)
(a) Oxygen, Nitrogen, Carbon Dioxide
(b) Helium, Lead, Carbon Dioxide
(c) **Methane, Ammonia, Hydrogen gas, and Water vapor**
(d) Nitrogen and Oxygen only

What two energy sources in the early environment were also part of the atmosphere? (Source: General science text book, BBISE)
(a) Heat and Wind
(b) **Sparks (Lightning) and Ultraviolet (UV) Rays**
(c) Gamma Rays and Infrared Waves
(d) Volcanic Eruptions and Gravity

Which statement describes the most basic type of life form believed to have first existed?(Source: General science text book, BBISE)
(a) Complex multicellular organisms
(b) Plants that photosynthesize
(c) **Single-celled organisms**
(d) Fish in the oceans

The existence of these first simple organisms was based on the functions of what fundamental substance? (Source: General science text book, BBISE)
(a) DNA
(b) The Cell Wall

(c) **Protoplasm**
(d) Enzymes

Which of the following is *not* listed as necessary for the activity and survival of protoplasm? (Source: General science text book, BBISE)
(a) Oxygen
(b) Water
(c) **Carbon Dioxide**
(d) Salts and Organic compounds

The formation of the first simple cell likely occurred after which critical environmental event? (Source: General science text book, BBISE)
(a) The atmosphere became rich in oxygen
(b) Animals started existing
(c) **Earth's temperature decreased and various organic compounds formed**
(d) The sun started shining brightly

A hypothetical early cell would possess the ability to perform which two functions for its survival? (Source: General science text book, BBISE)
(a) Move and communicate
(b) Absorb UV rays and produce sparks
(c) **Reproduce and synthesize/oxidize food**
(d) Create heavy water and graphite

The protoplasm of this hypothetical early cell was likely based on a combination of which substances? (Source: General science text book, BBISE)
(a) Water and salts only
(b) Methane and ammonia
(c) **A few proteins, enzymes, and nucleic acids**
(d) Lipids and carbohydrates only

What structure likely formed around this early protoplasm to protect it and give it a cell-like shape? (Source: General science text book, BBISE)
(a) A cell wall made of cellulose
(b) A protein-only layer
(c) **A membrane of fats (lipids) and proteins**
(d) A thick concrete block

The characteristics of this initial hypothetical cell most closely resemble which current types of cells? (Source: General science text book, BBISE)
(a) Only plant cells
(b) Only animal cells
(c) **Current plant and animal cells**
(d) Only bacterial cells

The existence of life on early Earth required conditions and components similar to those necessary for life's survival in the present time, such as: (Source: General science text book, BBISE)
(a) High temperatures and UV radiation
(b) Methane-rich atmosphere
(c) Unstable organic compounds
(d) **Presence of water, oxygen, and organic/inorganic compounds**

Which of the following is listed as the second "accommodating factor for the origin of life" in the concluding list? (Source: General science text book, BBISE)
(a) Decrease in Earth's temperature
(b) Presence of oxygen, nitrogen, and carbon dioxide in the air
(c) **Presence of water**
(d) Synthesis of organic compounds

Which of the following is listed as the third factor in the list? (Source: General science text book, BBISE)
(a) Synthesis of enzymes

(b) Synthesis of food
(c) **Presence of oxygen, nitrogen, and carbon dioxide in the air**
(d) Decrease in Earth's temperature
Explanation: Point 3 in the list refers to these atmospheric gases.

The transition from Inorganic Compounds to Organic Compounds is referred to by what scientific term in the list? (Source: General science text book, BBISE)
(a) Fusion
(b) **Synthesis**
(c) Oxidation
(d) Decomposition
Explanation: Point 4 in the list specifies "Synthesis" next to the Urdu description of creating organic compounds from inorganic ones.

What general class of organic compounds, vital for speeding up reactions, is mentioned specifically in point 5 of the list? (Source: General science text book, BBISE)
(a) Proteins
(b) Fats (lipids)
(c) Nucleic acids
(d) **Enzymes**
Explanation: Point 5 is the "Synthesis of Enzymes" (biological catalysts).

What complex molecules are listed in point 6 as necessary for the origin of life? (Source: General science text book, BBISE)
(a) Inorganic compounds
(b) Carbohydrates
(c) **Nucleic acids**
(d) Salts
Explanation: Point 6 is the "Synthesis of Nucleic Acids," which carry genetic information.

What process, listed in point 8, is necessary for organisms to produce energy for survival? (Source: General science text book, BBISE)
(a) Food synthesis
(b) Enzyme synthesis
(c) Decreasing temperature
(d) **Oxidation of food**

Which term is defined as the "material of life" found in all living beings? (Source: General science text book, BBISE)
(a) Nucleus
(b) Cell Wall
(c) **Protoplasm**
(d) Cytoplasm

What percentage of a cell is typically composed of water, an inorganic? (Source: General science text book, BBISE)
(a) 33%
(b) 50%
(c) **96%**
(d) 100%

In a plant cell, what structure is found surrounding the cell membrane? (Source: General science text book, BBISE)
(a) Nucleus
(b) **A thin wall (Cell Wall)**
(c) Protoplasm
(d) Chromatin network

Which type of cells are described as primarily having the function of producing chemical secretions (secretions)? (Source: General science text book, BBISE)
(a) Eye cells

(b) Blood cells
(c) Brain cells
(d) **Gland cells**

Which organ's cells are specifically mentioned for their role in message transmission? (Source: General science text book, BBISE)
(a) Eyes
(b) Heart
(c) **Brain**
(d) Glands

What is the function of blood cells? (Source: General science text book, BBISE)
(a) Message transmission
(b) Chemical secretions
(c) **Carrying oxygen to different parts of the body**
(d) Carrying light signals

Which description of protoplasm's appearance is given? (Source: General science text book, BBISE)
(a) Hard and colorless
(b) Transparent and solid
(c) **Soft, semi-solid, and thick**
(d) Liquid and thin

What specific term is used for the "thread-like" structure found within the nucleus (when the cell is not dividing)? (Source: General science text book, BBISE)
(a) Nucleolus
(b) Cell Wall
(c) **Chromatin Network**
(d) Cytoplasm

The Chromatin Network condenses during cell division to form which structures? (Source: General science text book, BBISE)
(a) Genes
(b) Nuclei
(c) Ribosomes
(d) **Chromosomes**

Which class of organic compounds makes up a significant portion of animal bodies? (Source: General science text book, BBISE)
(a) Carbohydrates
(b) Fats
(c) Nucleic Acids
(d) **Proteins**

In addition to Carbon, Hydrogen, and Oxygen, which element is also used in the composition of proteins? (Source: General science text book, BBISE)
(a) Helium
(b) Sulfur
(c) **Nitrogen**
(d) Phosphorus

Which specific protein is found in the pancreas gland? (Source: General science text book, BBISE)
(a) Hemoglobin
(b) Casein
(c) **Insulin**
(d) Keratin

Which protein is responsible for carrying oxygen in the blood? (Source: General science text book, BBISE)
(a) Insulin
(b) Casein
(c) Keratin
(d) **Hemoglobin**

What three elements compose Carbohydrates? (Source: General science text book, BBISE)
(a) Carbon, Nitrogen, Oxygen
(b) Nitrogen, Hydrogen, Oxygen

(c) Carbon, Nitrogen, Hydrogen
(d) **Carbon, Hydrogen, Oxygen**

What is an important example of a Carbohydrate that provides energy through oxidation? (Source: General science text book, BBISE)
(a) Insulin
(b) Hemoglobin
(c) **Glucose**
(d) Glycerol

Where is the material that life functions depend on found? (Source: General science text book, BBISE)
(a) In the cytoplasm
(b) On the cell wall
(c) **On the chromosomes**
(d) In the nucleus fluid

What is the definition of a Gene? (Source: General science text book, BBISE)
(a) A type of chromosome
(b) A type of protein
(c) **A hereditary unit**
(d) A type of chemical reaction

What is the function of a Gene? (Source: General science text book, BBISE)
(a) To create chemical reactions
(b) To determine the sex of an individual
(c) To ensure cell division
(d) **To transfer a character from generation to generation**

Approximately how many genes are found on each chromosome? (Source: General science text book, BBISE)
(a) Only one
(b) A few hundred
(c) **Several thousand**
(d) Millions

Explanation: The human genome project estimated that humans have between 20,000 and 25,000 genes, distributed across 46 chromosomes.

What is the branch of biology that studies the transfer of characteristics based on genes? (Source: General science text book, BBISE)
(a) Biology
(b) Cell Biology
(c) Genetic Engineering
(d) **Genetics**

Explanation: This is the scientific study of genes, heredity, and variation in living organisms.

In the modern era, what technique is used to make changes in the genes of bacteria using chemical reactions? (Source: General science text book, BBISE)
(a) Genetics
(b) Chemical synthesis
(c) **Genetic Engineering**
(d) Chromosomal analysis

Explanation: Genetic engineering is the deliberate modification of the characteristics of an organism by manipulating its genetic material.

What organism is specifically mentioned as having its genes modified using chemical reactions in the modern era? (Source: General science text book, BBISE)
(a) Humans
(b) Plants
(c) Animals
(d) **Bacteria**

Explanation: Bacteria (like E. coli) are often used in labs for genetic engineering experiments due to their rapid reproduction rate and simple genetic structure.

The process that provides energy from the breakdown of glucose is called: (Source: General science text book, BBISE)

(a) Synthesis
(b) Fission
(c) Reduction
(d) **Oxidation**

Fats (Lipids) are formed chemically by the combination of Fatty Acids with which molecule? (Source: General science text book, BBISE)

(a) Glucose
(b) Insulin
(c) Hemoglobin
(d) **Glycerol**

Fats primarily consist of which elements? (Source: General science text book, BBISE)

(a) Nitrogen, Carbon, Hydrogen
(b) **Carbon, Hydrogen, Oxygen**
(c) Oxygen, Nitrogen, Carbon
(d) Hydrogen, Oxygen, Phosphorus
Explanation: Like carbohydrates, fats are composed of carbon, hydrogen, and oxygen.

How many types of Nucleic Acid compounds are mentioned in the text? (Source: General science text book, BBISE)

(a) One
(b) **Two**
(c) Three
(d) Four
Explanation: Two main types are mentioned: DNA and RNA.

What does the acronym DNA stand for in the context of Nucleic Acids? (Source: General science text book, BBISE)

(a) Direct Nucleic Acid
(b) Deuterium Nucleic Acid
(c) **Deoxyribonucleic Acid**
(d) Deoxyribose Acid

What does the acronym RNA stand for? (Source: General science text book, BBISE)

(a) Ribose Nucleic Acid
(b) **Ribonucleic Acid**
(c) Radiant Nucleic Acid
(d) Replication Nucleic Acid
Explanation: RNA is the second type of nucleic acid mentioned.

Where is DNA found exclusively within the cell? (Source: General science text book, BBISE)

(a) In the cytoplasm
(b) In the cell wall
(c) **In the Chromatin network**
(d) In the nucleus only

Where is RNA found mostly in the cell? (Source: General science text book, BBISE)

(a) Only in the nucleus
(b) In the chromatin network
(c) **In the Cytoplasm and Nucleus**
(d) In the cell membrane

What is the term for the overall chemical activity that occurs within the cell's main body (excluding the nucleus)? (Source: General science text book, BBISE)

(a) Oxidation
(b) Reproduction
(c) Synthesis
(d) **Metabolism**

What is the description of the Cell Membrane's permeability? (Source: General science text book, BBISE)

(a) Fully permeable
(b) Impermeable

(c) **Semi-permeable**
(d) Non-existent

How do materials pass through the Cell Membrane? (Source: General science text book, BBISE)
(a) Through an active pump
(b) Through large holes
(c) Through explosion
(d) **Through the process of osmosis/diffusion**

Where is the Nucleus typically located within the cell? (Source: General science text book, BBISE)
(a) Near the edge
(b) Near the vacuoles
(c) **Generally in the center**
(d) Scattered throughout the cytoplasm

What type of membrane encloses the nucleus? (Source: General science text book, BBISE)
(a) Fully permeable
(b) Impermeable
(c) Non-existent
(d) **Semi-permeable**

What does the Chromatin Network eventually form? (Source: General science text book, BBISE)
(a) Genes
(b) Nucleoli
(c) Protoplasm
(d) **Chromosomes**

How many nucleoli are typically found within the nucleus? (Source: General science text book, BBISE)
(a) Three or four
(b) Zero
(c) **One or two**
(d) Many scattered ones

Where is Cytoplasm located within the cell? (Source: General science text book, BBISE)
(a) Inside the nucleus
(b) Inside the vacuoles
(c) **Between the nucleus and the cell membrane**
(d) Outside the cell membrane

What does the thick, semi-transparent granular fluid (cytoplasm) contain? (Source: General science text book, BBISE)
(a) Only inorganic compounds
(b) Only organic compounds
(c) Only water
(d) **Both inorganic and organic compounds**

Besides compounds, what other structures are found within the cytoplasm? (Source: General science text book, BBISE)
(a) Genes
(b) Chromosomes
(c) The nucleus
(d) **Cell organelles**

What is the shape of a Vacuole? (Source: General science text book, BBISE)
(a) Thread-like
(b) Irregular
(c) **Round or square**
(d) Semi-solid

What typically fills a Vacuole? (Source: General science text book, BBISE)
(a) Cytoplasm and nucleus fluid
(b) Proteins and carbohydrates
(c) **Water, food particles, or nitrogenous compounds**
(d) Cell membrane material

What is a primary difference between an animal cell and a plant cell concerning

their outer boundary? ((Source: General science text book, BBISE)

(a) Animal cells have a thick wall; plant cells do not.

(b) **Plant cells have a cellulose cell wall outside the membrane; animal cells do not.**

(c) Both have a cell wall made of protein.

(d) Animal cells have a larger nucleus.

Which organelle is present in animal cells but absent in plant cells? (Source: General science text book, BBISE)

(a) Plastids

(b) Nucleus

(c) Vacuole

(d) **Centrosome**

Explanation: Point 3 states that the centrosome is present in animal cells but not in plant cells. This organelle is vital for organizing microtubules during cell division in animal cells.

Which organelle is absent in animal cells but found in plant cells? (Source: General science text book, BBISE)

(a) Centrosome

(b) Nucleus

(c) **Plastids**

(d) Mitochondria

What is the Nucleoplasm? (Source: General science text book, BBISE)

(a) The material outside the nucleus.

(b) The main material of life.

(c) **The fluid substance inside the nucleus.**

(d) The cell wall material.

Explanation: The text defines the fluid substance inside the nucleus as the nucleoplasm, which houses the cell's genetic material.

What two main types of structures are found within the Nucleoplasm? (Source: General science text book, BBISE)

(a) Protons and Neutrons

(b) DNA and RNA

(c) **Chromosomes and Nucleolus**

(d) Genes and Proteins

Explanation: The text lists chromosomes and the nucleolus as the two primary structures found within the nucleoplasm.

Chromosomes are formed from which material? (Source: General science text book, BBISE)

(a) Nucleolus

(b) Cytoplasm

(c) **Chromatin**

(d) Genes

Explanation: Chromosomes are condensed forms of chromatin, the genetic material state when the cell is not dividing.

Chromatin is composed of which two components? (Source: General science text book, BBISE)

(a) Fat and Carbohydrates

(b) DNA and RNA

(c) **Protein and DNA**

(d) Genes and Nucleoplasm

Explanation: Chromatin is a complex made of both protein and DNA, allowing for efficient packaging of genetic material.

Which molecule holds all the characteristics of chromosomes? (Source: General science text book, BBISE)

(a) Protein

(b) RNA

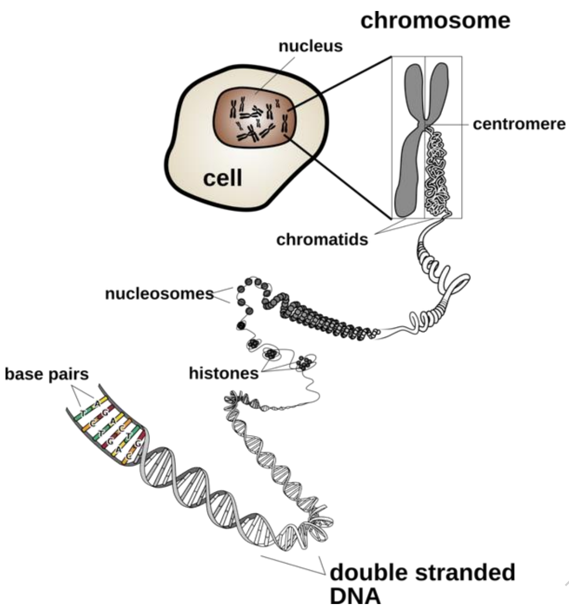
(c) Chromatin

(d) **DNA**

Explanation: All characteristics of chromosomes are based on the DNA molecule, which acts as the cell's genetic blueprint.

What is the structural description of a DNA molecule? (Source: General science text book, BBISE)

(a) A long straight line



(b) A simple protein
(c) **A coil/helix**
(d) A sphere
Explanation: Structurally, DNA is a coil or

When are chromosomes clearly visible? (Source: General science text book, BBISE)

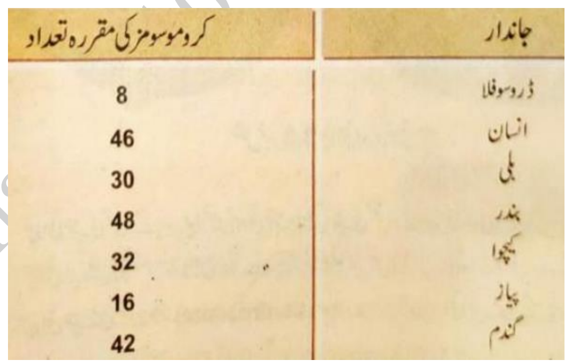
(a) When the cell is at rest
(b) When the organism grows
(c) **When the cell is dividing**
(d) When DNA is extracted
Explanation: Chromosomes are only clearly visible during cell division when the chromatin condenses into distinct structures.

Chromosomes control all cell functions through which contained molecule?) (Source: General science text book, BBISE)

(a) Protein
(b) Nucleolus
(c) Centrosome
(d) **DNA**
Explanation: The DNA within chromosomes acts as the central command system, controlling all cellular functions.

What happens if a genetic material defect occurs in chromosomes? (Source: General science text book, BBISE)

(a) It improves the individual's health
(b) It helps in the cell division process
(c) **It is harmful/damaging to the individual**
(d) It makes the individual resistant to diseases



کروموسومز کی مقررہ تعداد	جاندار
8	ڈروسوفلا
46	انسان
30	بلی
48	بندر
32	کیچوا
16	پیاز
42	گندم

How many chromosomes does a human have? (Source: General science text book, BBISE)

(a) 12
(b) 16
(c) 38
(d) **46**

Which organism listed has the highest number of chromosomes? (Source: General science text book, BBISE)

(a) Frog (26)
(b) Cow (60)
(c) Potato (48)
(d) **Dog (78)**

What is the function of the chromosomes after cell division? (Source: General science

text book, BBISE)
(a) They disappear completely
(b) They combine to form one massive chromosome
(c) They leave the nucleus for the cytoplasm
(d) **They are transferred into the new "daughter cell"**

The study of chromosome characteristics shows their extreme importance in:
(a) The weather cycle
(b) Industrial processes
(c) **The survival and evolution of life**.

How many chromosomes does the cell of "Rice" have? (Source: General science text book, BBISE)
(a) 12
(b) 16
(c) **24**
(d) 46

What is the chromosome count for an onion? (Source: General science text book, BBISE)
(a) 8
(b) **16**
(c) 32
(d) 42

How many chromosomes does a monkey have? (Source: General science text book, BBISE)
(a) 32
(b) 42
(c) 46
(d) **48**
Explanation: The table specifies that a monkey has 48 chromosomes.

What is the number of chromosomes found in wheat (گندم)? (Source: General science text book, BBISE)
(a) 32
(b) **42**
(c) 46
(d) 48

How many chromosomes does a turtle possess? (Source: General science text book, BBISE)
(a) 16
(b) 30
(c) **32**
(d) 42
Explanation: The table lists the turtle as having 32 chromosomes.

Which pair of sex chromosomes is typical for a female Drosophila fly? (Source: General science text book, BBISE)
(a) XY
(b) XO
(c) XXY
(d) **XX**

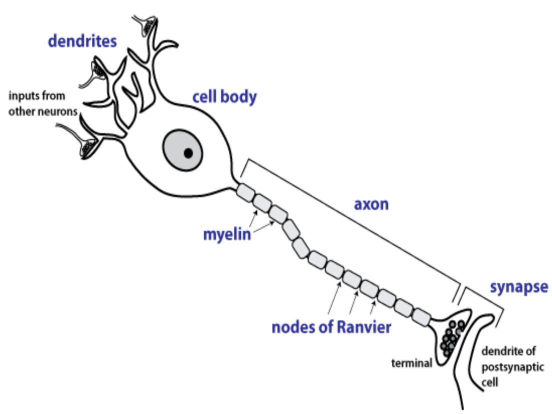
Which pair of sex chromosomes is typical for a male Drosophila fly? (Source: General science text book, BBISE)
(a) XX
(b) YO
(c) XXX
(d) **XY**.

Which organism listed in the table has exactly three dozen chromosomes? (Source: General science text book, BBISE)
(a) Human (46)
(b) Onion (16)
(c) **Turtle (32) (Approx 3 dozen/36, closest option)**
(d) Cat (30)

Which system establishes a very fast information delivery mechanism within the body using specialized cells? (Source: General science text book, BBISE)
(a) The hormonal system
(b) The muscular system
(c) The circulatory system
(d) **The nervous system**

What is the name of the specialized cell that makes up the nervous system? (Source: General science text book, BBISE)
(a) Muscle cell
(b) Gland cell
(c) **Neuron**
(d) Blood cell

What are the thread-like extensions that receive information from receptors? (Source: General science text book, BBISE)
(a) Axons
(b) Synapses
(c) Nuclei
(d) **Dendrites**



What is the function of "Receptors" in the nervous system? (Source: General science text book, BBISE)
(a) To transmit messages to muscles
(b) To send information away from the cell body
(c) **To receive stimuli/information**
(d) To insulate the axon

What part of the neuron transmits the impulse away from the cell body toward the muscles or other neurons? (Source: General science text book, BBISE)
(a) Dendrites
(b) Receptors
(c) **Axon**
(d) Synapse

What is the small gap or junction between two neurons called? (Source: General science text book, BBISE)
(a) Axon terminal
(b) Dendrite
(c) Receptor
(d) **Synapse**

What can the stimuli/excitations that enter the body be? (Source: General science text book, BBISE)
(a) Only chemical
(b) Only electrical
(c) Only mechanical
(d) **Chemical, electrical, or mechanical**

The nervous system sends instructions to muscles via which specific pathway? (Source: General science text book, BBISE)
(a) Through the synapse
(b) Through hormones
(c) Through dendrites
(d) **Through nerves**

What is the term for the message that travels along a neuron? (Source: General

science text book, BBISE)
(a) Stimulus
(b) Hormone
(c) **Nerve Impulse (**
(d) Synapse

How is a nerve impulse generally transmitted across the synapse gap? (Source: General science text book, BBISE)
(a) It jumps the gap as electricity.
(b) **Via a chemical process.**
(c) Via a mechanical shock.
(d) Via a hormonal signal.

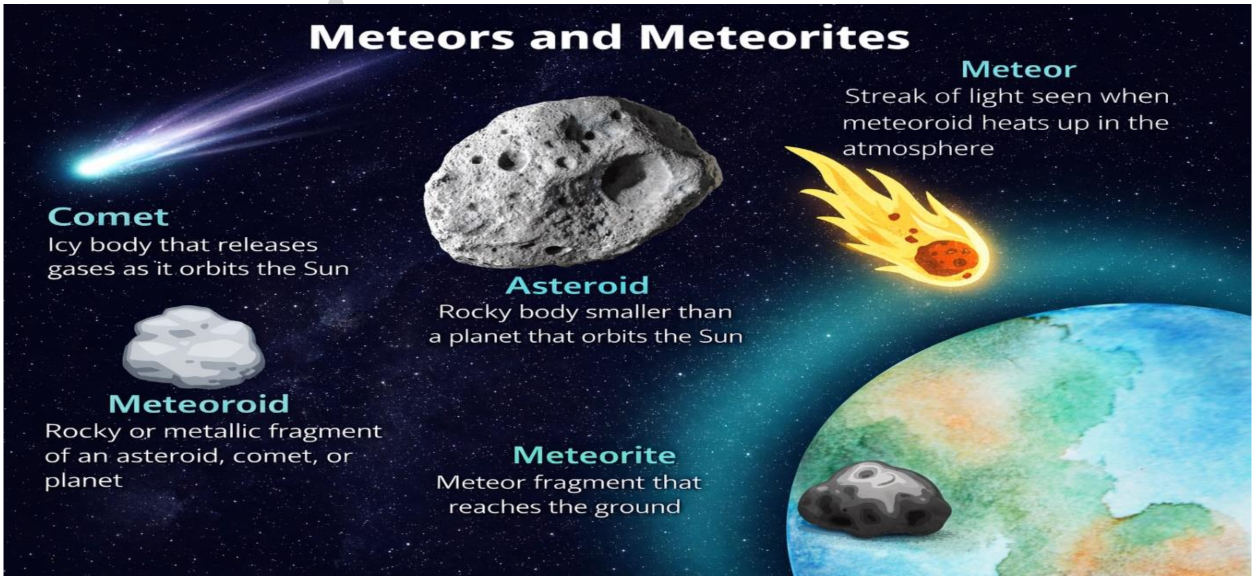
What happens when we feel pain? (Source: General science text book, BBISE)
(a) Hormones take over
(b) The muscles stop working
(c) **The brain receives instructions via nerves**
(d) The synapses close up

How does the speed of the nervous system's message transmission compare to the hormonal system's message transmission? (Source: General science text book, BBISE)
(a) Slower than hormonal
(b) Equal speed
(c) **Much faster than hormonal**
(d) Depends on the stimulus type

What specific type of object has been observed to hypothesize the possibility of life in space? (Source: General science text book, BBISE)
(a) Planets
(b) Comets
(c) **Meteorites**
(d) Asteroids

What are meteorites described as? (Source: General science text book, BBISE)
(a) Large stars moving in space
(b) **Solid bodies moving in space that fall towards planets**
(c) Types of asteroids that are very large
(d) Gaseous bodies found in space

How do meteorites compare in size to planets or asteroids? (Source: General science text book, BBISE)
(a) They are much larger than planets.



(b) They are similar in size to asteroids.
(c) They are usually the same size as planets.
(d) **They are much smaller than planets or asteroids.**

What two types of organic compounds were found when analyzing the composition of certain meteorites? (Source: General science text book, BBISE)
(a) Proteins and Water
(b) **Hydrocarbons and Amino Acids**
(c) Methane and Ammonia
(d) DNA and RNA

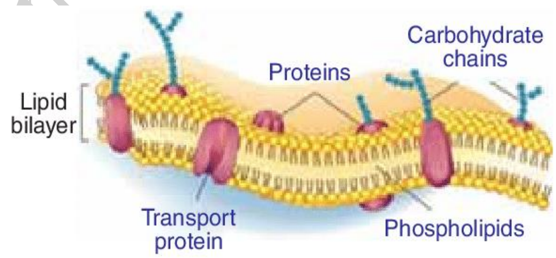
Explanation: Analysis of meteorites revealed the presence of "Hydrocarbons" and "Amino Acids". Amino acids are the building blocks of proteins, fundamental to life as we know it.

Has concrete evidence for the existence of life beyond Earth been found so far? (Source: General science text book, BBISE)
(a) Yes, many examples exist.
(b) Only on Mars.
(c) Yes, in the form of DNA.
(d) **No, there is currently no specific evidence.**

Which organelle is known as the "Powerhouse of the Cell"? (PakMcqs)
A) Nucleus
B) Ribosome
C) **Mitochondria**
D) Lysosome
Explanation: Mitochondria are responsible for cellular respiration, where they convert nutrients into ATP (Adenosine Triphosphate).

The basic structural and functional unit of all living organisms is the: (Testbook)
A) Tissue
B) Organ
C) **Cell**
D) Organelle

The fluid mosaic model of the plasma membrane was proposed by: (Examveda)
A) Robert Hooke
B) **Singer and Nicolson**
C) Watson and Crick
D) Schleiden and Schwann
Explanation: Proposed in 1972, this model describes the membrane as a "mosaic" of proteins, lipids, and carbohydrates that are fluid, meaning they can move laterally within the membrane.



Which organelle is primarily responsible for protein synthesis? (GeeksforGeeks)
A) **Ribosome**
B) Golgi Apparatus
C) Smooth ER
D) Vacuole
Explanation: Ribosomes are the sites where amino acids are assembled into proteins based on the genetic instructions from mRNA. They can be found floating in the cytoplasm or attached to the Rough ER.

Lysosomes are often referred to as "suicide bags" because they contain: (Byju's)

A) Food particles
B) **Digestive enzymes**
C) Genetic material
D) Pigments
Explanation: Lysosomes contain hydrolytic enzymes that break down waste and cellular debris. If a cell is badly damaged, the lysosome can burst and digest the cell.

The DNA of a prokaryotic cell is found in an irregularly shaped region called the: (Khan Academy)
A) Nucleus
B) **Nucleoid**
C) Nucleolus
D) Capsule

Which organelle is involved in the modification, sorting, and packaging of proteins? (Source: General science text book, BBISE) (GoPract)
A) **Golgi Apparatus**
B) Ribosome
C) Centriole
D) Chloroplast
Explanation: The Golgi apparatus receives proteins from the Endoplasmic Reticulum (ER), processes them (like adding sugar chains), and packages them into vesicles for transport.

The movement of water molecules across a selectively permeable membrane is called: (PakMcqs)
A) Diffusion
B) **Osmosis**
C) Active Transport
D) Phagocytosis
Explanation: Osmosis is a passive process that doesn't require energy. Water moves from an area of high-water concentration (low solute) to an area of low water concentration (high solute).

Which organelle is responsible for detoxifying drugs and poisons in liver cells? (Quizlet)
A) Rough ER
B) **Smooth ER**
C) Lysosome
D) Nucleus
Explanation: The Smooth Endoplasmic Reticulum lacks ribosomes and is involved in lipid synthesis and the detoxification of harmful chemicals and metabolic byproducts.

The cell wall of a fungus is primarily made of: (Examveda)
A) Cellulose
B) **Chitin**
C) Peptidoglycan
D) Starch

The process by which a cell engulfs large particles or whole cells is known as: (Biology Online)
A) Pinocytosis
B) **Phagocytosis**
C) Exocytosis
D) Facilitated Diffusion
Explanation: Often called "cell eating," phagocytosis is a type of endocytosis where the plasma membrane folds around a large object to form a vacuole.

Microtubules, microfilaments, and intermediate filaments together form the cell's: (LibreTexts Biology)
A) Cytoplasm
B) **Cytoskeleton**
C) Cell Wall
) Nucleoplasm

The folds of the inner mitochondrial membrane are called: (Britannica)
A) Grana
B) **Cristae**
C) Stroma
D) Matrix

The process of nuclear division that results in two daughter cells with the same number of chromosomes as the parent cell is: (PakMcqs)
A) Meiosis
B) **Mitosis**
C) Cytokinesis
D) Binary Fission
Explanation: Mitosis is used for growth, repair, and asexual reproduction. It ensures that the genetic consistency of a tissue is maintained by producing identical clones of the parent cell.

The exchange of genetic material between non-sister chromatids of homologous chromosomes is called: (Byju's)
A) Mutation
B) **Crossing over**
C) Independent Assortment
D) Segregation

A cell with two complete sets of chromosomes is referred to as: (IndiaBix)
A) Haploid
B) **Diploid**
C) Triploid
D) Polyploid
Explanation: In humans, somatic (body) cells are diploid (2n), containing 46 chromosomes. Gametes (sperm and egg) are haploid (n), containing only 23 chromosomes.

Which organelle is responsible for capturing light energy during photosynthesis? (Examveda)
A) Mitochondria
B) **Chloroplast**
C) Chromoplast
D) Leucoplast
Explanation: Chloroplasts contain the pigment **chlorophyll**, which absorbs light (mostly blue and red wavelengths) to drive the synthesis of glucose from CO_2

The division of the cytoplasm following nuclear division is known as: (Khan Academy)
A) Karyokinesis
B) **Cytokinesis**
C) Interphase
D) Meiosis
Explanation: While karyokinesis is the division of the nucleus, cytokinesis is the physical splitting of the cell body. In plants, this involves a "cell plate," whereas in animals, a "cleavage furrow" forms.

Programmed cell death is scientifically known as: (Biology Online)
A) Necrosis
B) **Apoptosis**
C) Autophagy
D) Senescence
Explanation: Apoptosis is a highly regulated process that removes old, unnecessary, or damaged cells without harming the surrounding tissue.

HUMAN HEALTH AND PHYSICAL DEVELOPMENT

What is the general term for the process of physical changes occurring within a living organism? (Source: General science text book, BBISE)
(a) Digestion
(b) Reproduction
(c) **Growth and Development**
(d) Excretion

The World Health Organization (WHO) defines health as a state of complete: [WHO Constitution]
A) Absence of disease
B) **Physical, mental, and social well-being**
C) Physical fitness and strength
D) Mental stability only

What specific process is mentioned that causes a person to feel hungry after a few hours? (Source: General science text book, BBISE)
(a) Digestion of water
(b) Excretion of food
(c) **Chemical reactions in the body**
(d) Hormone production

Which function of food involves repairing or replacing damaged parts of the body? (Source: General science text book, BBISE)
(a) Providing energy
(b) Maintaining temperature
(c) **Growth and repair of cells**
(d) Causing hunger

How does food contribute to maintaining body temperature? (Source: General science text book, BBISE)
(a) By providing water
(b) By creating hormones
(c) **By releasing heat energy during chemical reactions**
(d) By cooling the body down

Which term is used for chemical substances that control specific processes and the growth of the body? (Source: General science text book, BBISE)
(a) Enzymes
(b) Carbohydrates
(c) **Hormones**
(d) Protoplasm
Explanation: Hormones are identified as chemical substances that regulate body processes and growth.

Which substances act as catalysts to speed up chemical reactions within the body? (Source: General science text book, BBISE)
(a) Hormones
(b) Carbohydrates
(c) **Enzymes**
(d) Proteins
Explanation: Enzymes are described as organic compounds that facilitate and speed up necessary chemical reactions in the body.

Which category of food components includes "Starch" and "Sucrose"? (Source: General science text book, BBISE)

(a) Fats
(b) Proteins
(c) Vitamins
(d) **Carbohydrates**

What is the general term for substances in food that dissolve easily in water and provide immediate energy? (Source: General science text book, BBISE)
(a) Starch
(b) Cellulose
(c) **Sugars**
(d) Fiber
Explanation: In Carbohydrates "Sugars" as substances that dissolve easily in water and are absorbed quickly to provide energy.

Which type of carbohydrate is found in grains like wheat, rice, and maize? (Source: General science text book, BBISE)
(a) Sucrose
(b) Sugar
(c) Glucose
(d) **Starch**
Explanation: Starch is identified as a major component of these common grains.

What is the common table sugar used in homes called in a chemical context? (Source: General science text book, BBISE)
(a) Glucose
(b) Starch
(c) Cellulose
(d) **Sucrose**
Explanation: Sucrose is identified as the common "domestic sugar" used in homes.

Which class of food components are oily substances that do not dissolve in water? (Source: General science text book, BBISE)
(a) Carbohydrates
(b) Proteins
(c) Water
(d) **Fats (Lipids)**

What is the composition of fats? (Source: General science text book, BBISE)
(a) Only Carbon and Hydrogen
(b) Only Hydrogen and Oxygen
(c) Nitrogen, Carbon, and Hydrogen
(d) **Carbon, Hydrogen, and Oxygen**
Explanation: Fats primarily consist of the elements Carbon, Hydrogen, and Oxygen.

Fats are formed from the chemical combination of which two molecules? (Source: General science text book, BBISE)
(a) Glucose and Fatty Acids
(b) **Fatty Acids and Glycerol**
(c) Glycerol and Carbohydrates
(d) Proteins and Lipids

Where are fats stored in the body of humans and animals? (Source: General science text book, BBISE)
(a) In the bloodstream
(b) In the muscles
(c) **Under the skin and between organs**
(d) In the brain cells

What function does the layer of fat under the skin perform in the human body? (Source: General science text book, BBISE)
(a) Helps in muscle movement
(b) Aids in digestion
(c) **Prevents body heat from escaping (insulation)**
Which form of energy provides the highest amount of calories per gram?

(Source: General science text book, BBISE)
(a) Carbohydrates
(b) Proteins
(c) Water
(d) **Fats**
Explanation: The text states, "the energy amount (Calories) is very high in fats". Fats provide approximately 9 calories per gram, nearly double that of proteins or carbohydrates (around 4 calories per gram).

What is the function of fats when the body runs out of carbohydrates? (Source: General science text book, BBISE)
(a) They are immediately excreted.
(b) They turn into proteins.
(c) **They provide the necessary energy to the body.**
(d) They turn into water.

Which substance is essential for life but does not provide the body with energy? (Source: General science text book, BBISE)
(a) Fats
(b) Proteins
(c) Carbohydrates
(d) **Water**

Approximately what percentage of a cell's weight is composed of water? (Source: General science text book, BBISE)
(a) 50%
(b) **96%**
(c) 100%
(d) 4%

Water acts as a medium for what essential processes within the body? (Source: General science text book, BBISE)
(a) Energy storage
(b) Insulation
(c) **Chemical reactions (metabolism)**
(d) Protein synthesis

Water helps to maintain which physiological balance within the body? (Source: General science text book, BBISE)
(a) Blood pressure only
(b) Hormonal balance
(c) **Body temperature and normal functions**
(d) Fat storage levels

Which compound type is water classified as, despite its importance for life functions? (Source: General science text book, BBISE)
(a) Organic compound
(b) Protein
(c) Fat
(d) **Inorganic compound**

The process of "Oxidation of Food" is the primary way living beings produce what? (Source: General science text book, BBISE)
(a) Water
(b) Fats
(c) Hormones
(d) **Energy**

Which food item provides the highest amount of energy per 100g? (Source: General science text book, BBISE)
(a) Bread
(b) Ghee
(c) Sugar
(d) **Peanuts**
Explanation: Peanuts are listed at 555-648 MJ per 100g, the highest range in the table.

How much energy does 100g of Milk provide? (Source: General science text book, BBISE)
(a) 360 MJ
(b) 348 MJ
(c) **153 MJ**
(d) 100 MJ

How is a "balanced diet" defined? (Source: General science text book, BBISE)
(a) A diet consisting of only meat and vegetables.
(b) A diet that is expensive.
(c) **A diet that meets all the body's requirements.**
(d) A diet that only contains carbohydrates.

A balanced diet should include which main categories of food components? (Source: General science text book, BBISE)
(a) Only carbohydrates and fats
(b) Only proteins and water
(c) Only vitamins and salts
(d) **Carbohydrates, fats, proteins, vitamins, mineral salts, and water**

Why are the energy needs (calories) for infants and old people generally lower than those for young adults? (Source: General science text book, BBISE)
(a) They do more physical work.
(b) They need more fat insulation.
(c) **Their growth rate is different and metabolic needs are lower.**
(d) They only eat proteins.

What is recommended as the protein requirement per kilogram of body weight for youth? (Source: General science text book, BBISE)
(a) 0.5-1 gram
(b) **1.5-2 grams**
(c) 3-4 grams
(d) 1 gram

What fundamental function does mineral salt perform in the body? (Source: General science text book, BBISE)
(a) Store fat
(b) Generate immediate energy
(c) Act as hormones
(d) **Help in regulating normal bodily functions**

Explanation: "Mineral salts play an important role in the body's normal functions." They are crucial electrolytes and structural components (e.g., calcium in bones).

Which vitamins are classified as "Fats Soluble" according to the heading? (Source: General science text book, BBISE)
(a) Vitamin C and B complex
(b) Only Vitamin A and C
(c) **Vitamins A, D, E, and K**
(d) Vitamins A, B1, and C

Which vitamins are classified as "Water Soluble"? (Source: General science text book, BBISE)
(a) Vitamins A, D, E, and K
(b) **Vitamins of the B-complex group and Vitamin C**
(c) Only Vitamin A
(d) Only Vitamin C

The discovery that Vitamin A deficiency causes eye disease (xerophthalmia/night blindness) occurred in what year? (Source: General science text book,

BBISE)
(a) 1912
(b) 1941
(c) 1952
(d) **1906**

Explanation: It was discovered by Hopkins in 1906 that Vitamin A is vital for vision and immune function.

What is the chemical name for Vitamin A? (Source: General science text book, BBISE)
(a) Ascorbic acid
(b) Biotin
(c) Naphthalene
(d) **Retinol**

From which organic compound can Vitamin A be synthesized in the body? (Source: General science text book, BBISE)
(a) Retinol
(b) Naphthalene
(c) **Carotene**
(d) Alcohol

Which type of food is Carotene commonly found in? (Source: General science text book, BBISE)
(a) Meat and fish
(b) Dairy products only
(c) **Green vegetables, carrots, and fruits**
(d) Grains and pulses

What specific disease does Vitamin A prevent besides eye issues? (Source: General science text book, BBISE)
(a) Goiter
(b) Cholera
(c) Scurvy
(d) **Night blindness (nyctalopia)**

Which vitamin is essential for the body to absorb calcium and phosphorus for strong bones and teeth? (Source: General science text book, BBISE)
(a) Vitamin E
(b) Vitamin K
(c) Vitamin C
(d) **Vitamin D**

Which food item is a major natural source of Vitamin D? (Source: General science text book, BBISE)
(a) Meat
(b) Vegetables
(c) **Fish liver oil**
(d) Grains

Explanation: "Fish liver oil is a major source of Vitamin D." Sunlight exposure is also a primary source of this vitamin.

What is the function of Vitamin E in the body? (Source: General science text book, BBISE)
(a) Helps in calcium absorption
(b) Prevents night blindness
(c) Aids in blood clotting
(d) **Assists in the healthy creation of red blood cells**

A deficiency of which vitamin can lead to slow blood clotting or excessive bleeding? (Source: General science text book, BBISE)
(a) Vitamin A
(b) Vitamin C
(c) **Vitamin K**
(d) Vitamin D

Explanation: "A deficiency of Vitamin K causes the blood to clot slowly, and if a wound occurs, blood flow does not stop easily." Vitamin K is essential for synthesizing proteins involved in the coagulation process.

What is the chemical name for Vitamin B1? (Source: General science text book, BBISE)
(a) Riboflavin
(b) Ascorbic acid
(c) Naphthalene
(d) **Thiamine**

What term is used for substances that speed up chemical reactions, a function attributed to Vitamin B2? (Source: General science text book, BBISE)
(a) Hormones
(b) Protoplasm
(c) **Catalyst**
(d) Genes

Explanation: Vitamin B2 as acting as a "Catalyst" in the body's chemical reactions. Catalysts increase the rate of reactions without being consumed.

Vitamin A	Retinol
Vitamin B1	Thiamine
Vitamin B2	Riboflavin
Vitamin B3	Niacin
Vitamin B6	Pyridoxine
Vitamin B9	Folic Acid
Vitamin B12	Cobalamin
Vitamin C	Ascorbic Acid
Vitamin D2	Ergocalciferol
Vitamin D3	Cholecalciferol
VItamin E	Tocopherol

Which vitamin's deficiency is linked to a severe form of anemia (Pernicious Anemia)? (Source: General science text book, BBISE)
(a) Vitamin B1
(b) Vitamin B2
(c) Vitamin C
(d) **Vitamin B12**

Explanation: Vitamin B12 deficiency is specifically linked to "Pernicious Anemia," a condition where the body cannot make enough healthy red blood cells.

What is the chemical name for Vitamin C? (Source: General science text book, BBISE)
(a) Thiamine
(b) Retinol
(c) **Ascorbic acid**
(d) Riboflavin

What type of compounds are mineral salts classified as? (Source: General science text book, BBISE)
(a) Organic compounds
(b) Proteins
(c) Hormones
(d) **Inorganic compounds**

Hormones are chemical substances secreted by which specific structures? (Source: General science text book, BBISE)
(a) Nerve cells
(b) Muscles
(c) **Endocrine glands**
(d) Brain cells

Explanation: Hormones are secreted by "Endocrine Glands" into the bloodstream to regulate various body processes.

What is the function of hormones once they enter the bloodstream? (Source: General science text book, BBISE)
(a) They immediately cause an electric

shock.
(b) **They regulate functions like growth and metabolism.**
(c) They are filtered out by the kidneys immediately.
(d) They only affect the brain.

How many glands make up the Para-Thyroid system, and where are they located? (Source: General science text book, BBISE)
(a) One gland located in the brain
(b) Two glands located above the kidneys
(c) **Four small glands located behind the thyroid gland in the neck**
(d) Four glands located in the pancreas

Which two specific elements are regulated by the hormone secreted by the Para-Thyroid glands? (Source: General science text book, BBISE)
(a) Sodium and Potassium
(b) Iron and Iodine
(c) Oxygen and Nitrogen
(d) **Calcium and Phosphorus**

Where are the Adrenal glands located within the human body? (Source: General science text book, BBISE)
(a) In the neck area
(b) Below the brain
(c) In the pancreas
(d) **Above the kidneys**

Which specific hormone is secreted by the Adrenal glands? (Source: General science text book, BBISE)
(a) Insulin
(b) Glucagon
(c) **Adrenaline**
(d) Growth hormone

What is the primary function of the hormone Adrenaline? (Source: General science text book, BBISE)
(a) To control body growth
(b) To regulate calcium levels
(c) To control the functions of other glands
(d) **To control body functions during emergency situations**

Which gland is commonly referred to as the "Master Gland"? (Source: General science text book, BBISE)
(a) Adrenal Gland
(b) Thyroid Gland
(c) **Pituitary Gland**
(d) Thymus Gland

Explanation: The Pituitary Gland is explicitly named the "Master Gland" because it controls many other endocrine glands in the body.

Where is the Pituitary Gland located? (Source: General science text book, BBISE)
(a) Above the kidneys
(b) In the neck
(c) In the lower abdomen
(d) **Below the brain**

Explanation: The text states its location is "below the brain." It is a small pea-sized gland crucial for overall bodily regulation.

The Pituitary gland hormone primarily controls two things related to the body. What are they? (Source: General science text book, BBISE)
(a) Calcium levels and blood pressure
(b) **Body growth and the functions of other glands**

(c) Digestion and excretion
(d) Emergency response and immunity

Which two glands are responsible for producing hormones that control sexual functions in humans? (Source: General science text book, BBISE)
(a) Pituitary and Thyroid
(b) Adrenal and Pancreas
(c) **Ovary (female) and Testis (male)**
(d) Thymus and Adrenal

Which specific hormone is secreted by the pancreas gland and is vital for blood sugar regulation? (Source: General science text book, BBISE)
(a) Adrenaline
(b) Glucagon
(c) **Insulin**
(d) Retinol

Besides Insulin, which other hormone is secreted by the Pancreas? (Source: General science text book, BBISE)
(a) Thyroxin
(b) Adrenaline
(c) **Glucagon**
(d) Testosterone

What happens to the Thymus Gland as a person's age increases? (Source: General science text book, BBISE)
(a) It increases in size and function.
(b) It moves to a different location.
(c) **It stops working and disappears/reduces in size.**
(d) It starts producing adrenaline.

What causes the decay and decomposition of a dead body? (Source: General science text book, BBISE)
(a) Lack of oxygen
(b) Sunlight exposure
(c) **Bacteria, germs, and worms**
(d) Chemical reactions only

What structures remain mostly intact for a longer period after the soft parts have decomposed? (Source: General science text book, BBISE)
(a) Eyes and hair
(b) Muscles and skin
(c) **Bones, teeth, and hair**
(d) All parts decompose at the same rate

How does decomposition of a dead body contribute to the environment's balance? (Source: General science text book, BBISE)
(a) It causes pollution in the air.
(b) It kills off all bacteria.
(c) **It allows the body's chemicals to return to the earth.**
(d) It creates more living organisms.

The eye's cornea is naturally transparent. What happens if this transparency is lost? (Source: General science text book, BBISE)
(a) The eye becomes stronger.
(b) The eye changes color.
(c) **Vision is lost or severely impaired.**
(d) The eye produces more tears.

What is the cause of "Cornea Disease"? (Source: General science text book, BBISE)
(a) Excessive sunlight exposure
(b) Lack of water consumption
(c) **Viruses, chemical substances, or wounds**
(d) Genetic factors only

When a person with "Cornea Disease" receives a cornea from a dead person (cornea transplant), what does it restore? (Source: General science text book, BBISE)
(a) Ability to hear
(b) Sense of smell
(c) **Vision**
(d) Immune system function

How is a donated cornea preserved before transplantation? (Source: General science text book, BBISE)
(a) Kept in plain water
(b) Frozen immediately
(c) **Kept in a specific solution/liquid**
(d) Dried out in the sun

According to the top table, which element makes up the second-highest percentage of the human body? (Source: General science text book, BBISE)
(a) Hydrogen
(b) Calcium
(c) Nitrogen
(d) **Proteins**
Explanation: Protein at 18%, while fats are 10%, highlighting proteins as the second most abundant biological material after water.

What percentage of the human body's composition is made up of fats? (Source: General science text book, BBISE)
(a) 3.0%
(b) 60%
(c) 18%
(d) **10%**

What percentage of the air is composed of carbon dioxide? (Source: General science text book, BBISE)
(a) 60%
(b) 0.941%
(c) **0.04%**
(d) 6.1%

Which category do elements that make up less than 0.941% of the human body belong to? (Source: General science text book, BBISE)
(a) Major Elements
(b) Main Elements
(c) Abundant Elements
(d) **Trace Elements**

Which shape describes the rod-like form of bacteria? (Source: General science text book, BBISE)
(a) Cocci
(b) Spirilla
(c) Vibrio
(d) **Bacilli**

Which shape describes round-shaped bacteria? (Source: General science text book, BBISE)
(a) Bacilli
(b) Spirilla
(c) Vibrio
(d) **Cocci**

Which shape describes spiral-shaped bacteria? (Source: General science text book, BBISE)
(a) Cocci
(b) Bacilli
(c) Vibrio
(d) **Spirilla**

Which shape describes comma-shaped bacteria? (Source: General science text book, BBISE)
(a) Cocci

(b) Spirilla
(c) **Vibrio**
(d) Bacilli

What term is used when bacteria are found in clusters or groups? (Source: General science text book, BBISE)
(a) Bacilli
(b) Spirilla
(c) Vibrio
(d) **Colonies**

Which structure provides the outer protective layer for a bacterium? (Source: General science text book, BBISE)
(a) Cell Membrane
(b) Cytoplasm
(c) Flagella
(d) **Cell Wall**

Which structures help some bacteria move or swim? (Source: General science text book, BBISE)
(a) Cell Walls
(b) Cytoplasm
(c) **Flagella**
(d) Pili

Which of the following is a primary source of energy for the human body? (BBISE Biology Textbook)
A) Vitamins
B) **Carbohydrates**
C) Minerals
D) Water

A balanced diet must contain how many main groups of nutrients? (PakMcqs)
A) Three
B) Four
C) **Six**
D) Eight

Explanation: A balanced diet includes Carbohydrates, Proteins, Fats, Vitamins, Minerals, and Water in correct proportions to maintain health.

Which of the following diseases is transmitted through contaminated water? (Source: General science text book, BBISE)
A) Malaria
B) **Cholera**
C) Tuberculosis
D) Scurvy
Explanation: Cholera is an acute diarrheal infection caused by ingestion of food or water contaminated with the bacterium *Vibrio cholerae*.

The process of physical growth and maturation of human body parts is called: (Britannica)
A) Evolution
B) **Physical Development**
C) Metamorphosis
D) Digestion.

The softest bone-like tissue found at the ends of long bones to reduce friction is: (BBISE Biology Textbook)
A) Ligament
B) Tendon
C) **Cartilage**
D) Marrow

Which mineral is essential for the formation of strong bones and teeth?
A) Iron
B) **Calcium**
C) Iodine
D) Sodium

The period of life during which the body undergoes changes leading to reproductive maturity is: (Khan Academy)
A) Childhood
B) **Puberty**
C) Senescence
D) Infancy

A disease that can spread from one person to another is called: (Testbook)
A) **Communicable disease**
B) Deficiency disease
C) Degenerative disease
D) Hereditary disease

Scurvy is a disease caused by the deficiency of: (PakMcqs)
A) Vitamin B
B) **Vitamin C**
C) Vitamin E
D) Iron

The normal body temperature of a healthy human being is: (IndiaBix)
A) 32°C
B) **37°C**
C) 40°C
D) 98°C

Which of the following is considered a "Body Building" food? (Source: General science text book, BBISE) (Byju's)
A) Fats
B) **Proteins**
C) Carbohydrates
D) Roughage
Explanation: Proteins are made of amino acids and are essential for the growth and repair of muscles and tissues.

The main function of white blood cells (WBCs) in the body is to: (Examveda)
A) Transport oxygen
B) Help in clotting
C) **Fight against infections**
D) Digest food

Anaemia is caused by the deficiency of which mineral in the diet? (Source: General science text book, BBISE) (Testbook)
A) Calcium
B) **Iron**
C) Magnesium
D) Phosphorus
Explanation: Iron is a key component of hemoglobin. Lack of iron reduces the blood's capacity to carry oxygen, leading to fatigue.

Vaccination is a method used to develop: (Khan Academy)
A) Digestion
B) **Immunity**
C) Intelligence
D) Respiration
Explanation: Vaccines stimulate the immune system to recognize and fight specific pathogens without the person getting sick first.

Which of the following is a non-communicable disease? (Source: General science text book, BBISE) (PakMcqs)
A) Measles
B) **Diabetes**
C) Typhoid
D) Malaria
Explanation: Non-communicable diseases are chronic conditions that cannot be transmitted from person to person, often caused by lifestyle or genetics.

Iodine deficiency in the diet can lead to: (IndiaBix)
A) Rickets
B) **Goitre**
C) Beri-beri
D) Pellagra

The involuntary action of breathing is controlled by which part of the brain? (Source: General science text book, BBISE) (BBISE Biology Textbook)
A) Cerebrum
B) Cerebellum
C) **Medulla Oblongata**
D) Thalamus

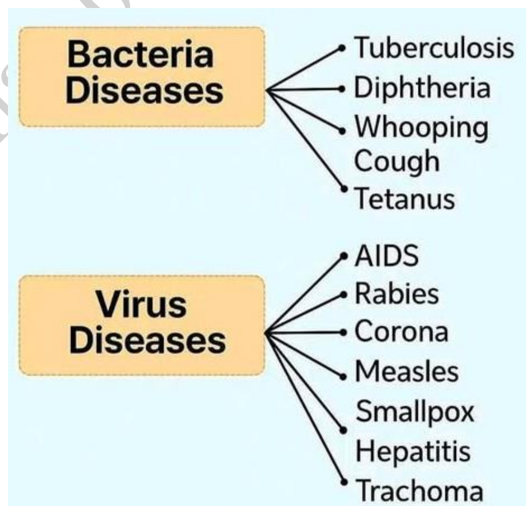
Malaria is a disease caused by a: (Examveda)
A) Virus
B) **Protozoan (Plasmodium)**
C) Bacterium
D) Fungus

The pigment that gives color to human skin and protects it from UV rays is: (BBISE Biology Textbook)
A) Hemoglobin
B) **Melanin**
C) Chlorophyll
D) Keratin

Beri-beri is caused by the deficiency of: (PakMcqs)
A) **Vitamin B1 (Thiamine)**
B) Vitamin B12
C) Vitamin C
D) Vitamin D
Explanation: Deficiency of Thiamine affects the nervous system and the heart.

What is the function of roughage (fiber) in our diet? (Source: General science text book, BBISE) (Byju's)
A) To provide energy
B) **To aid in digestion and prevent constipation**
C) To build muscles
D) To cure infections

Which of the following is a symptom of dehydration? (Source: General science text book, BBISE) (IndiaBix)
A) High energy
B) **Extreme thirst and dark urine**
C) Pale skin
D) Increased sweating
Explanation: Dehydration occurs when the body loses more water than it takes in, disrupting metabolic processes.



The largest organ of the human body is: (Testbook)
A) Liver
B) **Skin**
C) Brain
D) Small intestine

Which of the following is a viral disease? (Source: General science text book, BBISE) (PakMcqs)
A) Typhoid

B) **Hepatitis**
C) Tetanus
D) Syphilis

The exchange of gases in the lungs takes place in the: (Byju's)
A) Trachea
B) Bronchi
C) **Alveoli**
D) Diaphragm

Adolescence is the transition period between: (Britannica)
A) Infancy and Childhood
B) **Childhood and Adulthood**
C) Adulthood and Old age

The "Universal Donor" blood group is: (Examveda)
A) AB+
B) **O-**
C) A+
D) B-
Explanation: Group O-negative blood has no A, B, or Rh antigens on the surface of the red blood cells, so it can be given to anyone.

Rickets is a bone-weakening disease in children caused by lack of: (Testbook)
A) Vitamin A
B) Vitamin B
C) **Vitamin D**
D) Vitamin K
Explanation: Without Vitamin D, the body cannot absorb calcium, leading to soft and bowed bones.

Human physical development is influenced by both: (Britannica)
A) **Heredity and Environment**
B) Only Diet
C) Only Exercise
D) Only Weather

Which component of blood helps in the formation of a clot at the site of injury? (Source: General science text book, BBISE) (Examveda)
A) Red Blood Cells
B) **Platelets**
C) White Blood Cells
D) Plasma

A person suffering from high blood pressure is said to have: (PakMcqs)
A) Hypotension
B) **Hypertension**
C) Hyperglycemia
D) Anemia

The study of the heart and its diseases is called: (Testbook)
A) Neurology
B) **Cardiology**
C) Oncology
D) Dermatology
Explanation: Cardiologists specialize in the diagnosis and treatment of heart-related conditions.

Which organ in the human body can regenerate itself if part of it is removed? (Source: General science text book, BBISE) (Byju's)
A) Heart
B) **Liver**
C) Lung
D) Brain
Explanation: The liver has a unique ability to grow back to its original size from as little as 25% of its tissue.

ENERGY AND ITS PROCUREMENT

What is the general definition of energy that students learn first? (Source: General science text book, BBISE)
(a) The ability to create change
(b) The capacity to generate heat
(c) The movement of particles
(d) **The ability to do work**

How else can energy be described in a general sense? (Source: General science text book, BBISE)
(a) A factor that stops change
(b) The speed of an object
(c) **The factor that brings about any kind of change**
(d) The mass of an object

In the example of a boat moving down a flowing river, which type of energy is primarily at work? (Source: General science text book, BBISE)
(a) Potential energy
(b) Chemical energy
(c) Electrical energy
(d) **Kinetic energy (mechanical)**
Explanation: "If we leave a boat on flowing river water, it takes it from one place to another." This movement is an example of kinetic energy, which is energy due to motion.

When a person winds the spring of a watch with a key, which type of energy is stored? (Source: General science text book, BBISE)
(a) Kinetic energy
(b) Chemical energy
(c) Electrical energy
(d) **Potential energy (mechanical)**
Explanation: "mechanical potential energy" being stored in the wound spring, which is then released to run the watch. Potential energy is stored energy due to position or configuration.

Chemical energy is a form of stored energy found within what structures? (Source: General science text book, BBISE)
(a) Protons within the nucleus
(b) Moving electrons
(c) **The bonds of atoms and molecules**
(d) Gravity fields

When food is digested in the body, or fuel (like oil) is burned in cars, what process releases the chemical energy? (Source: General science text book, BBISE)
(a) Fusion
(b) **Chemical reaction/combustion**
(c) Electrical flow
(d) Magnetic attraction

The form of electrical energy related to the flow of electrons is categorized as:
(a) Potential energy
(b) Magnetic energy
(c) **Kinetic energy (motion of charges)**
(d) Nuclear energy
Explanation: The text links electrical energy to the flow of electrons ("electron flow"). When these charged particles are moving through a conductor (like a wire), it is a form of kinetic energy or "current electricity".

Static electricity in charged clouds represents which form of electrical energy? (Source: General science text book, BBISE)
(a) Kinetic energy
(b) Chemical energy
(c) **Potential energy**
(d) Nuclear energy

What phenomenon converts the potential energy in charged clouds into kinetic electrical energy? (Source: General science text book, BBISE)
(a) Wind currents
(b) **Lightning/Thunderclap**
(c) Magnetic attraction
(d) Nuclear fusion
Explanation: "lightning and a thunderclap" when they meet, at which point the potential energy becomes moving (kinetic) electrical energy.

In a nuclear reactor, the heat produced from nuclear fission is used for what immediate purpose? (Source: General science text book, BBISE)
(a) To generate electricity directly
(b) To cause fusion in the sun
(c) **To convert water into steam**
(d) To break more atoms

The energy produced in the Sun primarily originates not from fission, but from which process? (Source: General science text book, BBISE)
(a) Heat energy
(b) Chemical reactions
(c) **Fusion**
(d) Electrical discharge

Magnetic energy exists within a magnet in what primary form? (Source: General science text book, BBISE)
(a) Kinetic energy
(b) **Potential energy**
(c) Electrical energy
(d) Heat energy

Magnetic energy is used in applications like running trains, lifting objects, and rotating which device? (Source: General science text book, BBISE)
(a) A battery
(b) A heat exchanger
(c) **A generator**
(d) An electric bulb

Heat energy in an object is produced due to the motion of its:
(a) Protons only
(b) Electrons only
(c) Nuclei only
(d) **Molecules/Atoms**

Explanation: Heat energy is produced in that body due to the motion of the molecules of a body. Heat is essentially the total kinetic energy of a system's constituent particles.

Heat is transferred from an object with a higher temperature to one with a lower temperature because of:
(a) The flow of coldness
(b) Electromagnetic waves
(c) **Differences in temperature**
(d) Chemical reactions

Light energy is an important form of energy obtained when electrons move between different orbits within an atom. This is a form of:
(a) Chemical energy
(b) Heat energy

(c) **Electromagnetic radiation/energy**
(d) Mechanical energy

Which essential biological process requires light energy to occur? (Source: General science text book, BBISE)
(a) Respiration in animals
(b) Digestion of food
(c) **Photosynthesis in plants**
(d) Cell division

Explanation: Plants convert light energy into chemical energy (glucose) during this process, forming the basis of most food chains on Earth.

What term is used for the energy associated with the movement of an object? (Source: General science text book, BBISE)
(a) Potential energy
(b) Chemical energy
(c) Mechanical Potential Energy
(d) **Kinetic energy**

Kinetic energy is directly proportional to which two physical quantities of the object? (Source: General science text book, BBISE)
(a) Mass and speed
(b) **Mass and velocity squared**
(c) Mass squared and velocity
(d) Mass and acceleration

Explanation: kinetic energy is proportional to its mass and the square of its velocity (as in the formula

$$KE = \frac{1}{2}mv^2$$

When the movement of a moving object is stopped, what other forms of energy does the kinetic energy convert into? (Source: General science text book, BBISE)
(a) Light and chemical energy
(b) Electrical and magnetic energy
(c) Nuclear and heat energy
(d) **Sound and heat energy**

Explanation: When the movement of a moving thing is stopped, the kinetic energy converts into sound and heat energy.

What type of energy is stored within a tensed clock spring or a stretched catapult? (Source: General science text book, BBISE)
(a) Kinetic energy
(b) Gravitational potential energy
(c) Sound energy
(d) **Elastic potential energy**

What allows a heavy object to possess energy when it is lifted to a height from the ground? (Source: General science text book, BBISE)
(a) Elastic potential energy
(b) Kinetic energy
(c) The object's weight
(d) **Gravitational potential energy**

In the example of a hammer hitting a nail, which energy type is converted into heat and sound? (Source: General science text book, BBISE)
(a) Potential energy of the hammer's height
(b) Heat energy of the nail
(c) Sound energy of the hammer
(d) **Kinetic energy of the falling hammer**

Explanation: The potential energy due to height is converted to kinetic energy during the fall, which is then dissipated as heat and sound upon impact with the nail.

What general law of physics? (Source: General science text book, BBISE)
(a) Law of Gravity
(b) Law of Motion
(c) **Law of Conservation of Energy**
(d) Law of Thermodynamics

Explanation: Law of Conservation of Energy. This fundamental law states that energy can change form but is neither created nor destroyed.

Can energy be destroyed? (Source: General science text book, BBISE)
(a) Yes, it turns into mass.
(b) Yes, it is used up by work.
(c) **No, energy is only conserved, not destroyed.**
(d) Only in uncontrolled reactions.

Which two sources provide a large amount of energy today and are considered "traditional" sources? (Source: General science text book, BBISE)
(a) Wind and natural gas
(b) Geothermal and tidal
(c) Bio-gas and nuclear
(d) **Oil (Petroleum) and water**

Which conventional source of energy is used to generate mechanical energy from potential energy by falling from a height? (Source: General science text book, BBISE)
(a) Coal
(b) Petroleum
(c) Natural Gas
(d) **Water**

How is the electrical energy produced from the kinetic energy of water? (Source: General science text book, BBISE)
(a) The water is burned
(b) The water is used in a chemical reaction
(c) **The water runs turbines connected to generators**
(d) The water directly creates electricity

Which of the following elements is primarily found in coal? (Source: General science text book, BBISE)
(a) Nitrogen
(b) Sulfur
(c) Oxygen
(d) **Carbon**

Coal is formed over millions of years from what? (Source: General science text book, BBISE)
(a) Petroleum deposits
(b) Volcanic ash
(c) **Plants and trees buried underground**
(d) Marine animal remains

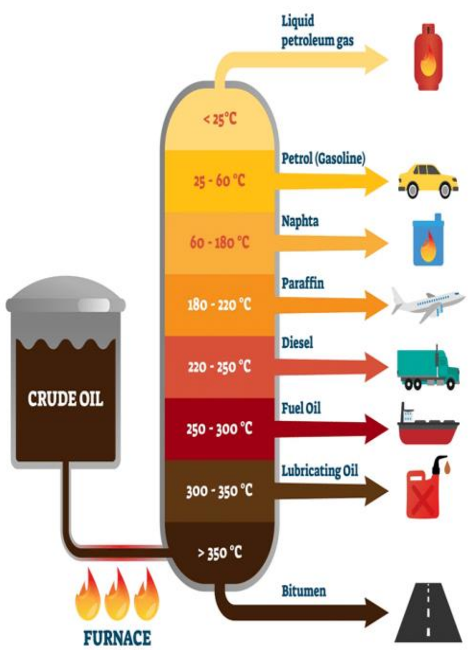
What type of coal is found in Pakistan, which has a higher moisture content? (Source: General science text book, BBISE)
(a) Anthracite
(b) Bituminous
(c) **Lignite**
(d) Sub-bituminous

Which valuable organic compound can be obtained from coal? (Source: General science text book, BBISE)
(a) Glycerol
(b) Methane
(c) **Naphthalene**
(d) Keratin

What industry commonly uses coke (charred coal) produced from coal? (Source: General science text book, BBISE)
(a) Agriculture
(b) Medicine
(c) Electricity generation in Karachi
(d) **Steel and metal manufacturing**

Petroleum is formed from the remains of which type of organism? (Source: General science text book, BBISE)
(a) Land plants and trees
(b) Dinosaurs
(c) **Marine organisms**
(d) Bacteria

In what form is crude petroleum found before refining? (Source: General science text book, BBISE)
(a) A black solid rock
(b) A light gas
(c) **A black, thick liquid**
(d) A clear, transparent liquid



What industrial process is used to separate crude petroleum into different useful parts? (Source: General science text book, BBISE)
(a) Filtration
(b) Chemical reaction
(c) **Fractional Distillation**
(d) Combustion

What is the important oil derived from petroleum used for reducing friction in machinery? (Source: General science text book, BBISE)
(a) Kerosene oil
(b) Diesel
(c) **Lubricating Oil**
(d) Petrol

Where is natural gas usually found? (Source: General science text book, BBISE)
(a) In the atmosphere above ground
(b) **Along with petroleum deposits**
(c) Deep within plant remains
(d) In solid rock formations

Natural gas is used in steam power stations and generators to produce what final form of energy? (Source: General science text book, BBISE)
(a) Potential energy
(b) Chemical energy
(c) Heat energy
(d) **Electrical energy**

Natural gas is also utilized as a raw material for which purpose? (Source: General science text book, BBISE)
(a) Making airplanes fly
(b) Lubricating machinery
(c) **Industrial use/fertilizer production**
(d) Running submarines

What is the fundamental relationship between the units of energy and work? (Source: General science text book, BBISE)
(a) Energy units are different from work units.
(b) Energy units are double the work units.
(c) Energy units are half the work units.
(d) **Energy and work have unit (Joule).**

What is the special name given to the unit Newton-meter (Nm.) in honor of British physicist James Joule? (Source: General science text book, BBISE)
(a) Newton
(b) Watt
(c) **Joule**
(d) Calorie

How much energy in Joules is required to pass a 1 Ampere current through a 1 Ohm resistance for one second? (Source: General science text book, BBISE)
(a) 4.2 Joules
(b) 0.941 Joules
(c) **1 Joule**
(d) 159 Joules

$$J = kg.m^2.s^{-2} = N.m = Pa.m^3 = W.s = C.V$$

What common units are typically used to measure electrical energy? (Source: General science text book, BBISE)
(a) Newtons or meters
(b) Calories
(c) Joules
(d) **Watt-seconds or Kilowatt-hours**
Explanation: However, electrical energy is usually measured in Watt-seconds or Kilowatt-hours. The Watt is a unit of power equal to Joules per second.

One Watt-second is exactly equivalent to which another unit? (Source: General science text book, BBISE)
(a) One Calorie
(b) One Ampere
(c) One Kilowatt-hour
(d) **One Joule**

What unit was commonly used to measure heat energy? (Source: General science text book, BBISE)
(a) Joule
(b) Watt
(c) Barrel
(d) **Calorie**

One Calorie is approximately equal to how many Joules? (Source: General science text book, BBISE)
(a) 1 Joule
(b) 0.941 Joules
(c) 159 Joules
(d) **4.2 Joules**

What is the specific definition of a Calorie in terms of water and temperature? (Source: General science text book, BBISE)
(a) The heat to boil 1 gram of water.
(b) The heat to warm 1 liter of water by 4.2 degrees.
(c) **The amount of heat required to warm 1 gram of water by 1 degree Celsius.**
(d) The amount of heat in a barrel of oil.

Which international unit is used for crude petroleum? (Source: General science text book, BBISE)
(a) Liter
(b) Ton
(c) Kilowatt-hour
(d) **Barrel**

RENEWABLE ENERGY AND ITS SOURCES

Energy sources that can be replenished naturally over a short period of time are called

a) Non-renewable Energy
b) **Renewable Energy**
c) Fossil Fuels
d) Nuclear Energy

Explanation: **Renewable Energy** comes from infinite or fast-growing sources like the sun, wind, and water.

Which category do sources like solar, bio-gas, wind, tidal, and biomass energy belong to? (Source: General science text book, BBISE)

(a) Non-renewable energies
(b) Traditional sources
(c) Fission energies
(d) **Renewable energies**

Which traditional energy source has been used in grinding mills for centuries? (Source: General science text book, BBISE)

(a) Bio-gas
(b) Solar energy
(c) **Flowing water**
(d) Wind energy

In villages, what has animal dung been used for as fuel for a long time? (Source: General science text book, BBISE)

(a) Making bio-gas
(b) Fertilizer
(c) **As a fuel source for burning**
(d) Generating electricity directly

What process allows natural gas to be obtained from animal dung? (Source: General science text book, BBISE)

(a) Combustion
(b) **Bio-gas process**
(c) Solar heating
(d) Nuclear reaction

Besides natural gas, what other useful byproduct is formed during the bio-gas process? (Source: General science text book, BBISE)

(a) Water vapor
(b) Kerosene oil
(c) Methane gas
(d) **A powerful fertilizer (manure)**

What kind of cells are used to convert solar energy into electrical energy? (Source: General science text book, BBISE)

(a) Animal cells
(b) Plant cells
(c) Bio-gas cells
(d) **Solar cells (Photovoltaic cells)**

In the fusion process, which elements combine to form helium? (Source: General science text book, BBISE)

(a) Oxygen
(b) Lead
(c) Uranium
(d) **Hydrogen**

Explanation: The text specifically mentions hydrogen nuclei combining to form a helium nucleus during fusion reactions. This is the process that powers stars.

Which non-conventional energy source is based on the movement of water during high and low tides? (Source: General

science text book, BBISE)
(a) Wind Energy
(b) Geothermal Energy
(c) **Tidal Energy**
(d) Solar Energy

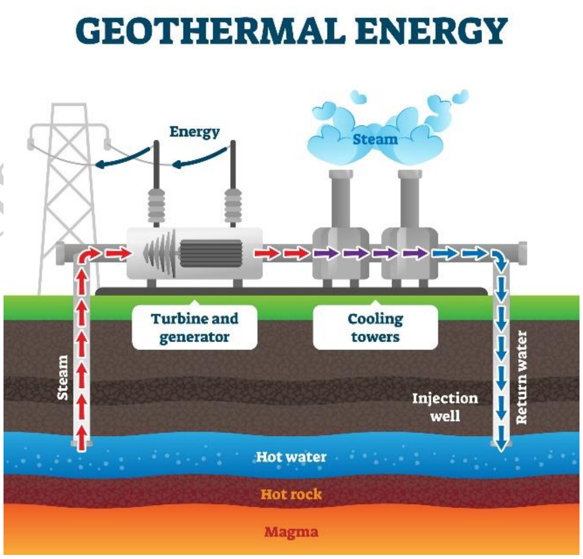
How does tidal energy produce electricity? (Source: General science text book, BBISE)
(a) By heating water to create steam
(b) By using solar panels
(c) **By installing turbines that rotate with the water flow**
(d) By using nuclear reactions
Explanation: The energy from the flowing tide water is used to run turbines installed in the sea, which then operate electric generators.

How far from the ground are wind turbines generally installed for efficiency? (Source: General science text book, BBISE)
(a) 5 meters
(b) 10 meters
(c) 15 meters
(d) **25-30 meters**

Which source of energy uses the heat produced naturally inside the Earth? (Source: General science text book, BBISE)
(a) Solar Energy
(b) Wind Energy
(c) Tidal Energy
(d) **Geothermal Energy**

What substance transfers the internal heat of the Earth towards the surface? (Source: General science text book, BBISE)
(a) Oil deposits
(b) **Magma (molten rock)**
(c) Underground air
(d) Water currents.

In the geothermal energy process, what happens to the water heated by the Earth's internal heat? (Source: General science text book, BBISE)
(a) It is discarded as waste.
(b) It is consumed as drinking water.
(c) **It turns into hot steam that runs a turbine.**
(d) It is used to run an atomic bomb reaction.



Nuclear energy is described as a non-conventional source obtained from which process? (Source: General science text book, BBISE)
(a) Fusion of hydrogen atoms
(b) Burning coal underground
(c) **Splitting the nucleus of a heavy atom (Fission)**
(d) Using tides to turn turbines

What is the core difference between the fission reaction used for peaceful energy

generation and that used in an atomic bomb? (Source: General science text book, BBISE)
(a) The bomb uses fusion; the reactor uses fission.
(b) The bomb uses light atoms; the reactor uses heavy atoms.
(c) **The reactor controls the chain reaction; the bomb reaction is uncontrolled.**
(d) The reactor produces sound; the bomb produces only heat.

What are some examples of organic materials that can be burned for biomass energy? (Source: General science text book, BBISE)
(a) Oil and Natural Gas
(b) Coal and Petroleum
(c) **Wood, husk, and sugarcane residue (bagasse)**
(d) Metals and plastics

In sugar mills, which specific waste product is burned to produce heat energy? (Source: General science text book, BBISE)
(a) Sugarcane juice
(b) Sugar crystals
(c) **Bagasse (sugarcane residue)**
(d) Molasses
Explanation: After extracting juice from sugarcane factories, the remaining bagasse is burned to obtain heat energy.

What valuable chemical can be made from waste products like sugarcane residue, wheat, and other grains? (Source: General science text book, BBISE)
(a) Methane
(b) Naphthalene
(c) Graphite
(d) **Alcohol (Ethanol)**

What is the primary advantage of renewable energy over fossil fuels
a) It is cheaper to set up
b) **It produces little to no greenhouse gas emissions**
c) It is available 24/7 everywhere
d) It takes up less space
Explanation: **Renewable Energy** is "clean" because it does not release **CO_2**, helping to fight Climate Change.

The term 'Clean Energy' usually refers to energy that
a) Is washed with water
b) **Does not pollute the environment**
c) Is made from chemicals
d) Is invisible
Explanation: **Clean Energy** sources do not emit toxic waste or harmful gases during electricity production.

Solar cells are primarily made of which material
a) Copper
b) **Silicon**
c) Iron
d) Aluminum
Explanation: **Silicon** is a semiconductor that is the most efficient and common material for solar panels.

The process by which a solar cell generates electricity is known as the
a) Photoelectric Effect
b) **Photovoltaic Effect**
c) Thermionic Emission
d) Seebeck Effect

A group of solar cells connected together is called a
a) Solar Battery
b) **Solar Panel (Module)**
c) Solar Grid
d) Solar Wire
Explanation: Individual cells are small; they are grouped into **Panels** to produce usable amounts of power.

Wind energy is a form of kinetic energy caused by
a) Earth's gravity
b) **Uneven heating of Earth's surface by the Sun**
c) Volcanic eruptions
d) Ocean waves
Explanation: Heat differences create pressure changes, which cause air to move, creating **Wind**.

The machine used to convert the kinetic energy of wind into electrical energy is a
a) Wind Mill
b) **Wind Turbine**
c) Wind Pump
d) Fan
Explanation: While a windmill grinds grain, a **Wind Turbine** is specifically designed to generate electricity.

Where is the best location for a Wind Farm to be most effective
a) In a forest
b) **In open plains or offshore (ocean)**
c) Behind high mountains
d) Inside a city
Explanation: **Wind Turbines** need steady, high-speed wind without obstacles like buildings or trees.

The blades of a wind turbine are connected to a
a) **Generator**
b) Battery
c) Solar Panel
d) Boiler
Explanation: As the blades spin, they turn a shaft connected to a **Generator** to produce electricity.

Wind energy is considered 'Intermittent' because
a) It is too expensive
b) **The wind does not blow all the time**
c) It is dangerous
d) It pollutes the air
Explanation: **Intermittent** means the power source is not constant and depends on weather conditions.

Hydroelectric power is generated by using the energy of
a) Boiling water
b) **Falling or flowing water**
c) Salt water
d) Raindrops
Explanation: The potential energy of water stored in a dam is converted into kinetic energy as it falls.

In a hydroelectric dam water turns a device called a
a) Piston
b) **Turbine**
c) Gear
d) Lever
Explanation: Flowing water spins the **Turbine**, which then spins the generator.

The largest hydroelectric dam in Pakistan is
a) Mangla Dam
b) **Tarbela Dam**
c) Warsak Dam
d) Neelum-Jhelum
Explanation: **Tarbela Dam** on the Indus River is the largest earth-filled dam in the world and a major power source.

What is a major environmental disadvantage of large Hydroelectric Dams
a) Air pollution
b) **Destruction of local ecosystems and fish migration**
c) High fuel cost
d) Radiation
Explanation: Flooding large areas to create reservoirs can displace people and destroy wildlife habitats.

Which form of hydropower uses the natural flow of a river without a large dam
a) Tidal Power
b) **Run-of-the-River System**
c) Pumped Storage
d) Wave Power
Explanation: **Run-of-the-River** systems have a smaller environmental footprint as they don't require large reservoirs.

Energy derived from organic materials like plants and animal waste is called
a) Fossil Energy
b) **Biomass Energy**
c) Geothermal Energy
d) Nuclear Energy
Explanation: **Biomass** is a renewable resource because we can always grow more plants and trees.

Gas produced by the decomposition of organic matter in the absence of oxygen is
a) Natural Gas
b) **Biogas (Methane)**
c) Oxygen
d) Carbon Dioxide
Explanation: **Biogas** is an excellent fuel for cooking and heating in rural areas.

Which fuel can be produced from crops like corn and sugarcane
a) Petrol
b) **Ethanol (Biofuel)**
c) Diesel
d) Kerosene
Explanation: **Ethanol** is a liquid biofuel that can be mixed with petrol to reduce vehicle emissions.

Geothermal energy is obtained from
a) The Sun
b) **Heat trapped inside the Earth**
c) Ocean currents
d) Fossilized plants
Explanation: **Geothermal** energy uses steam or hot water from underground to generate power.

Which country is a world leader in using Geothermal Energy due to its volcanoes
a) Saudi Arabia
b) **Iceland**
c) Pakistan
d) Brazil
Explanation: **Iceland** gets nearly 25% of its electricity from geothermal sources.

Ocean and Tidal Energy

Tidal energy is produced by the gravitational pull of the
a) Sun

b) **Moon**
c) Mars
d) Earth's Core
Explanation: The **Moon's** gravity causes tides, and the movement of this water can be used to generate power.

Energy produced by the rise and fall of ocean waves is called
a) Tidal Energy
b) **Wave Energy**
c) Hydro Energy
d) Solar Energy
Explanation: **Wave Energy** uses the kinetic energy of surface waves to drive generators.

The main challenge of Tidal and Wave energy is
a) Lack of water
b) **High cost and harsh saltwater environment**
c) Air pollution
d) Global warming
Explanation: Saltwater causes corrosion, making the maintenance of underwater machinery very difficult.

Which of the following is a predictable source of renewable energy
a) Wind
b) Solar
c) **Tidal**
d) Cloud
Explanation: Unlike sun or wind, **Tides** follow a strict astronomical schedule and can be predicted years in advance.

What is 'Circular Debt' in Pakistan's energy sector
a) A type of battery
b) **A chain of unpaid dues between power companies and the government**
c) A rotating generator
d) A loan from the World Bank
Explanation: **Circular Debt** is a major financial crisis that hampers the efficiency of Pakistan's power system.

Which organization regulates the power sector in Pakistan
a) OGRA
b) **NEPRA**
c) WAPDA
d) NADRA
Explanation: The **National Electric Power Regulatory Authority (NEPRA)** sets electricity prices and issues licenses.

The process of moving electricity from a power plant to a city is called
a) Distribution
b) **Transmission**
c) Generation
d) Storage
Explanation: **Transmission** happens over high-voltage wires across long distances.

A 'Smart Grid' is an electricity network that
a) Is very small
b) **Uses digital technology to monitor and manage energy flow**
c) Does not use wires
d) Is only for solar power
Explanation: **Smart Grids** help reduce waste and integrate renewable energy more efficiently.

Why is energy storage (batteries) important for Solar and Wind power
a) To make them look better
b) **To provide power when the sun isn't shining or wind isn't blowing**

c) To increase the voltage
d) To change the colour of light
Explanation: Since renewable sources are intermittent, **Batteries** store excess energy for later use.

The 'Greenhouse Effect' is primarily caused by an increase in
a) Oxygen
b) **Carbon Dioxide (CO_2)**
c) Nitrogen
d) Hydrogen
Explanation: Burning fossil fuels increases **CO_2**, which traps heat in the atmosphere.

Which renewable energy source is considered the cleanest in terms of CO_2 emissions
a) Biomass
b) **Wind**
c) Natural Gas
d) Coal
Explanation: **Wind Turbines** produce zero emissions during operation.

Hydrogen fuel cells produce electricity by combining hydrogen with
a) Carbon
b) **Oxygen**
c) Nitrogen
d) Sulfur
Explanation: The only byproduct of a **Hydrogen Fuel Cell** is pure water (**H_2O**).

'Net Metering' allows a solar panel owner to
a) Buy more panels
b) **Sell excess electricity back to the national grid**
c) Clean their panels with a net
d) Use the internet for free
Explanation: **Net Metering** reduces electricity bills by giving credit for the energy sent to the grid.

The 'Carbon Footprint' refers to
a) A literal footprint in coal
b) **The total amount of greenhouse gases produced by human activities**
c) The size of a solar panel
d) A type of fossil
Explanation: Reducing your **Carbon Footprint** means using less energy and switching to renewables.

What is the main component of Biogas
a) Oxygen
b) **Methane**
c) Helium
d) Argon
Explanation: **Biogas** is roughly 60-70% **Methane**, which is highly flammable and useful for fuel.

Energy from the Sun reaches Earth in the form of
a) Sound waves
b) **Electromagnetic Radiation**
c) Electricity
d) Wind
Explanation: Sunlight consists of photons traveling as **Electromagnetic Waves**.

Which of the following is NOT a renewable resource
a) Geothermal
b) **Uranium (Nuclear)**
c) Biomass
d) Sunlight
Explanation: While nuclear is low-carbon, **Uranium** is a finite mineral and therefore not renewable.

ENERGY CONSUMING SECTORS IN PAKISTAN

What unit of measurement is used for measuring electricity consumption, as indicated in the text book? (Source: General science text book, BBISE)
(a) Watts
(b) Amperes
(c) Volts
(d) **Kilowatt-hour (kWh)**

One kilowatt-hour (1 kWh) is commonly referred to in simple terms as what? (Source: General science text book, BBISE)
(a) One Volt
(b) One Ampere
(c) One Watt
(d) **One Unit**

A consumer uses a 1000-watt (1 kW) electric iron for one hour. How much electricity is consumed? (Source: General science text book, BBISE)
(a) 1000 Units
(b) **1 Unit (kWh)**
(c) 1 Watt-hour
(d) 100 Watts

If a 200-watt bulb is used for five hours, what is the total electricity consumed in kilowatt-hours (units)? (Source: General science text book, BBISE)
(a) 200 Units
(b) 5 Units
(c) 1000 Units
(d) **1 Unit (kWh)**

What device is used specifically for measuring electricity consumption in kilowatt-hours? (Source: General science text book, BBISE)
(a) Voltmeter
(b) Ammeter
(c) Ohmmeter
(d) **Kilowatt-hour meter (Electricity meter)**

What determines the speed at which the disk in the meter rotates? (Source: General science text book, BBISE)
(a) The ambient temperature
(b) The magnetic field of the Earth
(c) **The amount of electricity passing through the meter**
(d) The number of hours the meter has been installed

How many types of electricity meters are commonly found in homes, according to the text book? (Source: General science text book, BBISE)
(a) One type
(b) **Two types (Dial-based and Digital/Numerical)**
(c) Three types
(d) Four types

Who is responsible for reading the meter every month? (Source: General science text book, BBISE)
(a) The homeowner
(b) The local police
(c) **A representative of the electricity**

supply company (meter reader)
(d) A computer system

How is natural gas typically measured? (Source: General science text book, BBISE)
(a) In liters
(b) In tons
(c) **In cubic feet**
(d) In barrels

What percentage of Pakistan's total energy needs are currently met by oil (Petroleum)? (Source: General science text book, BBISE)
(a) 10%
(b) 25%
(c) **50%**
(d) 90%

Which energy source's reserves in Pakistan are insufficient to meet national needs? (Source: General science text book, BBISE)
(a) Coal and water
(b) **Oil and gas**
(c) Nuclear and solar
(d) Wind and tidal

When did the commercial production of natural gas start in Pakistan? (Source: General science text book, BBISE)
(a) 1947
(b) 1950
(c) **1952**
(d) 1960

Natural gas is primarily used as a source of energy for which purpose? (Source: General science text book, BBISE)
(a) Running airplanes
(b) Lubricating machinery
(c) Generating kinetic energy
(d) **Heat energy (burning)**

Explanation: Natural gas is primarily burned to produce heat energy, which is then used for cooking, heating, and power generation.

Natural gas is also used as a raw material for which industrial product? (Source: General science text book, BBISE)
(a) Steel manufacturing
(b) Lubricating oil
(c) Aviation fuel
(d) **Fertilizers**

What type of coal is generally found in Pakistan? (Source: General science text book, BBISE)
(a) Anthracite
(b) Bituminous
(c) Sub-bituminous
(d) **Lignite**

What industrial process is used to separate crude petroleum into different useful parts? (Source: General science text book, BBISE)
(a) Filtration
(b) Combustion
(c) **Fractional Distillation**
(d) Condensation

Which petroleum product is mentioned for its use in reducing friction in machinery? (Source: General science text book, BBISE)
(a) Kerosene oil
(b) Diesel
(c) Petrol
(d) **Lubricating Oil**

What valuable organic compound can be prepared from good quality coal? (Source: General science text book, BBISE)
(a) Alcohol
(b) Methane
(c) Glycerol
(d) **Naphthalene**

Coke, a product made from coal, is commonly used in which industry? (Source: General science text book, BBISE)
(a) Agriculture
(b) Medicine
(c) Power generation
(d) **Steel and metal manufacturing**

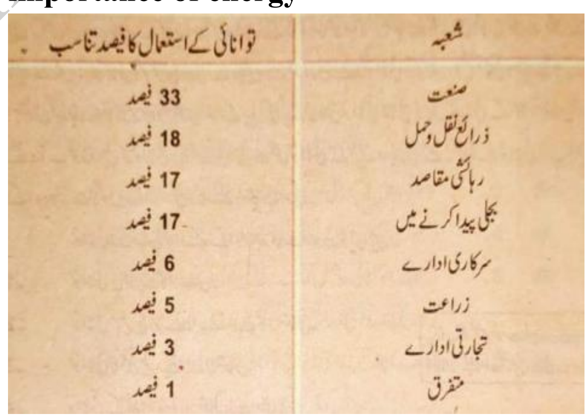
Approximately what percentage of Pakistan's total energy needs are met by natural gas? (Source: General science text book, BBISE)
(a) 10%
(b) **40%**
(c) 50%
(d) 90%

What percentage of energy needs are met by other sources like hydroelectric and coal in Pakistan? (Source: General science text book, BBISE)
(a) 50%
(b) 40%
(c) 15%
(d) **10%**

What major factor will cause fuel to become more expensive over time? (Source: General science text book, BBISE)
(a) Decreased demand from consumers
(b) Increased efficiency of power plants
(c) **The limited availability of traditional energy sources**
(d) Government regulations

The better use of energy is referred by which term? (Source: General science text book, BBISE)
(a) Energy generation
(b) Energy transformation
(c) Energy waste
(d) **Energy conservation**

Which of the following is listed as an important step to prevent energy wastage (Source: General science text book, BBISE)
(a) Using more electricity
(b) Ignoring energy consumption
(c) Buying expensive devices
(d) **Awareness/realization of the importance of energy**



شعبہ	توانائی کے استعمال کا فیصد تناسب
صنعت	33 فیصد
ذرائع نقل وحمل	18 فیصد
رہائشی مقاصد	17 فیصد
بجلی پیدا کرنے میں	17 فیصد
سرکاری ادارے	6 فیصد
زراعت	5 فیصد
تجارتی ادارے	3 فیصد
متفرق	1 فیصد

Which of the following is listed as an important step to prevent energy wastage? (Source: General science text book, BBISE)
(a) Using high-energy appliances
(b) Using old appliances
(c) **Using low-energy-consumption appliances**
(d) Avoiding the use of any appliances

Which sector is the largest consumer of Natural Gas in Pakistan
a) Transport
b) **Power (Electricity Generation)**
c) Industrial
d) Domestic
Explanation: A significant portion of Pakistan's **Natural Gas** is used as fuel in thermal power plants to produce electricity.

The total energy requirement of a country is often referred to as
a) Energy Waste
b) **Energy Demand**
c) Energy Surplus
d) Energy Grid
Explanation: **Energy Demand** must be balanced with supply to avoid load shedding and power outages.

In the domestic sector most energy is consumed during which season
a) Spring
b) **Summer**
c) Autumn
d) Winter
Explanation: In **Summer**, the use of fans, air conditioners, and coolers peaks, leading to the highest electricity demand.

Which home appliance typically consumes the most electricity in a Pakistani household
a) LED Bulb
b) **Air Conditioner**
c) Television
d) Laptop
Explanation: **Air Conditioners** have high wattage and run for long hours, significantly increasing the electricity bill.

Commercial sector energy consumption includes electricity used in
a) Farms
b) **Shops, Malls, and Offices**
c) Factories
d) Trains
Explanation: The **Commercial Sector** involves all business activities that are not related to manufacturing or agriculture.

What is the main source of energy for cooking in urban households of Pakistan
a) Coal
b) **Natural Gas (Sui Gas)**
c) Wood
d) Solar
Explanation: Most cities are connected to the gas grid managed by **SNGPL** or **SSGC**.

In rural areas of Pakistan the primary source of domestic energy is
a) Nuclear
b) **Biomass (Wood and Dung)**
c) Wind
d) Electricity
Explanation: Due to lack of gas infrastructure, many rural residents rely on **Biomass** for cooking and heating.

Which industry is one of the largest energy consumers in Pakistan
a) IT Sector
b) **Textile Industry**
c) Fishing
d) Education
Explanation: The **Textile Industry** requires massive amounts of electricity and gas for spinning, weaving, and processing.

The use of Natural Gas as a raw material in the Fertilizer industry is called
a) Fuel use
b) **Feedstock**
c) Waste
d) Lubricant
Explanation: **Feedstock** means the gas is used to provide the chemical components (hydrogen) for making urea.

Which heavy industry consumes large amounts of coal in Pakistan
a) Software
b) **Cement Industry**
c) Pharmaceutical
d) Tourism
Explanation: The **Cement Industry** uses coal-fired kilns to produce the high heat necessary for cement production.

The transport sector in Pakistan primarily consumes which form of energy
a) Electricity
b) **Petroleum Products (Petrol and Diesel)**
c) Coal
d) Solar
Explanation: Almost all trucks, buses, and private cars in Pakistan run on **Liquid Fuels**.

Which fuel was highly popular for cars in Pakistan but has declined due to shortages
a) Diesel
b) **CNG (Compressed Natural Gas)**
c) LPG
d) Ethanol
Explanation: **CNG** was introduced as a cheaper alternative to petrol, but supply has been restricted to prioritize other sectors.

Which part of the transport sector is the largest consumer of High Speed Diesel (HSD)
a) Rickshaws
b) **Heavy Trucks and Buses**
c) Aeroplanes
d) Cycles
Explanation: **HSD** is the primary fuel for the heavy transport industry and logistics.

The agriculture sector uses electricity primarily for
a) Lighting farms
b) **Operating Tubewells**
c) Driving tractors
d) Packaging fruit
Explanation: **Tubewells** are essential for irrigation, especially in areas where canal water is insufficient.

Which sector consumes the majority of the furnace oil imported by Pakistan
a) Transport
b) **Power Generation**
c) Agriculture
d) Domestic
Explanation: **Furnace Oil** is used in thermal power plants (IPPs) to generate electricity during peak demand.

Which sector uses 'Caking Coal' as a primary energy source
a) Households
b) **Steel Mills**
c) Schools
d) Hospitals
Explanation: **Steel Mills** use specialized coal to produce coke, which is necessary for smelting iron ore.

LPG (Liquefied Petroleum Gas) is mainly used in which areas
a) Where there is a gas pipeline
b) **Where there is no gas pipeline**
c) Only in large factories
d) Only in trains
Explanation: **LPG** cylinders are the main alternative for cooking in hilly areas and remote villages.

Which sector is responsible for the highest 'Carbon Emissions' in Pakistan
a) Agriculture
b) **Transport and Industrial**
c) Education
d) Banking
Explanation: The burning of fossil fuels in engines and furnaces makes these sectors the largest polluters.

Energy conservation means
a) Using more energy
b) **Reducing energy consumption through better habits**
c) Finding new oil
d) Increasing the price
Explanation: Simple habits like turning off unnecessary lights help in **Energy Conservation**.

Energy efficiency means
a) Doing less work
b) **Using less energy to perform the same task**
c) Using old machinery
d) Turning off all power
Explanation: Replacing an old bulb with an **LED** is an example of energy efficiency.

Which organization promotes energy efficiency and conservation in Pakistan
a) WAPDA
b) **NEECA**
c) K-Electric
d) OGRA
Explanation: The **National Energy Efficiency and Conservation Authority (NEECA)** works to reduce energy waste across all sectors.

Using a 'Building Code' helps reduce energy consumption in which sector
a) Transport
b) **Domestic and Commercial**
c) Agriculture
d) Fishing
Explanation: Better insulation and ventilation design mean less energy is needed for heating and cooling.

The 'Peak Hours' in Pakistan's energy consumption usually occur during
a) 3 AM to 6 AM
b) **6 PM to 10 PM**
c) 10 AM to 1 PM
d) Midday
Explanation: Demand is highest in the evening when everyone turns on lights and

'Electric Vehicles' (EVs) aim to shift energy consumption in transport from petrol to
a) Coal
b) **Electricity**
c) Wood
d) Natural Gas
Explanation: **EVs** run on batteries that can be charged from the national grid or solar power.

NATURAL GAS AND NATURAL GAS RESERVES IN PAKISTAN

Where is natural gas typically found? (Source: General science text book, BBISE)
(a) Deep inside volcanoes
(b) Exclusively under the sea bed
(c) **Along with petroleum deposits**
(d) In the Earth's atmosphere

What is the composition of natural gas? (Source: General science text book, BBISE)
(a) Pure methane only
(b) Only sulfur and nitrogen compounds
(c) **Various organic compounds including methane, ethane, propane, and butane**
(d) Water vapor and oxygen

Where are the harmful and useless compounds chemically separated from the natural gas? (Source: General science text book, BBISE)
(a) In the final factory where it is used
(b) **Near the gas wells/source**
(c) At the end of the pipeline
(d) In homes by the consumer
Explanation: These harmful components "are chemically separated near the gas wells themselves.

After purification, how is the clean natural gas transported to homes and factories? (Source: General science text book, BBISE)
(a) By truck
(b) By train
(c) **Through a pipeline system**
(d) In large barrels

In fertilizer factories, which specific product is made from natural gas? (Source: General science text book, BBISE)
(a) Methane
(b) Alcohol
(c) **Urea fertilizer**
(d) Naphthalene

How is electrical energy obtained using the heat energy from natural gas? (Source: General science text book, BBISE)
(a) By direct chemical reaction
(b) By heating oil deposits
(c) **By converting water into steam to run gas turbines and generators**
(d) By running diesel engines

Which of the following organic compounds is the primary component of natural gas? (Source: General science text book, BBISE)
(a) Ethane
(b) Propane
(c) Butane
(d) **Methane**

The process of burning natural gas to create steam for turbines converts chemical potential energy into which final form of useful energy? (Source: General science text book, BBISE)
(a) Potential energy

(b) Kinetic energy (of the turbine)
(c) Heat energy (intermediate step)
(d) **Electrical energy**

What is used for the measurement of natural gas? (Source: General science text book, BBISE)
(a) A thermometer
(b) A ruler
(c) A balance
(d) **A meter**

What unit of measurement is typically used in the meter to measure natural gas consumption? (Source: General science text book, BBISE)
(a) Liters
(b) Gallons
(c) **Cubic meters**
(d) Kilowatt-hours

In which year was the first major natural gas field discovered in Pakistan
a) 1947
b) **1952**
c) 1955
d) 1960
Explanation: The discovery was made at Sui, Baluchistan, just five years after Pakistan's independence.

The first and largest natural gas field in Pakistan was discovered at
a) Mari
b) **Sui**
c) Qadirpur
d) Kandhkot
Explanation: Sui remains the most famous gas field in Pakistan's history and gave the name "Sui Gas" to the fuel.

Which was the first city in Punjab to receive natural gas in 1958
a) **Multan**
b) Lahore
c) Faisalabad
d) Rawalpindi
Explanation: Multan was the gateway for the gas supply as the pipeline moved further north into the province of Punjab.

When was natural gas supply first provided to Quetta
a) 1955
b) 1960
c) 1975
d) **1983**

In which district of Balochistan is the Sui gas field located
a) Quetta
b) Gwadar
c) **Dera Bugti**
d) Mustang

Commercial production of natural gas from the Sui field began in
a) 1952
b) 1953
c) **1955**
d) 1960
Explanation: While discovered in 1952, it took three years to build the pipelines and infrastructure to start production.

Which company discovered the Sui gas field in 1952
a) OGDCL
b) **Pakistan Petroleum Limited (ppl)**
c) Shell
d) Mari Petroleum
Explanation: PPL was the pioneer in oil and

gas exploration in Pakistan and still operates the Sui field today.

Which city received the first supply of Sui gas in 1956
a) Lahore
b) Islamabad
c) **Karachi**
d) Quetta
Explanation: Karachi was the first to benefit from the gas because it was the main industrial hub and the capital at that time.

The second largest gas field in Pakistan by reserves is
a) **Qadirpur**
b) Badarpur
c) Uch
d) Dholak
Explanation: The Qadirpur field in Sindh is a massive reservoir primarily used to provide gas to the fertilizer industry.

Which province is the largest producer of natural gas in Pakistan
a) Balochistan
b) Punjab
c) **Sindh**
d) Khyber Pakhtunkhwa
Explanation: Today, Sindh produces over 60% of Pakistan's total gas output, surpassing Balochistan.

What is the primary component of natural gas in Pakistan
a) Ethane
b) **Methane**
c) Propane
d) Butane
Explanation: Natural gas is mostly Methane (CH_4), which is the simplest and cleanest burning fossil fuel.

What is the approximate percentage of methane in Sui gas
a) 50%
b) 75%
c) **90%**
d) 99%
Explanation: High methane content makes Sui gas very efficient for both domestic heating and industrial production.

Which company is the largest producer of natural gas in Pakistan
a) **OGDCL**
b) PPL
c) MPCL
d) SNGPL
Explanation: Oil and Gas Development Company Limited is a state-owned giant and the leading producer in the country.

The original recoverable reserves of the Qadirpur gas field were estimated at
a) 1 Tcf
b) 2 Tcf
c) **4 TCF**
d) 10 Tcf
Explanation: TCF stands for Trillion Cubic Feet, which is the standard unit used to measure large gas reserves.

The Uch gas field is located in which province
a) Sindh
b) Punjab
c) **Balochistan**
d) Khyber Pakhtunkhwa
Explanation: The Uch field is located in the Dera Murad Jamali region of Balochistan.

The Uch gas field is primarily used for
a) Cooking
b) **Power generation**

c) Fertilizer
d) Cars

The term TCF used in gas reserves stands for
a) Total Cubic Feet
b) **Trillion Cubic Feet**
c) Thousand Cubic Feet
d) Technical Cubic Feet

Which company is responsible for gas distribution in Punjab and KP
a) SSGC
b) **SNGPL**
c) OGDCL
d) PPL
Explanation: Sui Northern Gas Pipelines Limited (SNGPL) serves consumers in the northern half of Pakistan.

Which company manages gas distribution in Sindh and Balochistan
a) SNGPL
b) **SSGC**
c) MPCL
d) PARCO
Explanation: Sui Southern Gas Company (SSGC) handles the distribution network in the southern half of the country.

What is the chemical added to natural gas to give it a smell for safety
a) Methane
b) **Mercaptan**
c) Carbon Dioxide
d) Nitrogen
Explanation: Pure natural gas is odorless; Mercaptan adds a strong "rotten egg" smell so people can detect leaks easily.

Pakistan began importing Liquefied Natural Gas (LNG) primarily to
a) Replace Sui gas entirely
b) **bridge the gap between demand and supply**
c) Export it to China
d) Reduce the cost of gas
Explanation: Domestic production is falling while industrial needs are rising

The first LNG terminal in Pakistan was established at
a) Gwadar Port
b) **Port Qasim**
c) Karachi Port
d) Pasini

Which country is a major supplier of LNG to Pakistan
a) USA
b) **Qatar**
c) Iran
d) Russia.

What is the name of the proposed gas pipeline from Iran to Pakistan
a) TAPI
b) **IP pipeline**
c) Nord Stream
d) Indus Pipe
Explanation: IP stands for Iran-Pakistan

The TAPI pipeline project involves Turkmenistan Afghanistan Pakistan and
a) Iran
b) Iraq
c) **India**
d) China
Explanation: TAPI is an international project designed to bring gas from Central Asia down to South Asian markets.

Which sector in Pakistan is the largest consumer of natural gas

a) Domestic
b) **Power and industrial**
c) Transport
d) Fertilizer.

Compressed Natural Gas (CNG) is used in Pakistan primarily as
a) Cooking fuel
b) **Automotive fuel**
c) Industrial coolant
d) Fertilizer raw material

Natural gas is a source of which energy type
a) **Non-renewable**
b) Renewable
c) Inexhaustible
d) Tidal

What is the main cause of the gas shortage in winter in Pakistan
a) Exporting gas
b) **High domestic consumption for heating**
c) Low production in winter
d) Pipeline cleaning

Which province produces the least amount of natural gas
a) Sindh
b) Balochistan
c) **Punjab**
d) KP

The gas discovery at Razgir, Kohat in 2024 was made in which province
a) Sindh
b) **Khyber Pakhtunkhwa**
c) Balochistan
d) Punjab
Explanation: KP is becoming an increasingly important region for new oil and gas discoveries in Pakistan.

Which gas field has seen a major decline in pressure recently
a) Mari
b) **Sui**
c) Tal Block
d) Nashpa
Explanation: Because it has been producing since 1955, the Sui reservoir is naturally reaching the end of its life.

The pipeline that carries gas is usually made of
a) Plastic
b) **Steel**
c) Copper
d) Aluminum
Explanation: Large-diameter steel pipes are required to transport gas safely under high pressure over long distances.

Which city is known as the Gateway of Gas in Pakistan
a) Karachi
b) **sui**
c) Islamabad
d) Lahore
Explanation: Sui is the symbolic birthplace of the gas industry in Pakistan and the source of its first major energy boom.

What is the full form of OGRA
a) Oil and Gas Regional Authority
b) **Oil And Gas Regulatory Authority**
c) Organization for Gas Resource Area
d) Office of Gas Rate Assessment
Explanation: OGRA is the government body that regulates prices and issues licenses for the oil and gas sector.

The Tal Block is a major gas producing area in which province
a) Sindh

b) **Khyber Pakhtunkhwa**
c) Balochistan
d) Punjab
Explanation: The Tal Block, located near Kohat, is one of the most productive new areas for the energy sector.

Which is the largest oil and gas company listed on the Pakistan Stock Exchange
a) **OGDCL**
b) Mari Petroleum
c) PPL
d) Attock Oil
Explanation: OGDCL is often called the "Market Leader" because of its massive size and contribution to the economy.

Natural gas is found in which type of rock
a) Igneous
b) **Sedimentary**
c) Metamorphic
d) Volcanic
Explanation: Gas is trapped within the tiny pores of sedimentary rocks like sandstone or limestone.

The process of converting natural gas into a liquid for transport is called
a) Fractionation
b) **Liquefaction**
c) Gasification
d) Carbonization
Explanation: Cooling gas to -162°C turns it into a liquid, making it much easier to transport by ship.

What is the boiling point of Methane
a) 0C
b) **-161C**
c) 100C
d) -42C
Explanation: This extremely low temperature is what allows gas to be turned into liquid form (LNG).

Which regulatory body sets the price of gas for domestic consumers
a) NEPRA
b) **OGRA**
c) PEMRA
d) SECP
Explanation: OGRA calculates the gas prices based on production costs and submits them for government approval.

How much does the Sui gas field contribute to Pakistan production now
a) 50%
b) 25%
c) **less than 10%**
d) 1%
Explanation: Its share has dropped significantly over the decades as newer fields in Sindh and KP have been developed.

The Nashpa gas field is located in which district
a) **Kohat**
b) Ghotki
c) Jhelum
d) Sui
Explanation: The Kohat and Karak districts in KP have become major hubs for oil and gas production in recent years.

Which of the following is an inexhaustible resource
a) Natural Gas
b) Coal
c) **Solar energy**
d) Petroleum
Explanation: Unlike fossil fuels which will run out, solar energy is available as long as the sun shines.

What is the primary reason for the gas circular debt in Pakistan
a) High production
b) **unpaid bills and subsidies**
c) Low taxes
d) Exporting gas
Explanation: When the cost of gas is not fully recovered from consumers, debt builds up within the energy supply chain.

The Shahdadpur gas field is located in which province
a) Punjab
b) **Sindh**
c) Balochistan
d) KP
Explanation: Sindh remains the most vital province for Pakistan's current gas needs, hosting many active fields like Shahdadpur.

Which basin in Pakistan holds the majority of the country's gas reserves
a) Upper Indus Basin
B) **Lower Indus Basin**
C) Balochistan Basin
D) Potohar Basin
Explanation: The lower indus basin, covering parts of Sindh, is the most productive region for natural gas.

Which of the following is a major gas field located in the Ghotki district of Sindh
a) sui
b) **Qadirpur**
c) nashpa
d) dakhni
Explanation: Qadirpur is one of the largest gas-producing fields in Sindh and helps meet national demand.

What does the term BTU stand for in gas energy measurement
a) basic thermal unit
b) **British Thermal Unit**
c) burning temperature unit
d) binary thermal unit
Explanation: BTU is used to measure the heat content of gas; higher BTU means better quality gas.

In which year did the government of Pakistan first announce a Petroleum Policy
a) 1947
b) 1971
c) **1991**
d) 2012
Explanation: The 1991 policy was a major step to attract foreign investment in the oil and gas sector.

The process of searching for underground gas using sound waves is called
a) drilling
b) **seismic survey**
c) refining
d) mapping
Explanation: Companies use seismic surveys to create a map of underground rocks before they start drilling.

What is the name of the deep hole dug into the earth to find gas
a) canal
b) **well**
c) pit
d) trench
Explanation: An "exploratory well" is drilled to see if gas exists, while a "production well" is used to bring it out.

WATER RESOURCES OF PAKISTAN

What role have rivers historically played for human civilization, besides providing food? (Source: General science text book, BBISE)
(a) A source of electricity
(b) A place for lakes to form
(c) **A means of movement/transportation**
(d) A source of minerals

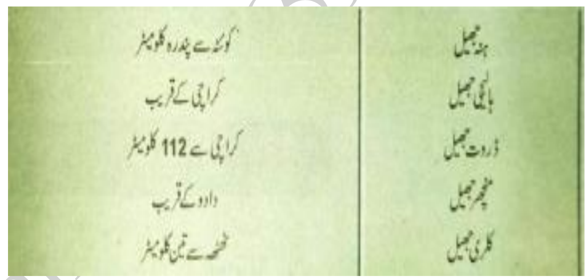
Where is the natural lake Saif-ul-Malook located? (Source: General science text book, BBISE)
(a) Near Quetta
(b) Near Karachi
(c) **In the Kaghan Valley**
(d) Near Peshawar

Which other beautiful lake is located in the northern areas, near Lulusar lake? (Source: General science text book, BBISE)
(a) Hub
(b) Kalri
(c) **Satpara and Khalti**
(d) Manchar

Where is the Hanna Lake located? (Source: General science text book, BBISE)
(a) Near Karachi
(b) Near Dadu
(c) 112 km from Karachi
(d) **15 km from Quetta**

Where is the Hadero Lake located? (Source: General science text book, BBISE)
(a) Near Dadu
(b) **Near Karachi**
(c) 3 km from Mangla
(d) Near Quetta

Where is the Hub Lake located? (Source: General science text book, BBISE)
(a) Near Dadu
(b) 3 km from Mangla
(c) **112 km from Karachi**
(d) Near Quetta



ہنہ جھیل	کوئٹہ سے پندرہ کلومیٹر
ہالیجی جھیل	کراچی کے قریب
دروت جھیل	کراچی سے 112 کلومیٹر
منچھر جھیل	دادو کے قریب
کلری جھیل	ٹھٹھہ سے تین کلومیٹر

Where is the Manchar Lake located? (Source: General science text book, BBISE)
(a) Near Karachi
(b) 112 km from Karachi
(c) **Near Dadu**
(d) 3 km from Mangla

Where are many natural springs commonly found in Pakistan? (Source: General science text book, BBISE)
(a) In Sindh and Punjab plains
(b) Near major cities like Karachi and Quetta
(c) **In the northern and western mountainous regions**
(d) In the deserts of Balochistan

What are the fountains of water that emerge in a hot state called? (Source: General science text book, BBISE)
(a) Springs

(b) Fountains
(c) **Geysers**
(d) Lakes

What materials are often added to strengthen large dams? (Source: General science text book, BBISE)
(a) Wood and plastic
(b) Mud and sand
(c) **Steel and concrete**
(d) Gold and iron

What are smaller dams often called in technical terms? (Source: General science text book, BBISE)
(a) Lakes
(b) Reservoirs
(c) **Barrages**
(d) Bunds

Where is the Rawal Dam located relative to Rawalpindi? (Source: General science text book, BBISE)
(a) 5 kilometers away
(b) **15 kilometers away**
(c) 800 kilometers away
(d) Near Islamabad

Which river is the Rawal Dam built on? (Source: General science text book, BBISE)
(a) Jhelum River
(b) Indus River
(c) Hub River
(d) **Korang Nadi (Korang River)**

In what year was the Mangla Dam constructed? (Source: General science text book, BBISE)
(a) 1955
(b) **1967**
(c) 1976
(d) 1988

On which river is the Mangla Dam built? (Source: General science text book, BBISE)
(a) Indus River
(b) Hub River
(c) **Jhelum River**
(d) Korang River

What is the electricity generation capacity of the Mangla Dam? (Source: General science text book, BBISE)
(a) 137 MW
(b) 240 MW
(c) **800 MW (Megawatts)**
(d) 6,627 MW

Where is the Tarbela Dam located relative to Attock? (Source: General science text book, BBISE)
(a) 15 km away
(b) 112 km away
(c) **65 km away**
(d) Near Karachi

Which river is the Tarbela Dam built on? (Source: General science text book, BBISE)
(a) Jhelum River
(b) Korang River
(c) Hub River
(d) **Indus River**

In what year was the Tarbela Dam constructed? (Source: General science text book, BBISE)
(a) 1967
(b) 1972
(c) **1976**
(d) 1988

What is the height and length of the Hub Dam? (Source: General science text book, BBISE)
(a) 15 meters high, 800 meters long
(b) 65 meters high, 500 meters long
(c) **48 meters high, 7 kilometers long**
(d) 90 meters high, 112 kilometers long

Besides providing irrigation for 81,000 acres of land near Karachi, what other major function does the Hub Dam perform? (Source: General science text book, BBISE)
(a) Generates 800 MW of power
(b) Provides drinking water to Islamabad
(c) **Protects Karachi from floods**
(d) Is a primary source of Naphthalene

How far is the Rawal Dam located from the city of Rawalpindi? (Source: General science text book, BBISE)
(a) 5 kilometers
(b) **15 kilometers**
(c) 50 kilometers
(d) 115 meters

What is notable about the Tarbela Dam's construction material? (Source: General science text book, BBISE)
(a) It is the largest concrete dam.
(b) **It is the largest dam in the world made of mud/earth-fill.**
(c) It is the highest steel dam.
(d) It is made of concrete and steel only.

What is the length and height of the Tarbela Dam? (Source: General science text book, BBISE)
(a) 7 km long, 48 meters high
(b) **2.75 km long, 120 meters high**
(c) 15 km long, 115 meters high
(d) 800 meters long, 65 meters high

Approximately how many acres of land does the Tarbela Dam's lake (reservoir) cover? (Source: General science text book, BBISE)
(a) 8,000 acres
(b) 81,000 acres
(c) 100,000 acres
(d) **75,000 acres**

How much electricity (approximately) is generated from the Tarbela Dam? (Source: General science text book, BBISE)
(a) 800 MW
(b) **3000 MW (annually)**
(c) 137 MW
(d) 75,000 MW

On which river is the Khanpur Dam built? (Source: General science text book, BBISE)
(a) Indus River
(b) Jhelum River
(c) Korang Nadi
(d) **Haro Nadi (Haro River)**

When was the Khanpur Dam built? (Source: General science text book, BBISE)
(a) 1967
(b) **1976**
(c) 1988
(d) 1955

How many acres of land does the Khanpur Dam irrigate in the Rawalpindi, Haro, and Attock areas? (Source: General science text book, BBISE)
(a) 8,000 acres
(b) 75,000 acres
(c) 81,000 acres
(d) **100,000 acres (approximately)**

What should pure drinking water be free from? (Source: General science text book, BBISE)
(a) Helpful salts
(b) **Harmful salts, chemicals, viruses, and radioactive materials**
(c) Only bacteria
(d) Only viruses and chemicals

A deficiency or excess of which specific element in water can cause problems in the throat (Goiter)? (Source: General science text book, BBISE)
(a) Calcium
(b) Iron
(c) Chlorine
(d) **Iodine**

Which diseases are common in Pakistan's regions due to the lack of clean drinking water? (Source: General science text book, BBISE)
(a) Heart attacks and diabetes
(b) **Typhoid, cholera, dysentery, flu, cold, and other diseases**
(c) Cancer and blood pressure issues
(d) Only flu and colds

In the water purification process, what chemical is added to settle down lighter, floating particles? (Source: General science text book, BBISE)
(a) Chlorine
(b) Ozone
(c) Charcoal
(d) **Alum**

Explanation: In the process, "alum (phatkari) is used to settle down the comparatively lighter and floating suspended particles."

What percentage of Pakistan's population has access to clean, purified water facilities? (Source: General science text book, BBISE)
(a) 50%
(b) **30%**
(c) 70%
(d) 90%

In which areas do more than half of the population typically rely on wells or hand pumps for water? (Source: General science text book, BBISE)
(a) Large cities
(b) Industrial zones
(c) Urban centers
(d) **Villages/Rural areas**

Why is water from deep wells or hand pumps generally cleaner than surface water? (Source: General science text book, BBISE)
(a) It is treated with chlorine underground.
(b) It flows faster than river water.
(c) It has high salt content which kills germs.
(d) **It passes through thick layers of earth, filtering impurities.**

Note: *For more MCQs on water resources of Pakistan read my Pakistan Studies Book for Patwari.*

MAJOR COMPONENTS OF COMPUTER AND ITS IMPORTANCE IN MODERN ERA

What is a computer described as in the text book? (Source: General science text book, BBISE)
(a) A slow calculation device
(b) A simple invention
(c) A device that solves simple problems
(d) **A powerful invention that solves complex problems quickly**

A computer operates according to what set of guidelines? (Source: General science text book, BBISE)
(a) Hardware
(b) Output Unit
(c) Data only
(d) **A Programme (program/instructions)**

How many basic parts does a computer has? (Source: General science text book, BBISE)
(a) Two
(b) Three
(c) **Four**
(d) Five
Explanation: The text book states, "A computer has four basic parts." These parts are the Input Unit, Control Unit, Output Unit, and Memory Unit.

Through which unit are data and instructions entered into the computer? (Source: General science text book, BBISE)
(a) Output Unit
(b) Control Unit
(c) Memory Unit
(d) **Input Unit**

Which of these is mentioned as a way to input data into a computer? (Source: General science text book, BBISE)
(a) Printer
(b) Monitor
(c) Magnetic Disk
(d) **Keyboard**

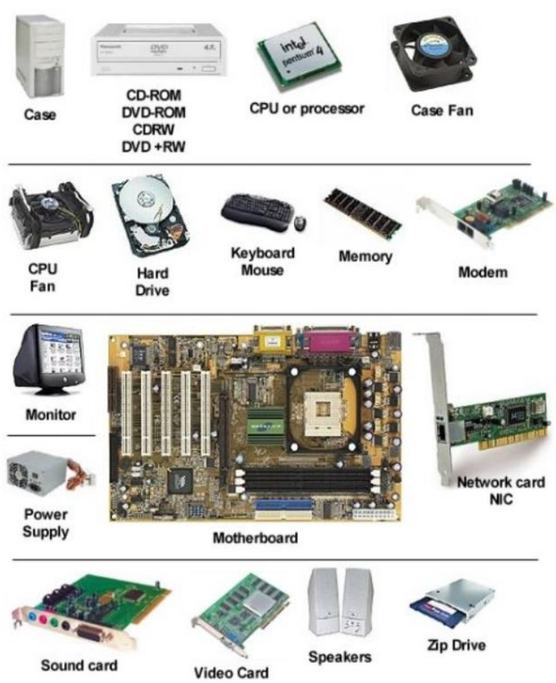
INPUT	OUTPUT
Keyboard	Monitor
Mouse	Printer
Scanner	Speaker
Joystick	Headphones
Microphone	Projector
Barcode Reader	Plotter
Light Pen	GPS
Gamepad	Braille Reader
MICR	Sound Card
Digital Camera	Video Card

Besides the Keyboard, what other physical input methods are? (Source: General science text book, BBISE)
(a) Monitor and Printer
(b) **Punched Cards and Magnetic Tape**
(c) Magnetic Disk and Processor
(d) Control Unit and ALU

After information is entered, where does it flow into next? (Source: General science text book, BBISE)

(a) The Output Unit
(b) The Memory Unit
(c) The Input Unit
(d) **The Processor**

What two major components are located inside the Processor? (Source: General science text book, BBISE)
(a) Input Unit and Output Unit
(b) Memory Unit and Input Unit
(c) **Control Unit and Arithmetic/Logic Unit (ALU)**
(d) Keyboard and Printer



What is the function of the Control Unit within the processor? (Source: General science text book, BBISE)
(a) Performs calculations
(b) Stores data permanently
(c) **Follows the instructions of the program**
(d) Displays output on a screen

What does the Arithmetic/Logic Unit (ALU) perform? (Source: General science text book, BBISE)
(a) Controls program flow
(b) Stores memory
(c) Displays information
(d) **Necessary calculations**

Where are the necessary data, instructions, and solutions stored within the computer system? (Source: General science text book, BBISE)
(a) In the processor
(b) In the input unit
(c) In the output unit
(d) **In the Memory Unit (Nerve Centre)**

Which unit is metaphorically referred to as the "Nerve Centre" of the computer? (Source: General science text book, BBISE)
(a) The Processor
(b) The Memory Unit
(c) The Output Unit
(d) **The Control Unit**

How is the result of the computer's process presented to the user? (Source: General science text book, BBISE)
(a) Through the Input Unit
(b) Through the Processor
(c) **Through the Output Unit**
(d) Through the Memory Unit

Which of these is an example of an Output Unit that looks like a television set? (Source: General science text book, BBISE)
(a) Printer
(b) Keyboard
(c) Magnetic Disk
(d) **Monitor/Screen**

Which output device can print results onto paper? (Source: General science text book, BBISE)

(a) Monitor
(b) Keyboard
(c) **Printer**
(d) Magnetic Tape

What characteristic of computers allows them to manage tasks like preparing electricity bills? (Source: General science text book, BBISE)

(a) Their small size
(b) Their high calculation speed
(c) Their ability to accept input
(d) **Their ability to store a large amount of information**

Which storage method is described as being like the tape used in a tape recorder? (Source: General science text book, BBISE)

(a) Punched Cards
(b) Magnetic Disk
(c) Processor Memory
(d) **Magnetic Tape**

Which storage method is mentioned as being used in some computers? (Source: General science text book, BBISE)

(a) Keyboard
(b) Printer
(c) Punched Cards
(d) **Magnetic Disc**

Information stored on Magnetic Tapes and Discs is entered into the computer when needed using which process? (Source: General science text book, BBISE)

(a) Direct processing
(b) Output unit
(c) Printer
(d) **Input mechanisms**

The Control Unit in the processor receives instructions from where? (Source: General science text book, BBISE)

(a) The Output Unit
(b) The Memory Unit
(c) **The Program (instructions in memory)**
(d) The ALU

The Control Unit supplies necessary information to which another unit? (Source: General science text book, BBISE)

(a) The Input Unit
(b) The Output Unit
(c) **The Arithmetic/Logic Unit (ALU)**
(d) The Magnetic Tape

The flow of information (data and instructions) is generally controlled by which component? (Source: General science text book, BBISE)

(a) The Memory Unit
(b) The ALU
(c) **The Control Unit**
(d) The Printer

A computer system based on which era of technology? (Source: General science text book, BBISE)

(a) Very modern (2020s)
(b) Recent (2000s)
(c) **Older (Punched cards/Magnetic tape era)**
(d) Futuristic (2050s)

Which part of the computer is known as the Brain of the computer

a) Monitor
b) **CPU**

c) Keyboard
d) Mouse
Explanation: The Central Processing Unit (CPU) is responsible for executing instructions and processing data.

What does the ALU in a CPU stand for
a) Always logical unit
b) **Arithmetic logic unit**
c) Array level utility
d) Auto logic unit
Explanation: The ALU performs all mathematical calculations and logical comparisons within the processor.

Which component of the CPU coordinates all computer activities
a) Alu
b) **Control unit**
c) Registers
d) Ram
Explanation: The Control Unit acts as a manager, directing data flow between the CPU and other devices.

The speed of a modern CPU is usually measured in
a) Meters
b) **Gigahertz**
c) Kilobytes
d) Watts
Explanation: Gigahertz (GHz) measures how many billions of instructions a processor can handle per second.

Which memory is known as volatile memory because it loses data when power is turned off
a) Rom
b) **Ram**
c) Hard disk
d) Pen drive
Explanation: Random Access Memory (RAM) is temporary storage; once the computer restarts, its data is cleared.

Which storage device has no moving parts and is much faster than a traditional hard disk
a) HDD
b) **SSD**
c) Floppy disk
d) CD-ROM
Explanation: Solid State Drives (SSD) use flash memory chips, making them extremely fast.

What is the primary function of the Hard Disk Drive
a) Temporary processing
b) **Permanent data storage**
c) Display images
d) Cooling the CPU
Explanation: The hard disk is the long-term storage where your files, photos, and apps are kept safely.

Which type of memory is used to bridge the speed gap between the CPU and RAM
a) **Cache memory**
b) Optical memory
c) Secondary memory
d) Auxiliary memory

Which input device is most commonly used for typing text and commands
a) Mouse
b) **Keyboard**
c) Scanner
d) Printer

A device used to convert hard copy images or text into digital soft copy is a
a) Printer

b) **Scanner**
c) Plotter
d) Monitor

Which of the following is an example of an output device
a) Microphone
b) Keyboard
c) **Monitor**
d) Joystick

The quality of a printer is measured in DPI which stands for
a) Data per inch
b) **Dots Per Inch**
c) Digital pixel input
d) Dots per index
Explanation: DPI measures the density of ink drops; higher DPI results in a clearer and sharper print.

Which output device is used to produce large architectural drawings and maps
a) Laser printer
b) **Plotter**
c) Inkjet printer
d) Speaker

The main circuit board to which all computer components connect is the
a) Hard drive
b) **Motherboard**
c) Power supply
d) Monitor
Explanation: The motherboard is the backbone of the computer that allows all parts to communicate.

The small battery on the motherboard that keeps the system date and time is the
a) Aa battery
b) **CMOS battery**
c) Li-ion battery
d) Lead acid battery
Explanation: The CMOS battery ensures the computer keeps the correct time even when it is unplugged.

What are the slots on the motherboard used to add extra features like graphics cards called
a) Holes
b) **Expansion slots**
c) Ports
d) Sockets
Explanation: Expansion slots allow users to customize their PC by adding video, sound, or network cards.

Which component is responsible for cooling the CPU to prevent overheating
a) **Heat sink and fan**
b) Motherboard
c) Ram
d) Smps
Explanation: Because the CPU gets very hot, a metal heat sink and a fan are used to blow heat away.

The set of instructions that tells the hardware what to do is called
a) Device
b) **Software**
c) Peripheral
d) Hardware
Explanation: Hardware is the physical body, while software is the set of "rules" or "instructions" it follows.

Which of the following is the most important system software in a computer
a) MS word
b) **Operating system**
c) Anti-virus

d) Media player
Explanation: The Operating System (like Windows or Linux) manages all hardware and other software programs.

Software designed to perform a specific task for the user like calculating or drawing is
a) System software
b) **Application software**
c) Utility software
d) Firmware
Explanation: Apps like Chrome, Photoshop, or WhatsApp are all examples of application software.

What is the full form of GUI
a) General User Index
b) **Graphical User Interface**
c) Global Unit Interface
d) Graphical Utility Icon
Explanation: GUI allows you to use a mouse to click icons instead of typing difficult text commands.

What is the first software that runs when you turn on a computer
a) Windows
b) **BIOS**
c) Browser
d) Calculator
Explanation: The BIOS performs a "Power On Self-Test" (POST) to make sure everything is working correctly.

The use of computers to perform tasks without human intervention is called
a) Manual labor
b) **Automation**
c) Calculation
What is the role of computers in the modern education system
a) Only for gaming
b) **E-learning and research**
c) Only for printing
d) To replace teachers entirely
Explanation: Computers enable students to take online classes and find information from anywhere in the world.

How have computers changed the modern banking system
a) By making it slower
b) **Through online banking and ATMs**
c) By increasing paper use
d) By closing all branches
Explanation: Computers allow you to check your balance or transfer money instantly using a phone or ATM.

The process of buying and selling goods over the internet is known as
a) E-mail
b) **E-commerce**
c) E-governance
d) E-banking
Explanation: Websites like Amazon or Daraz use computers to process orders and payments globally.

In the medical field computers are used for
a) Only billing
b) **MRI scans and patient records**
c) Making medicines at home
d) Only playing music
Explanation: Doctors use computers for advanced life-saving tools like MRI machines and robotic surgery.

Which component is required for a computer to connect to a network
a) Printer
b) **NIC (network interface card)**

c) Scanner
d) Speakers
Explanation: A Network Interface Card (NIC) provides the port or Wi-Fi connection needed to join a network.

What does the term URL stand for
a) United Resource Locator
b) **Uniform Resource Locator**
c) Universal Radio Link
d) Uniform Road Link
Explanation: A URL is the unique web address used to find a specific site on the internet.

The global network of computers that connects millions of people is the
a) Intranet
b) **Internet**
c) Lan
d) CPU
Explanation: The internet is the world's largest public network, allowing for global communication.

What is a 'Cloud' in modern computing
a) A weather pattern
b) **Storage and services provided over the internet**
c) A part of the CPU
d) A type of monitor
Explanation: Cloud computing lets you store files on the internet rather than on your physical computer.

A program designed to damage or steal data from a computer is called a
a) Bug
b) **Virus**
c) Driver
d) Utility
Explanation: Viruses are harmful programs that can delete your files or slow down your system.

What should you use to protect your computer from internet threats
a) Screen guard
b) **Firewall and anti-virus**
c) Mouse pad
d) Cooler fan
Explanation: Anti-virus software scans and removes threats, while a firewall blocks hackers from entering.

Making a copy of your important files to prevent data loss is called
a) Deleting
b) **Backup**
c) Scanning
d) Formatting
Explanation: Backing up ensures that if your computer crashes, you still have your important work saved elsewhere.

What is 'Spam' in the context of computers
a) A type of virus
b) **Unsolicited bulk emails**
c) A storage device
d) A programming language
Explanation: Spam refers to annoying junk mail often sent for advertising or scams.

Which key is used as a shortcut to refresh a webpage or computer folder
a) F1
b) **F5**
c) F12
d) Esc

Who is often called the Father of Computers
a) Bill Gates

b) **Charles Babbage**
c) Mark Zuckerberg
d) Steve Jobs
Explanation: Charles Babbage designed the Analytical Engine, the first concept of a general-purpose computer.

What is 'Laptop' primarily categorized as
a) Desktop computer
b) **Portable computer**
c) Supercomputer
d) Server
Explanation: Laptops are mobile computers designed to be used in different locations using a battery.

The physical parts of a computer that you can touch are called
a) Software
b) **Hardware**
c) Firmware
d) Data
Explanation: Anything you can physically hold, like the mouse or screen, is considered hardware.

Artificial Intelligence (AI) is mostly related to which generation of computers
a) First
b) Third
c) **Fifth**
d) Second
Explanation: Fifth-generation computing focuses on making machines "smart" using Artificial Intelligence.

Which computer is the most powerful and used for weather forecasting
a) Microcomputer
b) **Supercomputer**
c) Workstation
d) Tablet
Explanation: Supercomputers are massive machines used for complex tasks like space research and climate modeling.

What does 'Binary' refer to in computing
a) Base 10 system
b) **Base 2 system (0 and 1)**
c) Base 16 system
d) A type of software
Explanation: Computers process all information using a simple code of 1s and 0s.

A 'Bit' is the smallest unit of data and represents
a) A character
b) **0 or 1**
c) A word
d) A pixel
Explanation: Bit is short for "Binary Digit," which is the most basic building block of digital data.

How many bits make up one Byte
a) 4
b) **8**
c) 16
d) 1024

What is the importance of a UPS in a computer system
a) Increases speed
b) **Provides battery backup during power failure**
c) Connects to internet
d) Prints colors
Explanation: A UPS (Uninterruptible Power Supply) prevents the computer from shutting down it instantly.

USB stands for
a) Uniform service bus

b) **Universal serial bus**
c) Universal sector block
d) United serial bus
Explanation: USB is the standard interface used to connect almost all modern accessories to a computer.

The physical connector used to link a monitor to the CPU for high-definition video is
a) USB
b) **HDMI**
c) Serial Port
d) Power Cord
Explanation: **High-Definition Multimedia Interface (HDMI)** transmits both high-quality digital video and audio through a single cable.

Which component is often called the backbone of the computer
a) Hard Drive
b) **Motherboard**
c) Monitor
d) Mouse
Explanation: The **Motherboard** is the primary circuit board that allows communication between all other hardware parts.

The technology that allows computers to recognize human speech is
a) Graphics
b) **Voice Recognition**
c) Formatting
d) Scanning
Explanation: This modern feature enables tools like **Google Assistant** or **Siri** to understand spoken commands.

In the modern era GPS stands for
a) Global Path System
b) **Global Positioning System**
c) General Point Site
d) Global Planet Sensor
Explanation: **GPS** uses computer chips and satellites to provide accurate location and navigation data.

The use of computers to design buildings and machine parts is called
a) CAM
b) **CAD**
c) Chat
d) CPU
Explanation: **Computer-Aided Design (CAD)** allows engineers to create precise 3D models before manufacturing.

A collection of 4 bits is known as a
a) Byte
b) **Nibble**
c) Kilobyte
d) Word
Explanation: A **Nibble** is exactly half of a **Byte** (4 bits) and is a small unit of data storage.

How many Megabytes make up one Gigabyte
a) 100
b) **1024**
c) 500
d) 10
Explanation: In computer storage, units increase by powers of 2, so **1024 MB** equals **1 GB**.

Which storage media uses a laser beam to read and write data
a) Magnetic Tape
b) **Optical Disk**
c) Hard Drive
d) RAM

Explanation: **Optical Disks** like CDs, DVDs, and Blu-rays use light (lasers) to access information.

What is the main advantage of a Pen drive over a Hard Disk
a) More storage
b) **Portability and small size**
c) Faster processing
d) Better cooling
Explanation: **Flash Drives** (Pen drives) are tiny and can be carried in a pocket to move data between computers.

The small arrow or icon on the screen that moves with the mouse is the
a) Target
b) **Pointer**
c) Pixel
d) Marker
Explanation: The **Pointer** allows users to interact with items on the **Graphical User Interface (GUI)**.

A touch-sensitive screen is both an Input and an
a) Storage Device
b) **Output Device**
c) Processing Device
d) Power Device
Explanation: Since it shows images (Output) and accepts touches (Input), it is a **Dual-Purpose Device**.

Which software utility is used to reduce the size of a file
a) Anti-virus
b) **Compression Software**
c) Browser
d) Driver
like **WinZip** or **WinRAR** "squeeze" data to make it take up less storage space.

The process of starting or restarting a computer is called
a) Running
b) **Booting**
c) Walking
d) Kicking
Explanation: **Booting** is the process of loading the **Operating System** into the computer's memory.

Which component should you upgrade to run many programs at once
a) Monitor
b) **RAM**
c) Keyboard
d) Speakers
Explanation: More **RAM** provides a larger "workspace" for the computer to handle multiple tasks at the same time.

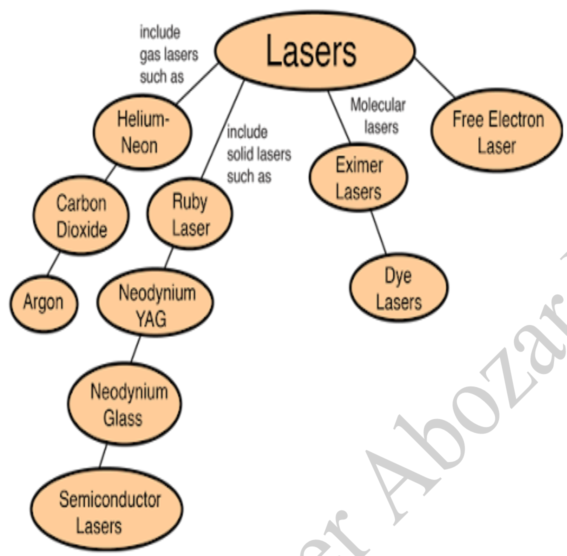
What is a Web Browser
a) A Search Engine
b) **A software used to view websites**
c) A type of computer
d) An internet cable
Explanation: Programs like **Chrome**, **Firefox**, and **Edge** are browsers used to access the internet.

What does WWW stand for
a) World Wide Wrestling
b) **World Wide Web**
c) World Web Window
d) World Wide Work

Which device converts digital signals to analog for internet over phone lines
a) Switch
b) **Modem**
c) Scanner
d) CPU

TYPES OF LASER AND USES OF LASER BEAMS

What is a fundamental characteristic of a laser beam compared to sunlight or light bulb radiation? (Source: General science text book, BBISE)
(a) It contains many different frequencies (colors).
(b) It is highly dispersed over distance.
(c) **It has only a single frequency (color).**
(d) It is very weak light.



What term is used to describe the intensity and focus of a laser beam? (Source: General science text book, BBISE)
(a) Dispersed
(b) Weak
(c) Multi-colored
(d) **Highly Concentrated**

Explanation: A laser beam is described as "a very powerful and highly concentrated ray of light," meaning the energy is focused into a narrow beam.

What property of laser beams allows them to travel long distances without spreading out? (Source: General science text book, BBISE)
(a) They are multi-directional.
(b) They are highly concentrated.
(c) They are slow-moving.
(d) **They are unidirectional**

Which application of lasers involves medical procedures on a specific part of the body? (Source: General science text book, BBISE)
(a) Removing nails
(b) Cutting metal plates
(c) Measuring distance of enemy planes
(d) **Surgery of eye patients**

Which application uses very thin laser beams for cosmetic purposes? (Source: General science text book, BBISE)
(a) Cancer treatment
(b) Eye surgery
(c) Measuring enemy speed
(d) **Removing body hair**

In industrial use, what can powerful lasers be used for with metallic plates? (Source: General science text book, BBISE)
(a) Measuring their thickness
(b) Cleaning them
(c) **Cutting and joining them (welding)**
(d) Measuring their distance

What specific type of laser is mentioned that uses a solid crystal? (Source: General science text book, BBISE)
(a) Gas laser
(b) Liquid laser
(c) **Solid State Laser**
(d) Diode laser

In military applications, lasers are used on guns, missiles, and bomber aircraft for what purpose? (Source: General science text book, BBISE)
(a) Causing immediate destruction
(b) Communicating with other planes
(c) **Measuring the speed and distance of enemy aircraft**
(d) Fusing their metal plates together

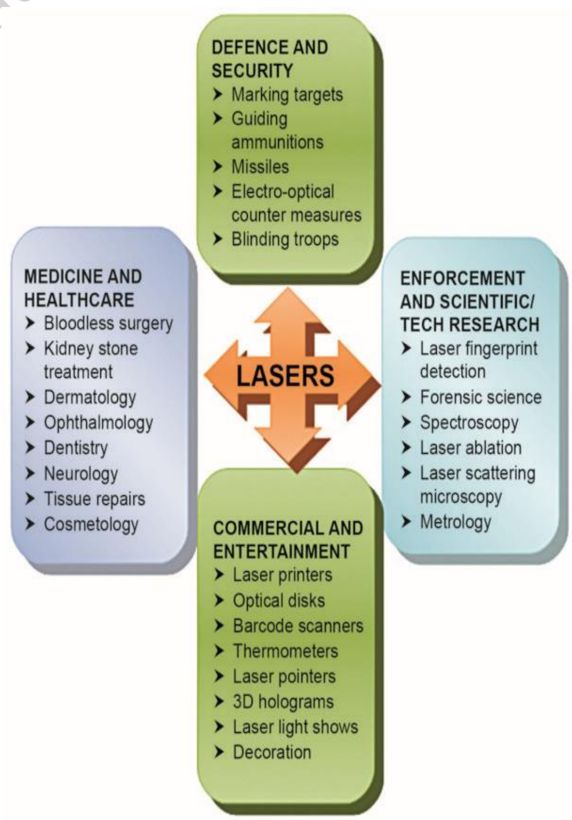
In the modern era, lasers come in two general power categories. What are they? (Source: General science text book, BBISE)
(a) Fission and Fusion
(b) **Very low power and extremely high power**
(c) Visible light and infrared
(d) Military and medical grade

What electrical and electronic machines are low-power lasers used in? (Source: General science text book, BBISE)
(a) High power generators
(b) Aircraft targeting systems
(c) **Different electrical and electronic machines**
(d) Uranium enrichment facilities

What extremely high temperature generating process are high-power lasers used to initiate? (Source: General science text book, BBISE)
(a) Fission
(b) Combustion
(c) **Fusion**
(d) Standard electricity generation

What specific nuclear material enrichment process uses extremely powerful lasers? (Source: General science text book, BBISE)
(a) Deuterium
(b) Plutonium
(c) Thorium
(d) **Uranium**

In basic terms, a laser beam is a highly concentrated ray of what? (Source: General science text book, BBISE)
(a) Sound
(b) Heat
(c) Electricity
(d) **Light**



Which characteristic makes laser light useful for long-distance targeting applications? (Source: General science text book, BBISE)
(a) High power
(b) Single color
(c) **Unidirectional travel**
(d) Heat generation

How does a conventional light bulb's light compare to a laser's light? (Source: General science text book, BBISE)
(a) A bulb's light is unidirectional
(b) A bulb's light has a single frequency
(c) **A bulb's light is a combination of many different frequencies**
(d) A bulb's light is more concentrated

What does the acronym LASER stand for
a) Light Amplification by Stimulated Emission of Radio
b) **Light Amplification by Stimulated Emission of Radiation**
c) Light Absorption by Stimulated Energy of Radiation
d) Light Amplitude by Simple Emission of Radiation

Who is credited with the first theoretical proposal of Stimulated Emission in 1917
a) Theodore Maiman
b) **Albert Einstein**
c) Charles Townes
d) Niels Bohr
Explanation: **Albert Einstein** established the theoretical foundations for the laser in his paper "On the Quantum Theory of Radiation."

Laser light is highly directional because it travels as a
a) Broad wave
b) **Parallel Beam**
c) Circular wave
d) Scattered light

Which was the first working laser ever built in 1960
a) Gas Laser
b) **Ruby Laser**
c) Helium-Neon Laser
d) Dye Laser
Explanation: The **Ruby Laser**, developed by **Theodore Maiman**, used a synthetic ruby crystal as the active medium.

The Ruby Laser belongs to which category of lasers
a) Gas Laser
b) **Solid-State Laser**
c) Liquid Laser
d) Semiconductor Laser

What is the active medium used in a He-Ne Laser
a) Pure Neon
b) **Mixture of Helium and Neon**
c) Hydrogen and Oxygen
d) Argon and Krypton
Explanation: The **He-Ne Laser** uses a mixture of Helium and Neon gas, typically in a 10:1 ratio, to produce red light.

Which laser is widely used in supermarket barcode scanners
a) Ruby Laser
b) **He-Ne Laser**
c) CO2 Laser
d) Ndoye Laser
Explanation: The **Helium-Neon (He-Ne) Laser** is preferred for scanners due to its continuous and stable visible red beam.

Which gas laser is highly efficient and used for heavy industrial cutting and welding
a) Argon Laser
b) **CO2 Laser**
c) He-Ne Laser
d) Excimer Laser
Explanation: The **Carbon Dioxide (CO2) Laser** produces high power in the infrared spectrum, making it ideal for cutting metal.

Semiconductor lasers are also commonly known as
a) Gas Lasers
b) **Laser Diodes**
c) Chemical Lasers
d) Fiber Lasers
Explanation: **Laser Diodes** are compact lasers found in CD players, laser pointers, and fiber-optic communication.

What material is commonly used to make a Semiconductor Laser
a) Silicon
b) **Gallium Arsenide**
c) Carbon
d) Germanium
Explanation: **Gallium Arsenide (GaAs)** is a direct band-gap semiconductor essential for light emission in diodes.

Which type of laser uses a liquid solution of organic dyes as the active medium
a) Gas Laser
b) **Dye Laser**
c) Excimer Laser
d) Diode Laser

Excimer lasers produce light in which part of the spectrum
a) Infrared
b) Visible
c) **Ultraviolet**
d) X-ray
Explanation: **Excimer Lasers** use reactive gases like **Chlorine or Fluorine** to produce high-energy **UV** light for eye surgery.

Lasers are used in eye surgery to repair a detached
a) Pupil
b) Lens
c) **Retina**
d) Cornea
Explanation: The laser beam acts as a "welding" tool to reattach the **Retina** to the back of the eye.

Which laser is most suitable for removing tattoos and skin spots
a) He-Ne Laser
b) **Nd:YAG Laser**
c) CO2 Laser
d) Ruby Laser
Explanation: Pulsed **Nd:YAG** lasers break down tattoo ink particles without damaging the surrounding skin.

In surgery lasers are used because they
a) Freeze the skin
b) **Cauterize blood vessels as they cut**
c) Are very cold
d) Do not touch the body
Explanation: The heat from the laser seals the blood vessels immediately, preventing bleeding during operations.

Which type of laser is used to shatter kidney stones in a process called Lithotripsy
a) **Holmium Laser**
b) He-Ne Laser

c) Dye Laser
d) Semiconductor Laser

Laser beams are used in LIDAR technology for
a) Heating food
b) **Measuring distances and mapping**
c) Printing photos
d) Cleaning water
Explanation: **Light Detection and Ranging (LIDAR)** is used in self-driving cars and aircraft to map terrain using laser pulses.

The process of using lasers to create 3D images is called
a) Photography
b) **Holography**
c) Radiography
d) Lithography
Explanation: **Holography** uses laser interference patterns to record and display three-dimensional images called holograms.

In CD and DVD players what is used to read data from the disk
a) A magnetic needle
b) **A Laser Diode**
c) An electron beam
d) A static charge

Lasers are essential in Fibre Optics for
a) Lighting up rooms
b) **High-speed data communication**
c) Charging batteries
d) Cooking
Explanation: Laser pulses carry digital information through glass fibres across oceans with very low loss.

Which property of laser allows it to drill holes in diamonds
a) Colour
b) **High Intensity and Sharp Focus**
c) Coherence
d) Speed
Explanation: Because lasers can be focused into an incredibly small spot, they concentrate enough energy to melt or vaporize the hardest materials.

Laser-guided missiles use a laser beam to
a) Explode the target
b) **Designate or paint the target for precision**
c) Fuel the engine
d) Communicate with the pilot

A Ring Laser Gyroscope is used in aircraft for
a) Measuring fuel
b) **Navigation without satellites**
c) Communicating with ground control
d) Entertainment
Explanation: **Laser Gyroscopes** detect rotation and changes in direction to help planes and missiles navigate.

What is the typical wavelength of a He-Ne Laser's red light
a) 400 nm
b) **632.8 nm**
c) 1000 nm
d) 10.6 um

Which laser operates in the Infrared region at a wavelength of 10.6 micrometers
a) Ruby Laser
b) He-Ne Laser
c) **CO2 Laser**
d) Argon Laser

Which laser is used in the treatment of glaucoma
a) **Argon Laser**
b) Ruby Laser
c) Nd:YAG Laser
d) Semiconductor Laser
Explanation: The **Argon Laser** is blue-green and is well-absorbed by eye tissues to treat pressure buildup in glaucoma.

Which of the following is a disadvantage of using high-power lasers
a) They are too fast
b) **They can cause permanent eye damage**
c) They are too quiet
d) They only work in the dark
Explanation: Even reflected laser light can burn the retina, which is why safety goggles are mandatory in labs.

Laser beams can be used to monitor the pollution of the
a) Ground
b) **Atmosphere**
c) Deep Ocean
d) Inner earth
Explanation: Lasers can detect chemical concentrations in the air from a distance through a method called LIDAR spectroscopy.

Which industry uses lasers for marking serial numbers on metal parts
a) Agriculture
b) **Manufacturing and Automotive**
c) Textile
d) Fishing
Explanation: Laser marking is permanent, fast, and does not involve physical contact or ink.

In an optical fibre the laser light stays inside because of
a) Refraction
b) **Total Internal Reflection**
c) Diffraction
d) Polarization
Explanation: **Total Internal Reflection (TIR)** keeps the laser signal trapped inside the core of the fibre optic cable.

Which laser is used for precisely cutting the semiconductor chips in computers
a) CO2 Laser
b) **Excimer Laser**
c) He-Ne Laser
d) Ruby Laser
Explanation: **Excimer Lasers** provide the precision needed for the fine patterns on microchips.

What colour is the light produced by an Argon Ion Laser
a) Red
b) **Blue-Green**
c) Yellow
d) Infrared
Explanation: **Argon Lasers** produce beautiful blue and green beams often seen in laser light shows and used in eye surgery.

Which of the following is an example of a high-power Gas Laser
a) Ruby Laser
b) **Carbon Dioxide (CO_2) Laser**
c) Semiconductor Laser
d) He-Ne Laser
Explanation: **CO_2 Lasers** are powerful gas lasers primarily used in heavy industry for cutting and welding.

TIDAL FORCE AND ASTRONOMICAL SCIENCES

The periodic rise and fall of sea levels caused by the combined effects of gravitational forces is called
a) Waves
b) **Tides**
c) Currents
d) Tsunamis
Explanation: **Tides** are the regular rise and fall of the ocean's surface caused by the **Moon**.

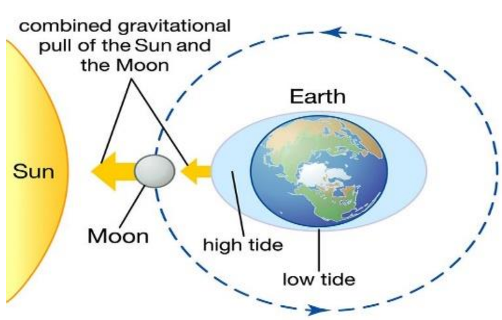
Which celestial body has the greatest influence on Earth's tides
a) Sun
b) **Moon**
c) Jupiter
d) Mars
Explanation: Even though the **Sun** is much larger, the **Moon** is closer to **Earth**, making its tidal force twice as strong.

The gravitational pull of the Moon is strongest on the side of Earth that is
a) Furthest from the Moon
b) **Closest to the Moon**
c) At the North Pole
d) In the center
Explanation: Gravity decreases with distance, so the **Moon** pulls more strongly on the water directly facing it.

Tidal forces are essentially a result of the difference in
a) Magnetic field
b) **Gravitational pull over a distance**
c) Atmospheric pressure
d) Solar heat

How many high tides usually occur in a 24-hour period on Earth
a) One
b) **Two**
c) Three
d) Four
Explanation: Earth experiences two high tides and two low tides daily due to its rotation through two tidal bulges.

When the Sun Moon and Earth are aligned in a straight line the tides are called
a) Neap Tides
b) **Spring Tides**
c) Ebb Tides
d) Low Tides
Explanation: **Spring Tides** occur during New Moon and Full Moon, resulting in the highest high tides.



During which phase of the Moon do Neap Tides occur
a) New Moon

b) Full Moon
c) **First and Third Quarter**
d) Blue Moon
Explanation: **Neap Tides** occur when the **Sun** and **Moon** are at right angles, partially cancelling each other's gravity.

Tidal bulging on the far side of Earth away from the Moon is caused by
a) Solar wind
b) **Inertia and Centrifugal Effect**
c) Magnetic repulsion
d) Global warming
Explanation: While gravity pulls water on the near side, **Inertia** causes a bulge on the opposite side of the earth.

The time interval between two successive high tides is approximately
a) 12 hours
b) **12 hours and 25 minutes**
c) 24 hours
d) 6 hours
Explanation
Because the **Moon** orbits **Earth** in the same direction Earth rotates, it takes a bit longer for a point to realign with the **Moon**.

What is the term for a tide that is falling or moving away from the shore
a) Flood Tide
b) **Ebb Tide**
c) Spring Tide
d) High Tide
Explanation: **Ebb Tide** refers to the period between high tide and low tide when water levels drop.

Tidal forces can cause a moon to be torn apart if it crosses the
a) Event Horizon
b) **Roche Limit**
c) Kuiper Belt
d) Oort Cloud
Explanation: The **Roche Limit** is the minimum distance to a planet where a celestial body held together only by gravity.

Which planet in our solar system has the most famous rings caused by tidal disruption
a) Jupiter
b) **Saturn**
c) Neptune
d) Earth
Explanation: **Saturn's** rings are believed to be the remains of moons or comets shattered by **Saturn's** massive tidal forces.

The slowing down of a planet's rotation due to tidal forces is known as
a) Tidal Heating
b) **Tidal Braking**
c) Orbital Decay
d) Solar Drift
Explanation: Friction from tides acts as a brake, gradually slowing down the rotation of planets like **Earth**.

Why does the same side of the Moon always face Earth
a) The Moon does not rotate
b) **Tidal Locking**
c) It is stuck in magnetic field
d) Earth's shadow

Which moon of Jupiter is the most volcanically active due to Tidal Heating
a) Europa
b) **Io**
c) Ganymede
d) Callisto
Explanation: **Io** is squeezed and stretched

by the gravity of **Jupiter** and other moons, creating internal heat and volcanoes.

What is the term for the height difference between high tide and low tide
a) Tidal Wave
b) **Tidal Range**
c) Tidal Current
d) Tidal Depth
Explanation: The **Tidal Range** varies depending on the location and the alignment of the **Sun** and **Moon**.

Which of the following is a potential source of renewable energy from the ocean
a) Salt extraction
b) **Tidal Power**
c) Underwater mining
d) Fishing
Explanation: **Tidal Power** uses the movement of water during tides to turn turbines and generate electricity.

The Bay of Fundy is famous for having the world's highest
a) Salt concentration
b) **Tidal Range**
c) Fish population
d) Depth
Explanation: The **Bay of Fundy** in Canada experiences a **Tidal Range** of up to 16 meters.

Galileo wrongly believed that tides were caused by Earth's
a) Gravity
b) **Motion and Rotation**
c) Atmosphere
d) Internal heat
Explanation: **Galileo** thought tides were proof of Earth's motion, ignoring the **Moon's** gravitational influence.

Who was the first scientist to mathematically explain tides using gravity
a) Johannes Kepler
b) **Isaac Newton**
c) Albert Einstein
d) Nicolaus Copernicus
Explanation: **Isaac Newton** explained tides in his "Principia" using his Law of Universal Gravitation.

What happens to the Moon's orbit as a result of tidal interaction with Earth
a) It is getting closer
b) **It is moving further away**
c) It is staying the same
d) It is becoming a circle
Explanation: The **Moon** drifts away from **Earth** at a rate of about 3.8 cm per year

Which type of tide occurs when the Moon and Sun are at right angles
a) Spring Tide
b) **Neap Tide**
c) Flood Tide
d) King Tide
Explanation: During **Neap Tide**, the high tides are lower than usual and low tides are higher than usual.

If Earth had no Moon would there still be tides
a) No
b) **Yes, but much smaller**
c) Yes, they would be bigger
d) Tides are only caused by wind
Explanation: The **Sun** would still cause

tides, but they would be about 44% the strength of current tides.

In which month is the Sun's tidal effect on Earth strongest
a) July
b) **January**
c) June
d) September
Explanation: Earth is at **Perihelion** (closest to the Sun) in **January**, making solar gravity slightly stronger.

The study of the relationship between tides and the position of stars is a part of
a) Meteorology
b) **Astrophysics**
c) Geology
d) Paleontology
Explanation: **Astrophysics** deals with the physical properties and interactions of celestial bodies like the **Moon** and **Sun**.

What is the 'Intertidal Zone'
a) The middle of the ocean
b) **The area between the high tide and low tide marks**
c) The bottom of the sea
d) The atmosphere above the sea

Large lakes like the Great Lakes have tides that are
a) Larger than the ocean
b) **Barely noticeable**
c) Non-existent
d) Caused by boats
Explanation: While gravity acts on all water, the volume in lakes is too small to show significant tidal bulging.

Tidal forces help stabilize Earth's
a) Magnetic field
b) **Axial Tilt**
c) Orbit around the Sun
d) Temperature
Explanation: The **Moon's** presence and tidal interaction prevent **Earth** from wobbling too much on its axis.

Which phase of the moon results in the largest tidal range
a) First Quarter
b) **New Moon**
c) Waxing Gibbous
d) Waning Crescent
Explanation: During a **New Moon**, the **Sun** and **Moon** pull in the same direction, creating the largest range.

Tidal forces can cause planets to become 'Oblate Spheroids' which means they are
a) Perfectly round
b) **Flattened at the poles and bulging at the equator**
c) Shaped like a triangle
d) Longer than they are wide
Explanation: Rotational and tidal forces cause planets like **Earth** and **Jupiter** to bulge at the middle.

The term 'Perigean Spring Tide' occurs when the Moon is
a) At its furthest point
b) **At its closest point (Perigee) and aligned with the Sun**
c) Eclipsed by Earth
d) Invisible
Explanation: These tides are exceptionally high because the **Moon's** gravitational pull is at its maximum.

A 'Diurnal' tide cycle consists of
a) Two high tides and two low tides
b) **One high tide and one low tide per day**

c) Only high tides
d) Constant water level
Explanation: Some areas, like the **Gulf of Mexico**, experience only one tidal cycle per day due to coastal geography.

The Earth's crust also experiences tides called
a) Ocean Tides
b) **Earth Tides**
c) Mantle Waves
d) Seismic Tides
Explanation: Just like water, the solid ground of **Earth** rises and falls by several centimeters due to tidal forces.

Which instrument is used to measure the height of tides
a) Barometer
b) **Tide Gauge**
c) Seismograph
d) Anemometer
Explanation: **Tide Gauges** or **mareographs**

What is a 'King Tide'
a) A tide during a storm
b) **An informal name for an exceptionally high Spring Tide**
c) A tide in the UK
d) A tide caused by planets
Explanation: **King Tides** often happen when the **Earth** is closest to the **Sun** and the **Moon** is closest to **Earth**.

A moon that orbits very close to its planet might experience 'Tidal Disruption' and become a
a) Star
b) **Ring System**
c) Black Hole
d) Gas Giant
Explanation: If a moon gets too close, tidal forces overcome its own gravity and pull it apart into pieces.

Which planet is known as the Red Planet
a) Venus
b) **Mars**
c) Jupiter
d) Saturn
Explanation: **Mars** appears red due to iron oxide (rust) on its surface.

The largest planet in our solar system is
a) Saturn
b) **Jupiter**
c) Neptune
d) Uranus
Explanation: **Jupiter** is so large that all other planets in the solar system could fit inside it.

Which planet has the shortest day in the solar system
a) Mercury
b) Earth
c) **Jupiter**
d) Mars
Explanation: **Jupiter** rotates very fast, with one day lasting only about 10 hours.

The hottest planet in our solar system is
a) Mercury
b) **Venus**
c) Mars
d) Jupiter
Explanation: **Venus** has a thick atmosphere of **CO_2** that traps heat (Greenhouse Effect), making it hotter than **Mercury**.

Between which two planets is the Asteroid Belt located
a) Earth and Mars

b) **Mars and Jupiter**
c) Jupiter and Saturn
d) Saturn and Uranus
Explanation: The **Asteroid Belt** is a region of rocky debris located between the orbits of **Mars** and **Jupiter**.

Which planet is known as the Evening Star or Morning Star
a) Mars
b) **Venus**
c) Mercury
d) Jupiter
Explanation: **Venus** is the brightest object in the sky after the **Sun** and the **Moon**.

The Great Red Spot is a giant storm located on
a) Mars
b) Saturn
c) **Jupiter**
d) Neptune
Explanation: The **Great Red Spot** is a high-pressure storm that has been raging on **Jupiter** for hundreds of years.

Which planet is famous for its prominent ring system
a) Jupiter
b) **Saturn**
c) Uranus
d) Neptune
Explanation: While other gas giants have rings, **Saturn's** rings are the most visible and complex.

Which planet rotates on its side (98-degree tilt)
a) Neptune
b) **Uranus**
c) Venus
d) Mars
Explanation: **Uranus** is unique because its axis of rotation is almost parallel to its orbital plane.

Which planet was demoted to the status of a Dwarf Planet in 2006
a) Ceres
b) **Pluto**
c) Eris
d) Makemake
Explanation:
The **IAU** reclassified **Pluto** because it has not cleared its neighboring region of debris.

The 'Blue Planet' refers to
a) Neptune
b) Uranus
c) **Earth**
d) Venus
Explanation: **Earth** is called the Blue Planet because over 70% of its surface is covered by water.

Which planet has the most moons (discovered so far)
a) Jupiter
b) **Saturn**
c) Uranus
d) Mars
Explanation: As of recent discoveries, **Saturn** leads with over 140 moons.

The smallest planet in the solar system is
a) Mars
b) **Mercury**
c) Venus
d) Pluto
Explanation: **Mercury** is the smallest "major" planet and is only slightly larger than the Moon.

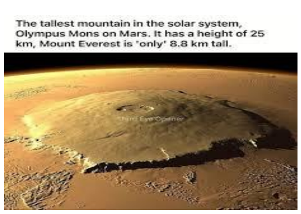
Which planet is known as Earth's Twin due to its similar size
a) Mars
b) **Venus**
c) Neptune
d) Mercury
Explanation: **Venus** and **Earth** are very similar in size, mass, and composition.

The 'Terrestrial Planets' include Mercury Venus Earth and
a) Jupiter
b) **Mars**
c) Saturn
d) Neptune
Explanation: **Terrestrial Planets** are rocky planets with solid surfaces.

The 'Gas Giants' are Jupiter and
a) Mars
b) **Saturn**
c) Uranus
d) Neptune
Explanation: **Jupiter** and **Saturn** are mostly made of hydrogen and helium.

The 'Ice Giants' are Uranus and
a) Jupiter
b) Saturn
c) **Neptune**
d) Pluto
Explanation: **Uranus** and **Neptune** contain more "ices" like water, ammonia, and methane.

Which planet has the highest mountain Olympus Mons
a) Earth
b) **Mars**
c) Venus
d) Jupiter



Titan is the largest moon of which planet
a) Jupiter
b) **Saturn**
c) Neptune
d) Mars

Phobos and Deimos are the moons of
a) Jupiter
b) **Mars**
c) Saturn
d) Neptune
Explanation: **Mars** has two small, irregularly shaped moons named after the Greek words for "Fear" and "Dread."

Which planet has the fastest winds in the solar system
a) Jupiter
b) Saturn
c) **Neptune**
d) Mars
Explanation: **Neptune** experiences winds that can reach speeds of over 2,000 kilometers per hour.

Which planet can float in water because its density is lower than water
a) Jupiter
b) **Saturn**
c) Uranus
d) Neptune

The time taken by Earth to complete one revolution around the Sun is
a) 24 hours
b) **365.25 days**
c) 30 days
d) 12 months
Explanation: The extra 0.25 day is why we have a **Leap Year** every four years.

The imaginary line around which Earth rotates is called it's
a) Orbit
b) **Axis**
c) Equator
d) Latitude

Day and night are caused by Earth's
a) Revolution
b) **Rotation**
c) Tilt
d) Gravity

Seasons on Earth are caused by its revolution and
a) Distance from the Sun
b) **Axial Tilt**
c) Speed of rotation
d) Moon's gravity
Explanation: The 23.5-degree tilt causes different parts of Earth to receive more or less sunlight throughout the year.

Which planet has a day longer than its year
a) Mercury
b) **Venus**
c) Mars
d) Jupiter
Explanation: **Venus** rotates so slowly that it takes more time to spin once than to orbit the Sun.

Which planet's atmosphere is mostly composed of Nitrogen and Oxygen
a) Venus
b) Mars
c) **Earth**
d) Jupiter
Explanation: **Earth's** atmosphere is approximately 78% Nitrogen and 21% Oxygen.

The 'Kuiper Belt' is located beyond the orbit of
a) Saturn
b) Uranus
c) **Neptune**
d) Jupiter
Explanation: The **Kuiper Belt** is home to many dwarf planets and icy objects like **Pluto**.

Comets are often called
a) Shooting stars
b) **Dirty Snowballs**
c) Space rocks
d) Fireballs
Explanation: **Comets** are made of ice, dust, and rock that vaporize as they get close to the Sun.

The glowing tail of a comet always points
a) Towards the Sun
b) **Away from the Sun**
c) Towards the Earth
d) In the direction of travel
Explanation: The **Solar Wind** pushes the gas and dust away from the Sun, creating the tail.

Halley's Comet appears in Earth's sky every
a) 10 years
b) **75-76 years**
c) 100 years
d) 500 years
Explanation: **Halley's Comet** is a "periodic" comet that last appeared in 1986.

Meteors are commonly known as
a) Comets

b) **Shooting Stars**
c) Planets
d) Moons
Explanation: A **Meteor** is a flash of light caused by a space rock burning up in Earth's atmosphere.

If a meteor survives the atmosphere and hits the ground it is called a
a) Meteoroid
b) **Meteorite**
c) Asteroid
d) Comet
Explanation: **Meteorites** are the actual rocks that land on the surface.

Which planet was named after the Roman king of the gods
a) Mars
b) Saturn
c) **Jupiter**
d) Mercury
Explanation: **Jupiter** is the largest planet, so it was named after the most powerful god.

Which planet has the most circular orbit
a) Mercury
b) **Venus**
c) Mars
d) Pluto

The distance between the Earth and the Sun is called one
a) Light year
b) **Astronomical Unit (AU)**
c) Parsec
d) Kilometer
Explanation: One **AU** is approximately 150 million kilometers.

Light from the Sun takes about how long to reach Earth
a) 1 second
b) **8 minutes and 20 seconds**
c) 1 hour
d) 1 year
Explanation: At the speed of light, it takes roughly 8 minutes and 20 seconds to travel 1 **AU**.

Triton is the only large moon that orbits in the opposite direction of its planet's rotation. It belongs to
a) Jupiter
b) Saturn
c) **Neptune**
d) Uranus
Explanation: **Triton** is believed to be a captured Kuiper Belt object.

Which planet has the most extreme temperature differences between day and night
a) **Mercury**
b) Venus
c) Mars
d) Earth
Explanation: **Mercury** has no atmosphere to trap heat, so it is scorching during the day and freezing at night.

Ganymede is the largest moon in the solar system. It orbits
a) Saturn
b) **Jupiter**
c) Neptune
d) Uranus
Explanation: **Ganymede** with a diameter of approximately **5,270 km**, is even larger than the planet **Mercury**.

Which planet's name comes from the Greek god of the sky
a) Mars
b) Jupiter
c) **Uranus**
d) Neptune
Explanation: Most planets are named after Roman gods, but **Uranus** is Greek.

Which planet has a surface pressure 90 times greater than Earth's
a) Mars
b) **Venus**
c) Jupiter
d) Mercury
Explanation: The thick atmosphere of **Venus** would crush a human instantly.

What is the 'Great Dark Spot'
a) A hole in the Sun
b) **A storm on Neptune**
c) A crater on Mars
d) A dark moon
Explanation: Similar to Jupiter's Red Spot

The Heliocentric model of the solar system was proposed by
a) Ptolemy
b) **Copernicus**
c) Aristotle
d) Galen
Explanation: **Copernicus** correctly stated that the planets revolve around the **Sun**, not the Earth.

What is the 'Solar Wind'
a) Oxygen blowing from the Sun
b) **A stream of charged particles released from the Sun**
c) The heat from the Sun
d) Light waves
Explanation: The **Solar Wind** interacts with Earth's magnetic field to create **Auroras**.

What is the primary fuel for a star like the Sun
a) Oxygen
b) **Hydrogen**
c) Helium
d) Carbon
Explanation: Stars produce energy by fusing **Hydrogen** into **Helium**.

The process by which stars produce energy is called
a) Nuclear Fission
b) **Nuclear Fusion**
c) Combustion
d) Chemical Reaction
Explanation: **Nuclear Fusion** occurs under extreme pressure and temperature in the star's core.

What is the name of our galaxy
a) Andromeda
b) **Milky Way**
c) Sombrero
d) Triangulum
Explanation: The **Milky Way** is a barred spiral galaxy containing hundreds of billions of stars.

The distance that light travels in one year is called a
a) Astronomical Unit
b) **Light Year**
c) Parsec
d) Kilometer
Explanation: A **Light Year** is a unit of distance, not time (approx. 9.46 trillion km).

Which star is the closest star to Earth
a) Alpha Centauri

b) **The Sun**
c) Sirius
d) Polaris
Explanation: The **Sun** is a star located at the center of our solar system.

The closest star system to our solar system is
a) Sirius
b) **Proxima Centauri**
c) Betelgeuse
d) Vega
Explanation: **Proxima Centauri** is about 4.24 light-years away from us.

Which color indicates the hottest stars
a) Red
b) Yellow
c) **Blue**
d) Orange
Explanation: **Blue** stars have the highest surface temperatures, while red stars are the coolest.

A massive explosion of a dying star is called a
a) Nebula
b) **Supernova**
c) Black Hole
d) Pulsar
Explanation: A **Supernova** can outshine an entire galaxy for a few weeks.

What remains after a massive star's core collapses if it is not heavy enough for a Black Hole
a) White Dwarf
b) **Neutron Star**
c) Red Giant
d) Planet
Explanation: **Neutron Stars** are incredibly dense objects with radius 10 to 15 km while mass is incredibly large.

A region of space where gravity is so strong that even light cannot escape is a
a) White Dwarf
b) **Black Hole**
c) Nebula
d) Quasar
Explanation: **Black Holes** are formed from the remains of the most massive stars.

The boundary surrounding a Black Hole is called the
a) Singularity
b) **Event Horizon**
c) Orbit
d) Core
Explanation: Once something crosses the **Event Horizon**, it can never return.

The 'Big Bang Theory' explains the
a) Death of a star
b) **Origin of the Universe**
c) Formation of the Moon
d) End of the world
Explanation: The **Big Bang** suggests the universe began as a single point and has been expanding ever since.

What is a 'Nebula'
a) A dying star
b) **A cloud of gas and dust in space**
c) A small planet
d) A type of telescope
Explanation: **Nebulae** are often called "Stellar Nurseries" because stars are born inside them.

The North Star is scientifically known as
a) Sirius
b) **Polaris**

c) Alpha Centauri
d) Rigel
Explanation: **Polaris** appears stationary in the sky because it is aligned with Earth's axis.

Which galaxy is the nearest large neighbor to the Milky Way
a) **Andromeda**
b) Triangulum
c) Large Magellanic Cloud
d) Whirlpool
Explanation: **Andromeda** is a spiral galaxy about 2.5 million light-years away.

What is the 'Sunspot'
a) A hole in the Sun
b) **A cooler, darker region on the Sun's surface**
c) An explosion on the Sun
d) A planet passing the Sun
Explanation: **Sunspots** are caused by intense magnetic activity.

The visible surface of the Sun is called the
a) Corona
b) **Photosphere**
c) Chromosphere
d) Core
Explanation: The **Photosphere** is the layer from which sunlight is emitted.

The outermost layer of the Sun's atmosphere, visible during an eclipse, is the
a) Photosphere
b) **Corona**
c) Core
d) Radiative Zone
Explanation: The **Corona** is a halo of plasma that extends millions of kilometers into space.

A star that has used up its fuel and collapsed to a small, dense size is a
a) Red Giant
b) **White Dwarf**
c) Supernova
d) Protostar
Explanation: Our **Sun** will eventually become a **White Dwarf** after its Red Giant phase.

The study of the universe as a whole is called
a) Astrology
b) **Cosmology**
c) Astronomy
d) Geology
Explanation: **Cosmology** looks at the origin, evolution, and eventual fate of the universe.

A group of stars that forms a pattern in the sky is a
a) Galaxy
b) **Constellation**
c) Nebula
d) Solar System

Which telescope was launched in 1990 and revolutionized our view of the universe
a) James Webb
b) **Hubble Space Telescope**
c) Keck Telescope
d) Galileo Telescope
Explanation: **Hubble** has provided the most detailed images of distant galaxies and nebulae.

PHOTOELECTRIC AND QUANTUM THEORY

The concept that light energy is emitted in discrete packets is called
a) Wave Theory
b) **Quantum Theory**
c) Particle Theory
d) Electromagnetic Theory

Quantum Mechanics is the branch of physics that deals with
a) Planets
b) **Subatomic particles**
c) Sound waves
d) Heat transfer
Explanation: It replaces classical mechanics for things at the scale of atoms and electrons.

The discrete packets of light energy are known as
a) Electrons
b) **Photons**
c) Protons
d) Neutrons

The energy of a photon is directly proportional to its
a) Wavelength
b) **Frequency**
c) Amplitude
d) Velocity
Explanation: According to **Planck's Equation**, higher frequency light carries more energy per photon.

What is the formula for the energy of a photon
a) $E = mc^2$
b) **$E = hf$**
c) $E = 1/2\ mv^2$
d) $E = h\lambda$
Explanation: In this equation, **h** is **Planck's Constant** and **f** is the frequency of light.

What is the approximate value of Planck's Constant (h)
a) 3×10^8 J. s
b) **6.63×10^{-34} J. s**
c) 9.1×10^{-31} J. s
d) 1.6×10^{-19} J. s

If the frequency of light is doubled the energy of the photon will
a) Remain the same
b) **Double**
c) Become half
d) Become four times
Explanation: Since **$E = hf$**, the energy is directly proportional to the frequency.

The momentum of a photon is given by the formula
a) $p = mv$
b) **$p = h/\lambda$**
c) $p = hf$
d) $p = mc^2$

Which phenomenon provides the strongest evidence for the particle nature of light
a) Interference
b) Diffraction
c) **Photoelectric Effect**
d) Polarization

Photons travel in a vacuum with a speed of
a) 330 m/s
b) **3×10^8 m/s**

c) 1.6×10^{-19} m/s
d) 0 m/s

The rest mass of a photon is
a) 9.1×10^{-31} kg
b) 1.6×10^{-27} kg
c) **Zero**
d) Infinite
Explanation: A **Photon** only exists while in motion; it has no mass at rest.

The energy of a photon in terms of wavelength (λ) is
a) $E = h\lambda$
b) $\mathbf{E = hc/\lambda}$
c) $E = \lambda/hc$
d) $E = h/\lambda$
Explanation: Since $\mathbf{f = c/\lambda}$, substituting this into $\mathbf{E = hf}$ gives $\mathbf{E = hc/\lambda}$.

Light of which colour has the most energetic photons
a) Red
b) Yellow
c) Green
d) **Violet**
Explanation: **Violet** light has the highest frequency and shortest wavelength in the visible spectrum.

Light of which colour has the least energetic photons
a) **Red**
b) Blue
c) Indigo
d) Violet
Explanation: **Red** light has the lowest frequency and longest wavelength in the visible spectrum.

An electron volt (eV) is a unit of
a) Potential
b) **Energy**
c) Charge
d) Force
Explanation: One **eV** is the energy gained by an electron accelerating through a potential difference of one volt.

One electron volt (1 eV) is equal to
a) 1.6×10^{-19} J
b) $\mathbf{1.602 \times 10^{-19}}$ **J**
c) 6.63×10^{-34} J
d) 9.1×10^{-31} J

The dual nature of light means it acts as both
a) Solid and Liquid
b) **Wave and Particle**
c) Positive and Negative
d) Heat and Light
Explanation: Light shows wave properties (Interference) and particle properties (Photoelectric Effect).

Who proposed the Quantum Theory of Radiation in 1900
a) Albert Einstein
b) **Max Planck**
c) Niels Bohr
d) Isaac Newton
Explanation: **Max Planck** introduced the idea of energy quantization to solve the Black Body Radiation problem.

According to Quantum Theory energy is not emitted continuously but in
a) Streams
b) **Quanta**
c) Rays
d) Waves
Explanation: **Quanta** refers to the minimum amount of any physical entity

involved in an interaction. ***Max Planck was awarded the Nobel Prize in Physics in 1918***

The intensity of light in the photon model depends on the

a) Energy of each photon
b) **Number of photons per unit area per second**
c) Velocity of photons
d) Wavelength of photons

Explanation: High intensity means more photons are hitting the surface, not that individual photons are stronger.

If the wavelength of light increases the energy of its photons

a) Increases
b) **Decreases**
c) Stays the same
d) Becomes zero

Explanation: Energy is inversely proportional to wavelength (**$E \propto 1/\lambda$**).

The momentum of a photon is inversely proportional to its

a) Frequency
b) **Wavelength**
c) Energy
d) Speed

Explanation: According to **$p = h/\lambda$**, as wavelength increases, momentum decreases.

Which of the following has zero charge

a) Electron
b) **Photon**
c) Proton
d) Alpha particle

Explanation: **Photons** are electrically neutral.

The emission of electrons from a metal surface when light shines on it is

a) Thermionic Emission
b) **Photoelectric Effect**
c) Secondary Emission
d) Field Emission

In the Photoelectric Effect light acts as a

a) Wave
b) **Particle**
c) Longitudinal Wave
d) Stationary Wave

Explanation: Only the **Particle (Photon)** model can explain why emission happens instantaneously.

The minimum energy required to remove an electron from a metal surface is

a) Kinetic Energy
b) **Work Function**
c) Threshold Frequency
d) Stopping Potential

Explanation: Each metal has a specific **Work Function (Φ)** based on its atomic structure.

The Work Function of a metal depends on

a) Intensity of light
b) **Nature of the metal**
c) Frequency of light
d) Wavelength of light

Explanation: Different metals (like Cesium vs. Tungsten) hold their electrons with different strengths.

The Work Function (Φ) is mathematically expressed as

a) $\Phi = hf$
b) **$\Phi = hf_0$**
c) $\Phi = 1/2\ mv^2$
d) $\Phi = h\ c\ \lambda$

Explanation: Here **f_0** is the **Threshold Frequency** specific to that metal.

Threshold Frequency is the minimum frequency required to
a) Stop an electron
b) **Cause photoelectric emission**
c) Increase the speed of light
d) Heat the metal
Explanation: If light frequency is below f_0, no electrons are emitted regardless of intensity.

If the frequency of incident light is less than threshold frequency then
a) Few electrons are emitted
b) **No electrons are emitted**
c) Many electrons are emitted
d) Metal melts
Explanation: This "All-or-Nothing" behavior is a key proof of the **Quantum Theory**.

The maximum Kinetic Energy of photoelectrons depends on
a) Intensity of light
b) **Frequency of light**
c) Time of exposure
d) Surface area
Explanation: Higher frequency photons give more "kick" to the electrons.

The number of photoelectrons emitted per second is proportional to
a) Frequency
b) **Intensity of light**
c) Work Function
d) Stopping Potential
Explanation: More light (Intensity) means more photons, which knock out more electrons.

Who explained the Photoelectric Effect using Quantum Theory in 1905
a) Max Planck
b) **Albert Einstein**
c) Robert Millikan
d) Werner Heisenberg
Explanation: **Einstein** extended Planck's idea to light itself, calling the packets "Light Quanta."

Einstein was awarded the Nobel Prize in 1921 for his work on
a) Theory of Relativity
b) **Photoelectric Effect**
c) Brownian Motion
d) Nuclear Fission
Explanation: While famous for Relativity, his Nobel Prize was officially for the law of the **Photoelectric Effect**.

Einstein's Photoelectric Equation is written as
a) $hf = \Phi$
b) $\mathbf{hf = \Phi + K.E._{max}}$
c) $hf = K.E.$
d) $\Phi = hf + K.E.$
Explanation: This shows that photon energy is split into removing the electron and giving it motion.

Photoelectric Effect is an instantaneous process because
a) Light is very fast
b) **One photon interacts with one electron**
c) Metals are good conductors
d) Electrons are very small
Explanation: Unlike wave heating, the energy transfer in a photon-electron collision happens at once.

Photoelectric cells are used in
a) Solar panels
b) **Automatic Street lights**
c) Heaters
d) Transformers

Explanation: They act as "Light Sensors" that can turn circuits on or off based on light.

Which metal is commonly used in photocells for visible light
a) Iron
b) **Cesium**
c) Copper
d) Tungsten
Explanation: **Cesium** has a very low work function, allowing it to emit electrons even with low-energy visible light.

The maximum wavelength of light that can cause photoelectric effect is called
a) Threshold Frequency
b) **Threshold Wavelength**
c) De Broglie Wavelength
d) Compton Wavelength
Explanation: Light must have a wavelength *shorter* than this value to have enough energy.

The formula for Threshold Wavelength (λ_0) is
a) $\lambda_0 = h/\Phi$
b) $\boldsymbol{\lambda_0 = hc/\Phi}$
c) $\lambda_0 = c/f$
d) $\lambda_0 = \Phi/hc$
Explanation: Derived from $\boldsymbol{\Phi = hf_0 = h(c/\lambda_0)}$.

Photoelectric current depends on
a) Frequency
b) **Intensity**
c) Speed of light
d) Stopping potential
Explanation: Current is a measure of the number of electrons flowing per second.

Who proposed that material particles like electrons also have wave properties
a) Einstein
b) **Louis de Broglie**
c) Schrodinger
d) Bohr
Explanation: In 1924, **De Broglie** suggested that nature is symmetrical and matter should behave like waves.

The wavelength associated with a moving particle is called
a) Electromagnetic Wavelength
b) **De Broglie Wavelength**
c) Planck Wavelength
d) Compton Wavelength
Explanation: Every moving object has a wave nature, though it is only noticeable for tiny particles.

The De Broglie Wavelength (λ) is given by the formula
a) $\lambda = hf$
b) $\boldsymbol{\lambda = h/p}$
c) $\lambda = mc^2/h$
d) $\lambda = h/mv^2$
Explanation: This equation links the wave property (λ) with the particle property (Momentum **p**).

If the velocity of a particle increases its De Broglie Wavelength
a) Increases
b) **Decreases**
c) Stays the same
d) Becomes zero
Explanation: Since $\boldsymbol{\lambda = h/mv}$, wavelength is inversely proportional to velocity.

De Broglie Waves are also known as
a) Light Waves
b) **Matter Waves**

c) Sound Waves
d) Shock Waves

Which experiment experimentally confirmed the wave nature of electrons
a) Young's Double Slit
b) **Davisson-Germer Experiment**
c) Millikan's Oil Drop
d) Michelson-Morley
Explanation: They observed electron diffraction from a nickel crystal, which is a wave property.

The wave nature of matter is not observable for large objects because
a) Their velocity is zero
b) **Their De Broglie Wavelength is too small**
c) They have no mass
d) They move too fast
Explanation: Because **h** is so small, the wavelength of a football is far too tiny to detect.

The resolution of a microscope depends on the
a) Intensity
b) **Wavelength**
c) Frequency
d) Speed
Explanation: Smaller wavelengths allow for the viewing of much smaller details.

If an electron and a proton move with the same velocity which has a longer wavelength
a) **Electron**
b) Proton
c) Both are same
d) Neither has a wavelength
Explanation: Since the **Electron** has a smaller mass, its wavelength is longer (**$\lambda \propto 1/m$**).

The concept of Wave-Particle Duality was applied to atoms by
a) Newton
b) **Niels Bohr**
c) Dalton
d) Rutherford
Explanation: **Bohr** used the idea of standing waves to explain why electrons only exist in certain orbits.

The uncertainty principle was proposed by
a) De Broglie
b) **Werner Heisenberg**
c) Erwin Schrodinger
d) Max Born

The Heisenberg Uncertainty Principle relates
a) Mass and Energy
b) **Position and Momentum**
c) Time and Distance
d) Force and Acceleration
Explanation: The more precisely you know where a particle is, the less you know about how fast it is moving.

The mathematical form of the Uncertainty Principle is
a) $\Delta x\ \Delta p = h$
b) **$\Delta x\ \Delta p \geq h/4\pi$**
c) $\Delta x\ \Delta p = 0$
d) $\Delta x/\Delta p = h$

The main equation of Wave Mechanics was developed by
a) Heisenberg
b) **Erwin Schrodinger**
c) Einstein

d) Maxwell
Explanation: The **Schrodinger Equation** describes how the wave function of a system changes over time.

The 'Wave Function' of a particle is represented by the symbol
a) Phi (φ)
b) **Psi (ψ)**
c) Lambda (λ)
d) Omega (ω)
Explanation: **Psi** contains all the information about the state of a quantum system.

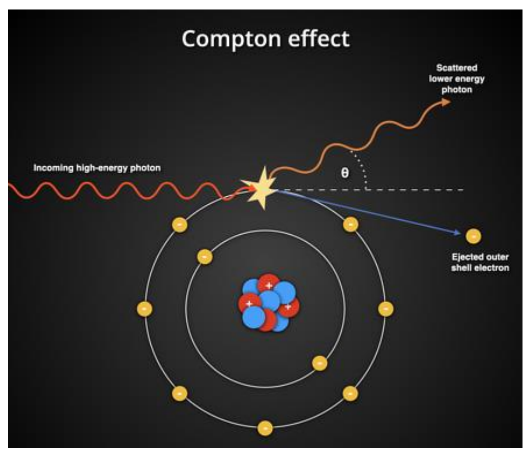
In Quantum Mechanics particles are described as
a) Solid balls
b) **Wave Packets**
c) Rays
d) Points
Explanation: A **Wave Packet** is a localized wave that represents a particle.

The Compton Effect involves the scattering of
a) Electrons by protons
b) **X-rays by electrons**
c) Light by mirrors
d) Sound by walls
Explanation: **Arthur Compton** showed that X-rays shift in wavelength when they hit electrons, proving light has momentum.

In the Compton Effect the scattered X-ray has
a) Smaller wavelength
b) **Larger wavelength**
c) Same wavelength
d) Higher frequency
Explanation: The X-ray loses energy to the electron, so its frequency drops and wavelength increases.

The change in wavelength in Compton scattering is called
a) Doppler Shift
b) **Compton Shift**
c) Redshift
d) Phase Shift
Explanation: The shift depends only on the angle of scattering, not on the material.

The Compton Shift is maximum when the scattering angle is
a) 0°
b) 90°
c) **180°**
d) 45°
Explanation: A "head-on" collision results in the maximum energy transfer and maximum wavelength change.



The Compton Effect proves the
a) Wave nature of X-rays
b) **Particle nature of X-rays**
c) Mass of light
d) Polarization
Explanation: It treats the interaction as a "billiard ball" collision between a photon and an electron.

The minimum energy required for Pair Production is
a) 0.51 MeV
b) **1.02 MeV**
c) 13.6 eV
d) 10 eV

Pair Production can only occur in the presence of a
a) Vacuum
b) **Heavy Nucleus**
c) Magnetic Field
d) Battery
Explanation: The nucleus is needed to conserve momentum during the process.

The opposite of Pair Production is called
a) Ionization
b) **Annihilation of Matter**
c) Fusion
d) Fission
Explanation: When an electron and positron meet, they disappear and turn back into two gamma-ray photons.

A Positron is an anti-particle of an
a) Proton
b) **Electron**
c) Neutron
d) Photon
Explanation: It has the same mass as an electron but a positive charge.

Which of the following has the shortest wavelength
a) Radio waves
b) **Gamma rays**
c) X-rays
d) Visible light
Explanation: **Gamma rays** are the most energetic part of the electromagnetic spectrum.

The wavelength of X-rays is approximately
a) 1 m
b) **10^{-10} m (1 Angstrom)**
c) 10^{-6} m
d) 10^{-15} m
Explanation: This size is comparable to the spacing between atoms in a crystal.

X-rays were discovered by
a) Becquerel
b) **Wilhelm Roentgen**
c) Marie Curie
d) J.J. Thomson
Explanation: He discovered them in 1895 and called them "X" because they were unknown.

X-rays are produced when fast-moving electrons hit a
a) Gas
b) **Metal target**
c) Liquid
d) Mirror
Explanation: The kinetic energy of electrons is converted into electromagnetic radiation.

The wavelength of X-rays depends on
a) Target material
b) **Accelerating Voltage**
c) Intensity of electrons
d) Cooling system
Explanation: Higher voltage creates faster electrons, which can produce X-rays with higher energy and shorter wavelengths.

An ideal black body is one which
a) Reflects all radiation
b) **Absorbs all radiation incident on it**
c) Is black in colour only
d) Does not emit radiation

Explanation: It is a perfect absorber and a perfect emitter.

The spectrum of an atom consists of
a) Continuous band of colours
b) **Discrete lines**
c) Only dark lines
d) Only one colour
Explanation: Every element has a unique "fingerprint" of specific light frequencies.

When an electron jumps from a higher orbit to a lower orbit it
a) Absorbs a photon
b) **Emits a photon**
c) Gains mass
d) Stops rotating
Explanation: The energy difference is released as light energy ($\mathbf{E_2 - E_1 = hf}$).

Which of the following is a property of laser light
a) Divergent
b) **Monochromatic**
c) Polychromatic
d) Incoherent
Explanation: **Monochromatic** means it consists of only one wavelength/colour.

The first working laser was a
a) Gas Laser
b) **Ruby Laser**
c) Diode Laser
d) Dye Laser
Explanation: It used a synthetic ruby crystal and a flash lamp.

A 'Photon' of light is emitted when
a) An electron is captured
b) **An excited electron falls to a lower state**
c) A nucleus splits
d) Light reflects
Explanation: This is the origin of all light we see from atoms.

X-rays are not deflected by electric or magnetic fields because they are
a) Very heavy
b) **Electrically neutral**
c) Moving too fast
d) Part of the nucleus
Explanation: Like all electromagnetic waves, they carry no charge.

Which of the following has the highest penetrating power
a) Alpha rays
b) Beta rays
c) **Gamma rays**
d) Visible light
Explanation: **Gamma rays** can pass through several centimeters of lead.

The phenomenon of splitting of spectral lines in a magnetic field is
a) Photoelectric Effect
b) **Zeeman Effect**
c) Stark Effect
d) Doppler Effect
Explanation: It shows that energy levels are further split by magnetic interaction with electron spin.

The splitting of spectral lines in an electric field is called
a) Zeeman Effect
b) **Stark Effect**
c) Raman Effect
d) Compton Effect
Explanation: It is the electrical version of the Zeeman Effect.